



O-RAN Fronthaul Working Group

Fronthaul Interoperability Test Specification (IOT)

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Contents

Foreword	5
Modal verbs terminology	5
1 Scope	6
2 References	6
2.1 Normative References	6
2.2 Informative References	7
3 Definition of terms, symbols and abbreviations	7
3.1 Terms	7
3.2 Abbreviations	9
4 Fronthaul Interoperability	10
4.1 General	10
4.2 Summary of Test Scenarios	11
4.3 Future Enhancements	11
4.4 Fronthaul M-Plane Architectural options	12
4.5 Fronthaul Synchronization options	12
5 Interoperability Measurements	13
5.1 Interoperability Standard Test Definitions	13
5.1.1 Standard Test Configurations	13
5.1.2 Device Under Test (DUT)	13
5.1.3 Network Management System (NMS)	14
5.1.4 Testing Tools	15
5.1.5 Time Synchronization	18
5.1.6 Assumptions	18
5.1.7 Specifications to be used for testing	18
5.1.8 Interoperability (IOT) Test Profiles	18
5.1.9 Measurements of interest	19
5.2 Standard Test Data Configurations	19
5.3 Standard Test Execution	19
5.4 Interoperability Test Cases	19
5.5 Reporting and validation of used capabilities	20
6 M-Plane IOT Test	21
6.1 Overview	21
6.2 Start-up in hierarchical mode	21
6.2.1 Test Description and Applicability	21
6.2.2 Minimum Requirements (Prerequisites)	21
6.2.3 Test Purpose and Scope	21
6.2.4 Testability requirements imposed on O-RU and O-DU	22
6.2.5 Methodology and Initial Conditions	22
6.2.6 Procedure: M-Plane start up test	22
6.2.7 Test Requirement (expected result)	25
6.2.8 Test Report (Failure)	25
6.3 Start-up in hybrid mode	26
6.3.1 Test Description and Applicability	26
6.3.2 Minimum Requirements (Prerequisites)	26
6.3.3 Test Purpose and Scope	26
6.3.4 Testability requirements imposed on O-RU, O-DU and NMS	27
6.3.5 Methodology and Initial Conditions	27
6.3.6 Procedure: M-Plane start up test	27
6.3.7 Test Requirement (expected result)	29
6.3.8 Test Report (Failure)	29
6.3.9 Supplementary test: NETCONF Access Control (Failure)	29

7	S-Plane IOT Test.....	29
7.1	Overview	29
7.2	Functional test of O-DU + O-RU using ITU-T G.8275.1 profile (LLS-C1)	32
7.2.1	Test Description and Applicability	32
7.2.2	Minimum Requirements (Prerequisites).....	32
7.2.3	Purpose and Scope	32
7.2.4	Testability Requirements imposed on O-RU and O-DU	32
7.2.5	Test Methodology	32
7.2.6	Test Requirements (expected results).....	33
7.3	Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.1 profile (LLS-C2).....	34
7.3.1	Test Description and Applicability	34
7.3.2	Minimum Requirements (Prerequisites).....	34
7.3.3	Purpose and Scope	34
7.3.4	Testability Requirements imposed on O-RU and O-DU	34
7.3.5	Test Methodology	35
7.3.6	Test Requirements (expected results).....	35
7.4	Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.1 profile (LLS-C3).....	37
7.4.1	Test Description and Applicability	37
7.4.2	Minimum Requirements (Prerequisites).....	37
7.4.3	Purpose and Scope	37
7.4.4	Testability Requirements imposed on O-RU, O-DU and bridged network	37
7.4.5	Test Methodology	37
7.4.6	Test Requirements (expected results).....	38
7.5	Functional test of O-DU + O-RU using ITU-T G.8275.2 profile (LLS-C1)	39
7.6	Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.2 profile (LLS-C2).....	39
7.7	Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.2 profile (LLS-C3).....	39
7.8	Functional test of O-DU + bridged network + O-RU (LLS-C4)	40
7.9	Performance test of O-DU + Two O-RUs using ITU-T G.8275.1 profile (LLS-C1)	40
7.9.1	Test Description and applicability	40
7.9.2	Minimum Requirements (Prerequisites).....	40
7.9.3	Purpose and scope	40
7.9.4	Testability Requirements imposed on O-RU and O-DU	40
7.9.5	Test Methodology	40
7.9.6	Test Requirement (expected result).....	41
7.10	Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.1 profile (LLS-C2).....	41
7.10.1	Test Description and applicability	41
7.10.2	Minimum Requirements (Prerequisites).....	41
7.10.3	Purpose and Scope	41
7.10.4	Testability Requirements imposed on O-RU and O-DU	41
7.10.5	Test Methodology	42
7.10.6	Test Requirement (expected result).....	42
7.11	Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.1 profile (LLS-C3).....	42
7.11.1	Test Description and applicability	42
7.11.2	Minimum Requirements (Prerequisites).....	42
7.11.3	Purpose and scope	43
7.11.4	Testability Requirements imposed on O-RU and O-DU	43
7.11.5	Test Methodology	43
7.11.6	Test Requirement (expected result).....	43
7.12	Performance test of O-DU + Two O-RUs using ITU-T G.8275.2 profile (LLS-C1)	43
7.13	Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.2 profile ((LLS-C2)	44
7.14	Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.2 profile (LLS-C3).....	44
7.15	Performance test of O-DU + bridged network + Two O-RUs (LLS-C4)	44
8	C/U-Plane IOT Test	44
8.1	Overview	44
8.2	Radio Layer 3 C-Plane establishment and Initial Radio U-Plane data transfer	44
8.2.1	Test Description and Applicability	44
8.2.2	Minimum Requirements (Prerequisites).....	45
8.2.3	Test Purpose and Scope	45
8.2.4	Testability requirements imposed on O-RU and O-DU	45
8.2.5	Test Methodology	46

8.2.6	Test Requirement (expected result).....	46
8.3	Radio U-Plane downlink data transfer (Downlink throughput performance).....	46
8.3.1	Radio U-Plane downlink data transfer performance with one UE.....	46
8.3.2	Radio U-Plane downlink data transfer performance with two UEs.....	48
8.4	Radio U-Plane uplink data transfer (Uplink throughput performance).....	50
8.4.1	Radio U-Plane uplink data transfer performance with one UE.....	50
8.4.2	Radio U-Plane uplink data transfer performance with two UEs.....	52
9	C/U-Plane Delay Management IOT Test.....	55
9.1	General.....	55
9.2	Test environment.....	56
9.3	Timing accuracy definition.....	57
9.4	Delay Management #1, minimum fronthaul latency.....	57
9.4.1	Test Description and Applicability.....	57
9.4.2	Minimum Requirements.....	57
9.4.3	Test Purpose.....	57
9.4.4	Testability requirements imposed on O-RU and O-DU.....	57
9.4.5	Test Methodology.....	57
9.4.6	Test Requirement (expected result).....	58
9.5	Delay Management #2, maximum fronthaul latency.....	58
9.5.1	Test Description and Applicability.....	58
9.5.2	Minimum Requirements.....	58
9.5.3	Test Purpose.....	58
9.5.4	Testability requirements imposed on O-RU and O-DU.....	58
9.5.5	Test Methodology.....	59
9.5.6	Test Requirement (expected result).....	59
9.6	Delay Management #3, normal fronthaul latency.....	59
9.6.1	Test Description and Applicability.....	59
9.6.2	Minimum Requirements.....	59
9.6.3	Test Purpose.....	59
9.6.4	Testability requirements imposed on O-RU and O-DU.....	60
9.6.5	Test Methodology.....	60
9.6.6	Test Requirement (expected result).....	60
9.7	Delay Management #4, larger fronthaul latency then supported.....	60
9.7.1	Test Description and Applicability.....	60
9.7.2	Minimum Requirements.....	60
9.7.3	Test Purpose.....	61
9.7.4	Testability requirements imposed on O-RU and O-DU.....	61
9.7.5	Test Methodology.....	61
9.7.6	Test Requirement (expected result).....	61
Annex A	(normative): Profiles used for Interoperability Testing.....	62
A.1	General.....	62
A.2	M-Plane IOT Profile.....	65
A.2.1	M-Plane IOT Profile 1 Hierarchical-sudo.....	65
A.2.2	M-Plane IOT Profile 2 Hybrid-sudo+nms.....	67
A.2.3	M-Plane IOT Profile 3 Hierarchical-sudo-IPv6.....	69
A.2.4	M-Plane IOT Profile 4 Hybrid-sudo+nms-IPv6.....	71
A.3	CUS-Plane IOT Profiles.....	74
A.3.1	NR TDD.....	74
A.3.2	NR FDD.....	168
A.3.3	LTE FDD.....	183
A.3.4	LTE TDD.....	191
A.4	Delay Sets.....	195
A.4.1	NR TDD FR1 Cat-B mMIMO WDBF Delay Sets for DMRS-BF profiles.....	195
A.4.2	NR TDD FR1 Cat-B mMIMO CIBF Delay Sets.....	196
A.4.3	NR TDD FR1 Cat-B mMIMO DMRS-BF Delay Sets.....	198
Annex	(informative): Change history/Change request (history).....	199

Foreword

This Technical Specification (TS) has been produced by WG4 of the O-RAN Alliance.

The content of the present document is subject to continuing work within O-RAN and may change following formal O-RAN approval. Should the O-RAN Alliance modify the contents of the present document, it will be re-released by O-RAN with an identifying change of version date and an increase in version number as follows:

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Modal verbs terminology

In the present document "**shall**", "**shall not**", "should", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the O-RAN Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in O-RAN deliverables except when used in direct citation.

1 Scope

The present document specifies the interoperability testing (IOT) for O-DU and O-RU from different vendors connected using the O-RAN fronthaul (FH) interface. It is noted however that the same content can be utilized for the IOT for O-DU and O-RU from the same vendor connected using the O-RAN FH interface.

A guiding principle defining the tests in the present document is that any test shall exercise the fronthaul interface to a greater extent than the test prerequisite so that the test is a proper test of some fronthaul functionality or performance.

All tests focus on testing the fronthaul interface, but due to the non-intrusive nature of the tests, system-level aspects of the O-DU and O-RU are inevitably part of the interoperability tests too. Because of this, it is recognized that positive test outcomes indicate successful interoperability, but negative results may not be attributed solely to problems in the FH interface and additional investigation may be required.

It is also noted that additional test scenarios are required for comprehensive testing of the FH interface functionality and performance, these are not addressed in the present document.

In general, unless otherwise stated, the tests cover LTE (Stand-Alone), NR Non-Stand-Alone (NSA) and NR Stand-Alone (SA). An O-DU and O-RU are considered interoperable under a Profile Test Configuration (PTC), if the device under test (DUT) can pass all mandatory and applicable conditional mandatory tests using the specific Profile Test Configuration.

The material contained within the present document is of stage 3 [25].

2 References

2.1 Normative References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies.
- In the case of a reference to a 3GPP document, a non-specific reference implicitly refers to the latest version of that document in Release 18, or the latest 3GPP release prior to Release 18 that includes that document. .
- The versions of the ITU-T documents reflect their publication dates at the time of the writing. For the most recent version, please refer to ITU-T <https://www.itu.int/rec/T-REC-G/e>.

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|-----|--|
| [1] | Void |
| [2] | O-RAN WG4 Control, User and Synchronization Specification Version 17.01, February 2025 |
| [3] | O-RAN WG4 Management Plane Specification Version 17.01, February 2025 |
| [4] | O-RAN Management Plane Yang Models Version 17.01, February 2025 |
| [5] | ITU-T G.8275.1, Precision time protocol telecom profile for phase/time synchronization with full timing support from the network, ITU, March 2020 |
| [6] | ITU-T G.8275.2, Precision time protocol telecom profile for phase/time synchronization with partial timing support from the network, ITU, March 2020 |
| [7] | ITU-T G.8271.1 “Network limits for time synchronization in packet networks”, ITU, March 2020 |
| [8] | ITU-T G.8271.2 “Network limits for time synchronization in packet networks with partial timing support from the network”, ITU, Aug 2017 |

- [9] ITU-T G.8273 “Framework of phase and time clocks”, March 2018
- [10] ITU-T G.8261. “Timing and synchronization aspects in packet networks”, August 2019
- [11] eCPRI Specification v1.0 “Common Public Radio Interface: eCPRI Interface Specification”, August 2017
- [12] 3GPP TS 36.104, “Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) radio transmission and reception”
- [13] 3GPP TS 38.104, “NR; Base Station (BS) radio transmission and reception”
- [14] 3GPP TS 36.211, “Evolved Universal Terrestrial Radio Access (E-UTRA); Physical channels and modulation”
- [15] 3GPP TS 38.211, “NR; Physical channels and modulation”, 3GPP, (Release 15.8.0)
- [16] 3GPP TS 36.331, “Evolved Universal Terrestrial Radio Access (E-UTRA); Radio Resource Control (RRC)”
- [17] 3GPP TS 38.331, “NR; Radio Resource Control (RRC); Protocol specification”
- [18] 3GPP TS 36.141, “Evolved Universal Terrestrial Radio Access (E-UTRA); Base Station (BS) conformance testing”, 3GPP, (Release 15.10.0)
- [19] 3GPP TS 38.141-2, “NR; Base Station (BS) conformance testing Part 2: Radiated conformance testing”, 3GPP, (Release 15.7.0)
- [20] 3GPP TS 23.401, “General Packet Radio Service (GPRS) enhancements for Evolved Universal Terrestrial Radio Access Network (E-UTRAN) access”, 3GPP, (Release 15.12.0)
- [21] 3GPP TS 23.502, “Procedures for the 5G System (5GS)”, 3GPP, (Release 15.11.0)
- [22] 3GPP TS 37.340, “Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Multi-connectivity”, 3GPP, (Release 15.10.0)
- [23] 3GPP TS 38.214, “NR; Physical layer procedures for data”, 3GPP, (Release 15.11.0)
- [24] O-RAN Fronthaul Working Group, Conformance Test Specification Version 12, February 2025
- [25] ITU-T Recommendation I.130 “Method for the characterization of telecommunication services supported by an ISDN and network capabilities of an ISDN”
- [26] 3GPP TS 38.306, “NR; User Equipment (UE) radio access capabilities”

2.2 Informative References

- [i.1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications"

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms and definitions given in 3GPP TR 21.905 [i.1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [i.1].

C-Plane	Control Plane: refers specifically to real-time control between O-DU and O-RU, and should not be confused with the UE's control plane
Customized IOT Configuration	A set of IOT parameters not matching any specific IOT Profile Test Configuration that may be defined by anyone wanting to execute an interoperability test defined in this specification but instead using a custom set of test parameters
DL	DownLink: data flow towards the radiating antenna

Delay Set	Collection of timing values specific to one (or more) beamforming methods. A Delay Set (DS) can include all or part of the timing values of a beamforming method
eNB	eNodeB (applies to LTE) <E-UTRAN NodeB / Evolved NodeB>
fm-pm	Fault Management, Performance Management role
gNB	gNodeB (applies to NR) <Next Generation NodeB>
IOT Profile	A high-level interoperability test configuration definition under which multiple more-detailed IOT Profile Test Configurations may fit, allowing the grouping of test cases in broad categories
IOT Profile Test	Configuration An interoperability test configuration defined within this specification that includes detailed parameter values that are representative of typical deployment cases, such that the results of interoperability tests using such a configuration may guide an understanding of the ability of a multi-vendor deployment to interoperate
M-Plane	Management Plane: refers to non-real-time management operations between the O-DU and the O-RU
NETCONF	Network Configuration Protocol. For details see RFC 6241, “Network Configuration Protocol (NETCONF)”, IETF, June 2011
NMS	A Network Management System dedicated to O-RU operations
nms	NMS role
NOT REQ	No requirement on the O-DU or O-RU as an entry criterion for IOT testing for a given functionality. The O-DU may support the functionality. The O-RU may support the functionality. See Table A.1-1
NSA	Non-Stand-Alone network mode that supports operation of SgNB attached to MeNB
O-CU	O-RAN Central Unit – a logical node hosting PDCP, RRC, SDAP and other control functions
O-DU	O-RAN Distributed Unit: a logical node hosting RLC/MAC/High-PHY layers based on a lower layer functional split. O-DU in addition hosts an M-Plane instance
O-RU	O-RAN Radio Unit <O-RAN Radio Unit: a logical node hosting Low-PHY layer and RF processing based on a lower layer functional split. This is similar to 3GPP’s “TRP” or “RRH” but more specific in including the Low-PHY layer (FFT/iFFT, PRACH extraction).>. O-RU in addition hosts M-Plane instance
O-RU REQ	The O-RU supports the functionality as an entry criterion for IOT testing. The O-DU may support the functionality. See Error! Reference source not found.
PLFS	Physical Layer Frequency Support, as per ITU-T G.8271.1. SyncE is one example
PTP	Precision Time Protocol (PTP) is a protocol for distributing precise time and frequency over packet networks. PTP is specified in the IEEE Standard 1588
PDCCH	Physical Downlink Control Channel applies for LTE and NR air interface
PBCH	Physical Broadcast Channel applies for LTE and NR air interface
REQ	The O-DU and O-RU support the functionality as an entry criterion for IOT testing. See Table A.1-1
SA	Stand-Alone network mode that supports operation of gNB attached to a 5G Core Network
SCS	OFDM Sub Carrier Spacing
SSH	Secure Shell protocol for remote login
SSB	Synchronization Signal Block, in 5G PBCH and synchronization signal are packaged as a single block
sudo	Super-User Do role
subordinate	The term “subordinate” is used as a replacement for “slave” and is consistent with its use in O-RAN WG4 CUS Specification [2]. When consensus emerges on how to eliminate the use of the term “slave” in the referenced standards organization, that approach will be applied to this specification

swm	Software fault management role
S-Plane	Synchronization Plane: Data flow for synchronization and timing information between nodes
SyncE	Synchronous Ethernet, is an ITU-T standard for computer networking that facilitates distribution of clock signals over the Ethernet physical layer
T-BC	Telecom Boundary Clock
TLS	Transport Layer Security protocol
TWAMP	Two-Way Active Measurement Protocol
UDP	User Datagram Protocol
UE	User Equipment terminology for a mobile device in LTE and NR
UL	UpLink: data flow from the UE towards the core network, that is from the O-RU towards in the O-DU in a Fronthaul context
U-Plane	User Plane: refers to IQ sample data transferred between O-DU and O-RU

3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [i.1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [i.1].

5GS	5G System, comprises the access network, core network and user equipment
ARFCN	Absolute Radio Frequency Channel Number
BFN	Beamforming network
CC	Component Carrier
CN	Core Network
C/U-Plane	C-Plane and U-Plane
DHCP	Dynamic Host Configuration Protocol
DL	DownLink
DS	Delay Set
DUT	Device Under Test
eNB	eNodeB
EARFCN	E-UTRA ARFCN
EVM	Error Vector Magnitude
FH	Fronthaul
GSCN	Global Synchronization Channel Number
gNB	gNodeB
MeNB	Master eNB
MIMO	Multiple Input Multiple Output
M-MIMO	Massive MIMO
mMIMO	M-MIMO
NETCONF	Network Configuration Protocol
NMS	Network Management System
NSA	Non-Stand-Alone

O-CU	O-RAN Central Unit
O-DU	O-RAN Distributed Unit
O-RU	O-RAN Radio Unit
OTA	Over the Air
PBCH	Physical Broadcast Channel
PCI	Physical Cell Identity
PRTC	Primary Reference Time Clock
PTC	Profile Test Configuration
RF	Radio Frequency
RSRP	Reference Signal Received Power
SA	Stand-Alone
SCS	Sub Carrier Spacing
SDAP	Service Data Adaptation Protocol
SgNB	Secondary gNB
SSB	Synchronization Signal Block
UE	User Equipment
UL	UpLink
U-Plane	User Plane

4 Fronthaul Interoperability

4.1 General

O-RAN WG4 has specified and published the O-RAN FH interface specifications (CUS-Plane [2] and M-Plane [3]) with the objective to enable interoperability of O-DU and O-RU from different vendors. The aim of the present document is to further facilitate such multi-vendor IOT using the O-RAN FH interface by defining standard test configurations, IOT profiles and interoperability test cases.

It shall be possible to perform interoperability testing in a non-intrusive manner; that is, in a manner in which the network elements under test are not required to support any functionality or mode of operation beyond that required for normal operation in a telecommunication network. However, making the endpoints of the FH interface between the O-DU and O-RU visible would require definition of new interfaces which may entail caching and transport of data. Furthermore, and more importantly, operators require that the total radio system functions and performs adequately when integrating O-DU and O-RU from different vendors. Consequentially, this specification approaches interoperability testing by means of system level testing.

FH interoperability testing by way of system test involves creation of a stimulus in the O-DU to O-RU direction using an actual or emulated O-CU, potentially with CN support/emulation, and measurement of the result at the output of the O-RU in the RF domain by an actual or emulated UE together with an RF signal/spectrum analyzer as required. Likewise, in the reverse direction, the stimulus to probe the FH in the O-RU to O-DU direction is provided by an actual or emulated UE and is measured at the output of the O-DU by an actual or emulated O-CU with CN support/emulation as required.

Inasmuch as the interoperability testing by way of system test involves configuring and collecting results from the O-RU it shall be effected by means of the M-Plane FH interface. For elements that fall outside of this scope, such as the O-DU, the configuration and collection of data and status information for testing purposes may be accomplished by NMS.

Interoperability involves testing the FH interface in terms of M-Plane, S-Plane, C-Plane and U-Plane. Some aspects of these planes may be tested independently. However, some tests, such as those that require the devices to be brought into service and a call established entail simultaneous activity across multiple planes.

The status of the test can be MANDATORY, CONDITIONAL MANDATORY or OPTIONAL. Note that test that is MANDATORY is required to be performed for interoperability testing. The test that is CONDITIONAL MANDATORY is only required to be performed if the specific condition(s) are met. If a test is OPTIONAL, it does not need to be included in interoperability testing but could be included if desired.

4.2 Summary of Test Scenarios

The following set of interoperability test cases are defined for the current version of WG4 FH IOT specification.

- FH tests focused on the M-Plane:
 - Start-up (O-RU start-up from the power-on of O-RU to the availability of service)
- FH tests focused on the S-Plane:
 - Synchronization status detection
 - Frequency and time error (performance)
- FH tests focused on the C/U-Planes:
 - Radio Layer 3 C-Plane establishment and Initial Radio U-Plane data transfer
 - Radio downlink U-Plane data transfer (Downlink throughput performance)
 - Radio uplink U-Plane data transfer (Uplink throughput performance)
- FH tests focused on the C/U-Plane Delay Management
 - Test with minimum, intermediate, maximum, and excess latency

4.3 Future Enhancements

Additional test cases are under consideration for the future versions of the present document. A non-exhaustive list of candidate test cases for future versions is provided below:

- FH tests focused on the M-Plane:
 - Software management (O-RU software update)
 - Fault management
 - File management
- FH tests focused on the C/U-Plane:
 - Beamforming (testing the actual characteristics of the radiated/received signals at O-RU)
 - Compression (testing different compression methods and IQ bitwidth and validating expected performance)

Note: beamforming and compression are tested as part of the IOT cases already specified in Clause 8 and the relevant IOT profiles (refer to Annex A for more details). The test cases above extend the capability to test the fronthaul interface in respect of these features.

- Multiple O-RUs
- Multiple UEs

4.4 Fronthaul M-Plane Architectural options

As described in the “O-RAN WG4 Management Plane Specification” [3], two architectural models are supported, namely a “hierarchical” and a “hybrid” model. As a general guideline the following apply,

1. Hierarchical model: The O-RU is entirely managed by one O-DU (sudo) using a NETCONF based M-Plane interface.
2. Hybrid model: The architecture allows one or more direct logical interface(s) between management system(s) (performing different roles, nms, fm-pm, swm, etc) and O-RU in addition to a logical interface between O-DU (sudo role) and the O-RU.

4.5 Fronthaul Synchronization options

Various synchronization options have been specified in the O-RAN WG4 CUS-Plane Specification [2] (LLS-C1, LLS-C2, LLS-C3 and LLS-C4). Depending on the specific O-RAN deployment being considered, not all of them might be relevant. When testing the S-Plane, the System Integrator shall identify which of the test cases are relevant depending on the specific deployment scenarios addressed. As a general guideline the following applies:

1. Direct connection between O-DU and O-RU:

LLS-C1 is generally the main sync option to be validated

LLS-C4 may be considered as an alternative or as a complement to LLS-C1

2. Bridged network between O-DU and O-RU

LLS-C2 for cases where the synchronization is delivered to the O-RU via the O-DU and over the bridged network. In this case the PRTC/PRC may be embedded in the O-DU or may be located anywhere in the network (connected via backhaul or FH transport).

LLS-C3 for cases when the synchronization is distributed to the O-RU without involving the O-DU In this case the PRTC/PRC may be located anywhere in the network (connected via backhaul or FH transport) and may also be co-located with the O-DU.

LLS-C4 may be considered as alternative or as a complement to LLS-C2/LLS-C3. LLS-C4 is not addressed in the present document.

The FH focused tests for S-Plane for the current version of this specification covers LLS-C1, LLS-C2 and LLS-C3 using the ITU-T G.8275.1 [5] profile (Full Timing Support). FH focused tests for S-Plane for LLS-C1, LLS-C2 and LLS-C3 using the ITU-T G.8275.2 [6] profile (Partial Timing Support), and LLS-C4 are not addressed in the present document.

5 Interoperability Measurements

5.1 Interoperability Standard Test Definitions

5.1.1 Standard Test Configurations

Interoperability testing is performed to prove that the end-to-end functionality between the O-DU and O-RU is as required by the O-RAN FH CUS-Plane [2] and M-Plane [3] specifications on which these components are based. This requires system level testing with O-DU and O-RU as an integrated system.

5.1.2 Device Under Test (DUT)

The case where the O-DU and O-RU are provided by different vendors is the focus of the present document, but the case where both are from the same vendor is also valid.

The O-DU and O-RU, with their interconnecting FH, is considered as the DUT. This is the same whether the O-CU and O-DU are implemented as an aggregated node, or as disaggregated nodes.

For simplicity and without prejudice the tests in the following sections are defined with reference to a disaggregated O-DU. However, the tests apply equally when the O-DU is replaced by the O-DU functionality of an aggregated O-CU/O-DU and where the Layer 2 and Layer 3 radio processing on the network side is provided by the O-CU functionality of the combined-node. Any differences to the test due to replacement of the disaggregated O-DU with an aggregated O-CU/O-DU are detailed as they arise.

The simplest test configuration involves a single O-DU and a single O-RU. In this configuration, the O-DU and O-RU are labelled as DUT1(O-DU) and DUT1(O-RU) respectively.

More advanced test configurations will involve defining the cardinality between the $1 \dots M$ O-DU(s) and $1 \dots N$ O-RU(s) as part of the test scenario which will determine the configuration required.

An example of such test configuration is to have a single O-DU connected to multiple N O-RUs. In this example, the O-DU is labelled as DUT1(O-DU) and the O-RU(s) are labelled as DUT1(O-RU) ... DUTN(O-RU) accordingly.

A second example is to have multiple M O-DUs connected to multiple N O-RUs. In this second example, the O-DUs are labelled as DUT1(O-DU) ... DUTM(O-DU) and the O-RU(s) are labelled as DUT1(O-RU) ... DUTN(O-RU) accordingly. However, deployment scenarios where a single O-RU is managed and used by more than one O-DU is not addressed in this version of the WG4 FH IOT Specification.

External physical and or logical connection between the O-DUs and O-RUs are to be specified in the IOT profiles (eg, number of 10/25/40GE).

The IOT profile specifies a complete DL and UL configuration to be used for the DUT subject to IOT. An O-DU and an O-RU are considered interoperable under a PTC if the DUT can pass all mandatory and applicable conditional mandatory tests using the specific PTC.

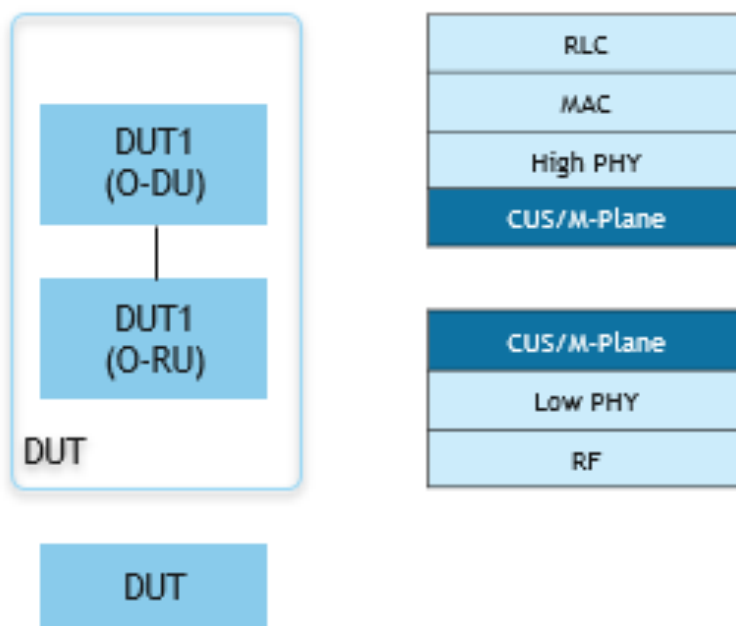


Figure 5.1.2-1: The simplest DUT configuration

5.1.3 Network Management System (NMS)

Network Management System (NMS) is required to support M-Plane testing particularly in the Hybrid architecture model as the O-RU has logical connection with the NMS [3]. In the hierarchical model, the NMS manages the O-DU and the O-DU manages the O-RU respectively.

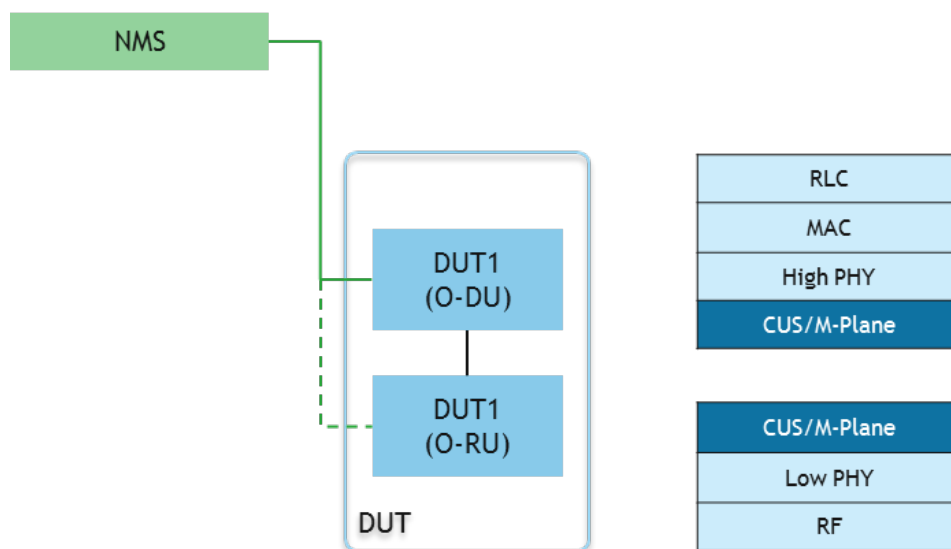


Figure 5.1.3-1: The relationship between DUT and NMS (hybrid mode connection to O-RU shown dashed)

External physical and or logical connection between the NMS and O-DUs and O-RUs are to be specified in the IOT profiles (eg, number of 10/25/40GE), see Annex A for details.

5.1.4 Testing Tools

One of the key objectives of interoperability testing is to validate the functionality of production grade DUTs. Hence it is important to ensure that the DUTs are not negatively impacted with the utilization of internal functions solely to support interoperability testing. ie, DUTs are not expected to be testing tools when deployed in production networks and therefore DUTs should not be used as testing tools during interoperability tests.

Interoperability tests are performed with a set of testing tools which are used to both apply active stimulus and as well as passive monitoring and measurements of the DUTs.

The test is applicable to deployment scenarios where the O-CU and O-DU are implemented as disaggregated nodes connected by an F1 interface and scenarios where they are implemented as an aggregated node without an accessible F1 interface.

The deployment scenario where the 4G LTE eNB is implemented as disaggregated 4G O-CU and 4G O-DU nodes connected by a W1 interface is not addressed in the present document.

Active stimulus testing tools:

- **5G NR O-CU or O-CU emulator:** either as a disaggregated node or as an aggregated node with O-DUs (DUT): used to provide Layer 2 and Layer 3 radio processing on the network side. In case of disaggregated nodes, terminates the 3GPP 5G F1 interface with the O-DU (DUT)
 - O-CU and O-CU emulator can be connected to either a real Core network or emulated Core. An emulated Core can be simpler to deploy for interoperability testing purposes
 - External physical and or logical connection between the O-CU or O-CU emulator with the O-DU (DUT), if any, will be lab setup dependent either through physical or wireless medium

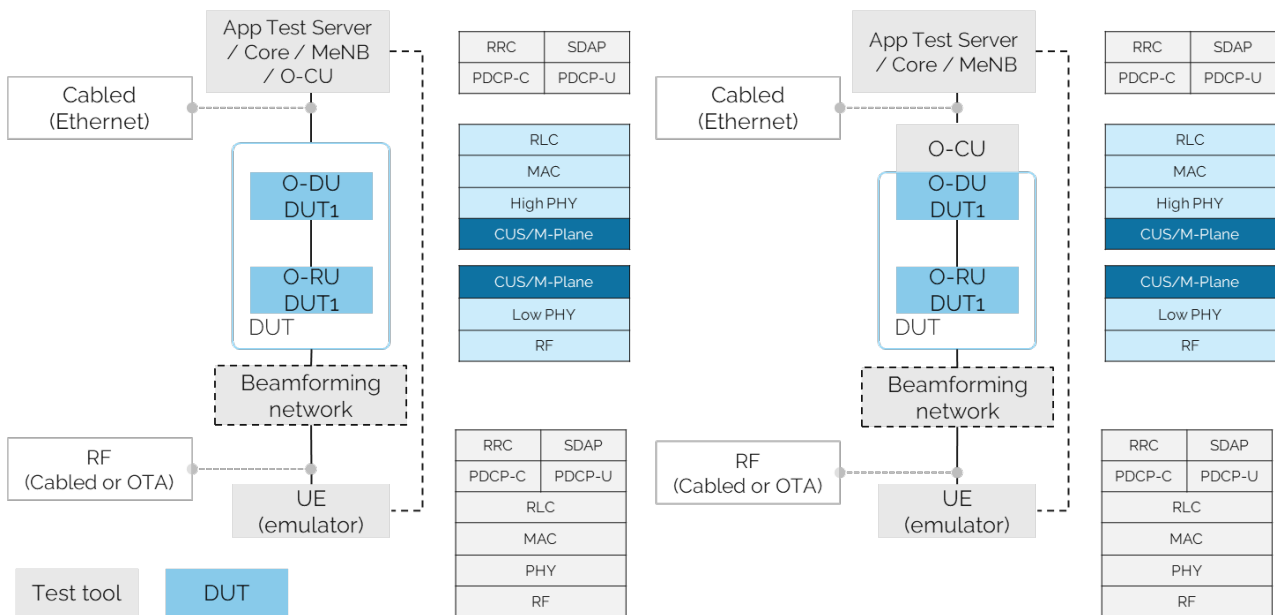


Figure 5.1.4-1: Test setup when O-CU is disaggregated from O-DU (left) and when O-CU is aggregated with O-DU (right).

Figure 5.4.1-1 illustrates the test set-up for disaggregated O-CU/O-DU and aggregated O-CU/O-DU. The beamforming network is optional, indicated by the grey box with dashed outline; it is used for beamforming IOT profiles.

- **4G LTE MeNB or MeNB emulator:** used to terminate the 3GPP EN-DC X2 interface with the 5G NR O-CU or O-CU emulator
 - Required when the DUTs (O-DU and O-RU) are configured to operate 5G NR NSA option 3/3a/3x

- 4G LTE MeNB or MeNB emulator can either have physical or logical connection with the Test UE or UE emulator. RF connection between the 4G LTE MeNB or MeNB emulator with the Test UE or UE emulator will be either through cabled connection or Over the Air (OTA)
- **Test UEs and/or UEs emulator and optional beamforming network (for beamforming IOT profiles):** used to generate stateful device connections and traffic to validate the O-DU and O-RU implementation of the O-RAN FH interface protocols as these are stimulated by the 3GPP upper layer protocols. The beamforming network (BFN), for example a Butler Matrix, is used to generate a static RF channel with rank not lower than the number of spatial streams required for the beamforming IOT profiles
 - Required so that the O-RU which is the DUT does not need to be put into a “test mode” which does not happen in live deployments
 - Test UEs will require SIM cards which are pre-provisioned with subscriber profiles. UEs used for testing can be simpler to setup but given that these Test UEs are designed to be commercial UEs with possibly certain diagnostic functions enabled for logging purposes, they are limited in terms of configurability
 - UEs emulator will require SIM profile configuration with the subscriber’s profiles. UEs emulator can be used in test scenarios which require multiple UEs’ sessions, more flexibility and configurability to help drive test scenarios
 - RF connection between the Test UEs or UEs emulator with the O-RU (DUT) will be either through cabled connection or OTA. However, in case BFN is used, the RF connection between the Test UEs or UEs emulator and the BFN is via cable and the RF connection between BFN and O-RU (DUT) is also via cable. The use of OTA for beamforming IOT profiles is not addressed in the present document

Passive monitoring and measurements testing tools:

- **Test UEs and/or UEs emulator:** used to produce measurements and logs
 - Measurements and KPIs logs for test case validation and reporting
 - Diagnostics logs for troubleshooting purposes which can help with test setup validation and root cause analysis for failed test cases
 - Diagnostics mode is enabled on the Test UEs for diagnostic logging purposes. Device logging tools are connected to the Test UEs for logging purposes
 - UEs used for testing can be simpler to setup but given that these Test UEs are designed to be commercial UEs, they are limited in terms of diagnostic logging capabilities due to limited processing and buffer space
 - UEs emulator can be used in test scenarios which require extensive diagnostics capabilities
- **Application Test Server:** An endpoint application test traffic emulator which can be used to generate and/terminate various traffic streams to or from the Test UEs/UEs emulator respectively. May provide one or more of these options,
 - Stateful traffic, eg, TCP, TWAMP
 - Stateless traffic eg, UDP
 - Required to place traffic load on the DUT
 - External physical and or logical connection between Application Test Server and O-CU/O-CU emulator is out of scope of the specification
- **Test results and KPIs reporting:** which can be provided through one or more of these options
 - UE logging tools which are connected to the Test UEs
 - Device emulator reporting dashboards typically built in as part of the Device emulator solution

- External dashboard and reporting applications
 - **FH Protocol Analyzer:** used for protocol analysis of O-RAN FH protocols to achieve the following:
 - Test case validation which can help with test setup validation and troubleshooting purposes for root cause analysis for test cases which fail
 - Understanding which capabilities were used during test execution
 - Decoding and validation of M-Plane flow sequencing prior to SSH/TLS secure connection establishment
 - Monitoring traffic from the O-RAN FH typically through a tap or span port. Taps are typically preferred as span ports are less reliable but can be used if taps are not readily available in the test lab. Connectivity specifics (eg, number of 10/25/40GE) are to be specified in the IOT profiles
 - **RF Spectrum and Beam Signal Analyzer:** used for RF and Beam power and quality analysis for:
 - Test case validation and troubleshooting which can help with test setup validation and root cause analysis for test cases which fail

Eg, the Beam Signal Analyzer can be used to validate that the O-RUs (DUT) are in-service, configured and operating correctly, which in case of 5G, includes validating that the SS/PBCH blocks (SSB) are configured with the correct Sub Carrier Spacing (SCS), transmitted at the correct frequency locations (can be offset from the center frequency), and in case of both 4G and 5G, burst periods with the correct Physical Cell Identities (PCI), Beam Identifiers and, expected power and quality

 - RF Spectrum and Beam Signal Analyzer performs OTA RF measurements and signal analysis
- O-CU or O-CU emulator
- MeNB or MeNB emulator
- Core or Core emulator:
 - Used to produce measurements and diagnostics logs for troubleshooting purposes which can help with test setup validation and root cause analysis for test cases which fail

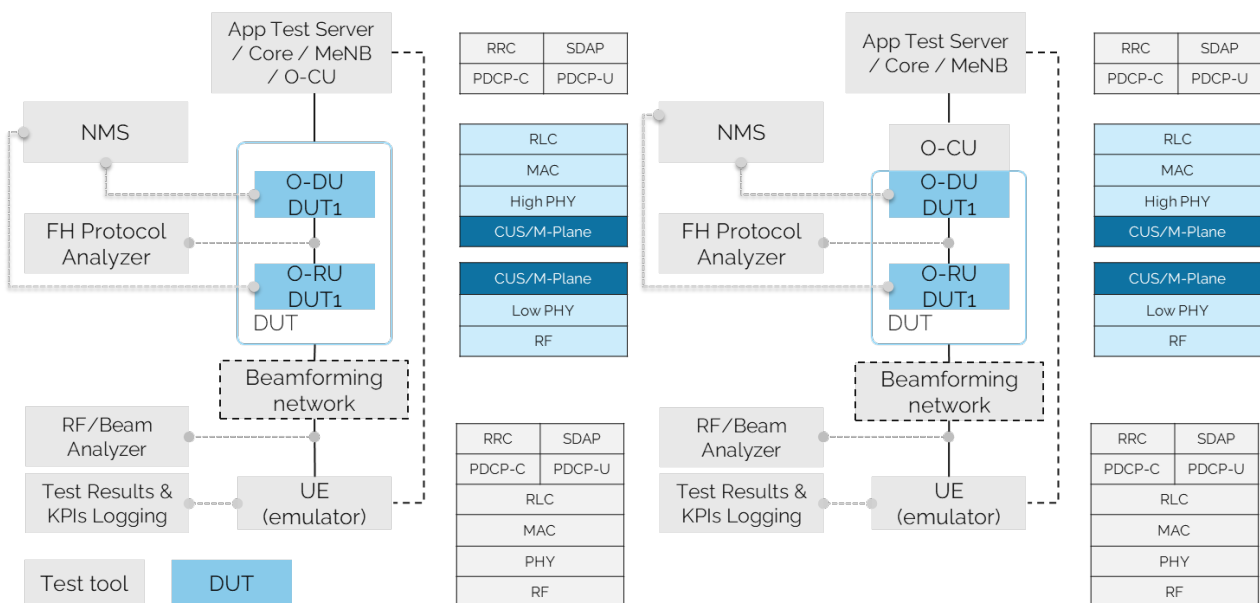


Figure 5.1.4-2: Test setup when O-CU is disaggregated from O-DU (left) and when O-CU is aggregated with O-DU (right).

Figure 5.4.1-2 illustrates the test set-up for disaggregated O-CU/O-DU and aggregated O-CU/O-DU. The beamforming network is optional, indicated by the grey box with dashed outline; it is used for beamforming IOT profiles.

5.1.5 Time Synchronization

All the components including the DUTs (O-DUs and O-RUs) and the Testing Tools are required to be synchronized to a common system time and master time source unless otherwise stated.

Synchronization Plane (S-Plane) configuration for the DUTs is specified in the IOT profiles (see Annex A for details).

Test tools shall be time synchronized to the master timing source with PTP.

5.1.6 Assumptions

In this version of the WG4 FH IOT specifications, the following assumptions apply

- All wireline transport interfaces and the air interface are assumed to conform to ideal conditions (no impairments).
- All wireless connection assumed to adopt 3GPP approaches for “ideal” RF environment for test setup.
- O-DU and O-RU comply to the same version of the O-RAN FH interface specifications.
- All elements in the interoperability test and the supporting test environment, where 3GPP support is relevant, comply to the same version of the 3GPP Specification.
- All O-RUs involved in a test are of the same category; all A or all B.

5.1.7 Specifications to be used for testing

In this version of the WG4 FH IOT specifications, the following sets of specifications and releases/versions shall be supported

- O-RAN FH
 - Control, User and Synchronization Plane Specification [2]
 - Management Plane Specification [3]
 - Management Plane Yang Models [4]
- eCPRI
 - eCPRI Specification V1.0 “Common Public Radio Interface: eCPRI Interface Specification” [11]

It is important to ensure that all DUTs (O-DU and O-RU) and Testing Tools use compatible release/version of the O-RAN FH and eCPRI specifications [11] which in this version of the WG4 FH IOT Specification is ensured by use of the same version.
- 3GPP
 - Release 15 December 2018 and later versions specifications

It is important to ensure that all DUTs (O-DU and O-RU) and Testing Tools use compatible release/version of the 3GPP specifications, which in this version of the WG4 FH IOT Specification is ensured by use of the same version.

5.1.8 Interoperability (IOT) Test Profiles

The IOT Profiles are specified in Annex A. Under each profile, one or more Profile Test Configurations is defined, which specify test entry criteria (parameter values supported) on O-DU and O-RU undergoing IOT testing. There may be cases where the precise values of parameters in an IOT Profile Test Configuration cannot be used, perhaps due to

equipment limitations or licensing rules. For example, a tester may only be licensed to utilize 90MHz of spectrum, for which no relevant IOT Profile Test Configuration is defined. In such a case the tester may choose to define a Customized IOT Configuration which will generally be similar to a defined IOT Profile Test Configuration but with some variations in parameter values. For example, a tester may choose to define a Customized IOT Configuration that would fall under the “NR-TDD-FR2-CAT-A-ABF” IOT Profile but use five 100MHz component carriers, which is not defined in any of the IOT Profile Test Configurations under the “NR-TDD-FR2-CAT-A-ABF” IOT Profile.

Some IOT profiles specify multiple beamforming methods, with each beamforming method specifying different C/U-Plane timing relationships (delay values). In such cases, allowed delay values can be specified separately from the profile table. The allowed delay values are specified as Delay Sets. A Delay Set (DS) is a collection of timing values specific to one beamforming method. If a separate delay table is specified for an IOT profile enumerating allowed Delay Sets, then the IOT test entry criterion includes that the O-DU and O-RU both can be configured with at least one common Delay Set for each beamforming method specified in the IOT profile.

- EXAMPLE 1: If an O-DU can configure DS1 and DS3 for a beamforming method and the O-RU supports DS2 for the same beamforming method, they cannot be tested together
- EXAMPLE 2: If an O-DU can configure DS1 and DS2 for a beamforming method and the O-RU supports DS2 for the same beamforming method, they can be tested (using DS2)
- EXAMPLE 3: If an O-DU can configure DS1 for beamforming method #1 and DS2 and DS3 for beamforming method #2, and the profile specifies both beamforming methods to be tested, the IOT testing using this profile requires an O-RU which can support DS1 for beamforming method #1 and either supports DS2 or DS3 (or both) for beamforming method #2

Normally when reporting the result of an interoperability test, the tester may cite only the IOT Profile Test Configuration name (along with the relevant version of the present document containing the IOT Profile Test Configuration definition), which can be used to understand the full test situation by consulting Annex A. In the case of reporting the results of an interoperability test using a Customized IOT Configuration, the tester should also provide a complete list of parameter values to allow a reader to understand the full test scenario. Table A.1-2 identifies whether a particular IOT profile parameter is modifiable or forbidden in definition of a Customized IOT Configuration.

5.1.9 Measurements of interest

- Availability (eg, are the DUTs in service)
- Accessibility (eg, can the device connect to the network)
- Retainability (eg, can the device connection be maintained)
- Mobility (eg, moving between two O-RUs)
- Integrity (eg, data transfers between the device and the network)

5.2 Standard Test Data Configurations

The test data configurations are specified as part of the IOT profiles in Annex A.

5.3 Standard Test Execution

Most interoperability tests will follow a standard execution plan although individual tests are expected to deviate from this in some way. By defining the standard execution plan an understanding of how tests are arranged can be gained, thereafter examining individual tests can reveal how deviations from the standard execution plan may be defined.

5.4 Interoperability Test Cases

As stated in the introduction, interoperability involves testing the FH interface in terms of M-Plane, S-Plane, C-Plane and U-Plane. Some aspects of these planes may be tested independently. However, some tests, such as those that require the devices to be brought into service and a call established entail simultaneous activity across multiple planes

Each interoperability test provides 1) test description and applicability, 2) the minimum requirements and prerequisites listing required testing tools as required, 3) test purpose and scope, 4) testability requirements imposed on the O-RU and O-DU, 5) test methodology with procedural description as required, 6) the test requirement and expected test result.

Failure conditions are not addressed.

The following set of IOT cases are defined in the present document.

M-Plane IOT Test

1. Startup Installation: O-DU and O-RU getting in Service

S-Plane IOT Tests

1. Functional test of O-DU + O-RU using ITU-T G.8275.1 profile (LLS-C1)
2. Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.1 profile (LLS-C2)
3. Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.1 profile (LLS-C3)
4. Performance test of O-DU + Two O-RUs using ITU-T G.8275.1 profile (LLS-C1)
5. Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.1 profile (LLS-C2)
6. Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.1 profile (LLS-C3)

C/U-Plane IOT Tests

1. Radio Layer 3 C-Plane establishment and Initial Radio U-Plane data transfer
2. Radio U-Plane downlink data transfer (Downlink throughput performance)
3. Radio U-Plane uplink data transfer (Uplink throughput performance)

C/U-Plane Delay Management Test

1. Test with minimum fronthaul latency
2. Test with maximum fronthaul latency
3. Test with a fronthaul latency value between maximum and minimum
4. Test larger fronthaul latency then supported

5.5 Reporting and validation of used capabilities

Since the interoperability tests are designed to be non-intrusive in nature, the tests may pass without invoking a certain capability supported by O-RU and O-DU, even if the capability is indicated by the PTC as test entry criterion. See Table A.1-1 for terminology of IOT test entry requirements. It is possible that a subset of capabilities might not be exercised during a test, and it may not be clear which capability was exercised or not during a test. For C/U-plane IOT tests, the utilization of “to be reported” capabilities associated with each PTC shall be reported. This aids in understanding which capabilities of the DUT were positively tested as part of the IOT tests. Furthermore, the utilization of “to be validated” capabilities shall be validated in C/U-plane IOT tests. “To be reported” capabilities and “to be validated” capabilities are described for each C/U-plane IOT test and for some PTCs in Annex A.

If the utilization of a “to be validated” capability cannot be confirmed, then the inconclusive outcome of the test shall be documented as a “fail”. The utilization of a “to be reported” capability does not change test outcome, ie, the result of a C/U-plane test is a “pass” if the performance acceptance criteria are met, even if the utilization of one or more “to be reported” capabilities are not confirmed.

How to validate a capability is outside the scope of the present document. Examples of validation methods include analysing FH protocol logs, DUT using the capability by design or demonstrated by DUT test logs, inferring from the support declaration and enablement via M-plane, etc.

NOTE: In the present document, “to be reported” capabilities and “to be validated” capabilities are listed only for NR TDD DMRS-BF IOT profiles and C/U-plane IOT tests.

6 M-Plane IOT Test

6.1 Overview

The M-Plane IOT test for hierarchical mode is described in Section 6.1 and the respective test for hybrid mode is described in Section 6.2. Two IOT profiles are defined for each mode, the first addressing IPv4 with SSH and the second addressing IPv6 with a choice of SSH or TLS. In this version of the specification NMS operation in hybrid mode is limited to read-only operation.

6.2 Start-up in hierarchical mode

6.2.1 Test Description and Applicability

This test case is MANDATORY.

The DUT is composed of single O-DU and single O-RU.

Test scenario refers to Chapter 6 “Start-up” in hierarchical architecture for M-Plane [3].

6.2.2 Minimum Requirements (Prerequisites)

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU))

1. Are connected through the O-RAN FH

Assumptions which are required for this test scenario

1. DHCP server is configured for test purposes (either function served by O-DU or external DHCP server with O-DU or router as relay – that should have no impact on test scenario). In case of external DHCP server, Configuration of DHCP server and DHCP procedures are excluded from validation of IOT.
2. IPv4 and IPv6 are conditional mandatory from v07 of [3] and either IPv4 or IPv6 transport connection for M-Plane is used in this test.
3. A combination of VLAN identity and MAC address is only used for C/U- Plane connectivity.
4. Network between O-DU and O-RU allows for connectivity between actors.
5. O-RU has account configured with sudo access privilege. O-DU knows credentials of sudo account available at O-RU.
6. Appropriate software file for O-RU is pre-installed in O-RU and corresponding manifest.xml is located in O-DU. No software upgrade is required during startup test sequence of O-RU for this IOT test case.

Testing tools which are required for this test scenario

- FH Protocol Analyzer monitoring traffic between O-DU and O-RU is used for validation of M-Plane procedural flows and contents of messages prior to the establishment of the SSH/TLS connection for M-Plane and when certain procedures do not use encryption (eg, loopback messages and responses of IEEE 802.1Q).

6.2.3 Test Purpose and Scope

Purpose of this test is to validate the startup sequence of O-RU and the interface to DHCP server and NETCONF client in O-DU for the start-up scenario.

The detailed steps outlined in this test case are for informational purposes and can be useful for troubleshooting purposes in the event that the test case fails. Procedures which are not encrypted can be observed with the FH Protocol Analyzer for validation of the test progress.

Notwithstanding this being an M-Plane test, correct evaluation of the success of the M-Plane operation requires also that the fronthaul C-Plane and U-Plane also be operational, at least in the DL direction.

6.2.4 Testability requirements imposed on O-RU and O-DU

The appropriate software runs on the O-RU and O-DU for the test purpose.

6.2.5 Methodology and Initial Conditions

1. There is physical connectivity between O-RU and O-DU
2. O-DU is powered up and in service or running
3. DHCP server is connected and available
4. O-RU is powered up for this start-up test scenario
5. The credential information (per the used M-Plane IOT profile) is commonly pre-installed in O-DU and O-RU as one of test assumptions in 6.2.2.

6.2.6 Procedure: M-Plane start up test

6.2.6.1 Step: Transport layer initialization

1. Depending on initial condition – either power-on to O-RU or physical network connection is enabled between O-RU and O-DU.
2. FH Protocol Analyzer observation: DHCP Discovery coming from O-RU's MAC address

Note: message is sent as part of VLAN scan procedure, hence it can be in serial sequence "VID after VID with timer in between" or in parallel sequence "burst to subset of VIDs, timer, another burst to different subset of VIDs". FH Protocol Analyzer confirms that the O-RU includes an appropriate option in the DHCP DISCOVER per the used IOT profile.
3. FH Protocol Analyzer observation: VLAN scan continues until response from DHCP server is received.
4. FH Protocol Analyzer observation: O-RU continues DHCP procedure using only the VID on which response from DHCP server has been received.
5. FH Protocol Analyzer observation: DHCP procedure is finished by DHCP Acknowledgement message sent towards O-RU. As a result of DHCP procedure, O-RU obtains its own IP details plus IP per the used IOT profile of corresponding NETCONF Client in O-DU.

6.2.6.2 Step: RU calls home to NETCONF client (TCP connection establishment)

1. FH Protocol Analyzer observes Call Home – a TCP session establishment initiated by O-RU towards O-DU per the used IOT profile

6.2.6.3 Step: SSH/TLS Secure connection establishment

1. SSH session/TLS connection establishment initiated by O-DU towards O-RU.

6.2.6.4 Step: NETCONF Capability discovery

1. NETCONF Hello message sent by O-RU towards O-DU. The message exposes capability of ietf-yang-library

2. NETCONF Hello message sent by O-DU towards O-RU
3. <rpc><get> message sent by O-DU to O-RU. The <rpc><get> has < yang-library xmlns = "urn:ietf:params:xml:ns:yang:ietf-yang-library"> subtree filter imposed
4. <rpc><reply> message sent by O-RU to O-DU. The message contains content of leaf "yang-library" of ietf-yang-library.yang module

6.2.6.5 Step: Optional provisioning of new management accounts

Note: this step is intentionally omitted as pre-provisioned O-RU is expected for IOT (meaning: no need to perform optional step and configure supplementary management accounts).

6.2.6.6 Step: Initial process of NETCONF Subscribe for each stream

1. NETCONF "create-subscription" RPC message(s) sent by O-DU towards O-RU. Number of NETCONF subscriptions is up to O-DU.

Note: O-DU can subscribe itself to default event stream or to specific streams of events. In case no subscription to default stream is performed, then O-DU shall subscribe itself at least to event streams "supervision-notification" and "alarm-notif". Subscription to other streams is optional, however not prohibited.

2. NETCONF "create-subscription" RPC Reply messages sent by O-RU for each RPC message

6.2.6.7 Step: Supervision of NETCONF connection

1. Periodic sequence of NETCONF messages:
 - a. supervision-notification sent by O-RU towards O-DU
 - b. RPC supervision-watchdog-reset sent by O-DU towards O-RU
 - c. RPC reply sent by O-RU towards O-DU

Note: in above sequence following timers shall be respected: "supervision-notification-interval", "guard-timer-overhead" sent as parameters in RPC supervision-watchdog-reset (O-DU -> O-RU) and "next-update-at" sent as parameter in RPC reply (O-RU -> O-DU).

6.2.6.8 Step: Retrieval of O-RU information and Additional configuration

1. O-DU sends <rpc><get><filter="subtree"> to get each yang module listed in ietf-yang-library.yang in O-RU, for example
 - ietf-hardware augmented by o-ran-hardware
 - ietf-interface augmented by o-ran-interfaces
 - o-ran-operations
 - o-ran-transceiver
 - o-ran-sync
 - o-ran-mplane-int
 - o-ran-lbm
 - o-ran-performance-management
 - o-ran-delay-management
 - o-ran-module-cap
 - o-ran-alarm-id

- o-ran-fan
 - o-ran-supervision
 - o-ran-user mgmt
2. O-RU responds with `<rpc-reply><data>` for each yang module per `<rpc><get>`
 3. O-DU sends `<rpc><edit-config>` to each configurable yang module for additional configuration to O-RU whenever it is necessary
 4. O-RU responds with `<rpc-reply><ok/>`

Note: The configurable yang modules are o-ran-sync, o-ran-lbm, o-ran-operations and others (up to O-DU implementation) except o-ran-uplane-conf, o-ran-processing-element, ietf-interface augmented by o-ran-interfaces. Additional configuration of step 3 and 4 can be examined between Chapters 6.2.6.9 and 6.2.6.10.

6.2.6.9 Step: Software management

1. O-DU sends `<rpc><get><filter="subtree">` to get o-ran-software-management.yang
2. O-RU responds with `<rpc-reply><data>` for `<software-inventory>< software-slot>`. At least 2 slots are contained in software-inventory

Note: software update is not applied as the pre-condition that appropriate software file for O-RU is pre-installed in O-RU and corresponding manifest.xml is located in O-DU.

6.2.6.10 Step: C/U-Plane transport connectivity check between O-DU and O-RU

1. O-DU configures by `<rpc><edit-config>` vlan-id for the usage of C/U-Plane in ietf-interface augmented by o-ran-interfaces to O-RU. O-RU responds with `<rpc-reply><ok/>`
2. O-DU sends loopback message to O-RU with MAC address and vlan-id periodically. The O-RU MAC address and vlan-id used in LBM is same as the one that set in ietf-interface augmented by o-ran-interfaces.
3. O-RU sends loopback response to O-DU per received loopback message respectively

6.2.6.11 Step: U-Plane configuration between O-DU and O-RU

1. O-DU sends `<rpc><get><user-plane-configuration>` to determine the presence of following instances: multiple static-low-level-[tr]x-endpoints, multiple [tr]x-arrays and the relations between them. O-RU replies `<rpc-reply><data>` including key information on number of endpoints, band number, number of arrays and polarization.
2. O-DU sends `<rpc><edit-config>` to create low-level-[tr]x-endpoints, with the same name as static-low-level-[tr]x-endpoints. The NETCONF Client is responsible for assigning unique values to the "eaxc-id" addresses of all low-level-rx-endpoint elements and low-level-tx-endpoint elements, within the O-RU when operating in the same direction (Tx or Rx), even when these operate across different named interfaces of the O-RU. Number of instances of low-level-[tr]x-endpoints depend on contents of selected test profile, eg, for number of CCs. It may be less than number of instances of static-low-level-[tr]x-endpoints. O-RU replies `<rpc-reply><ok/></rpc-reply>`.
3. O-DU sends `<rpc><edit-config>` to create [tr]x-array-carriers. Number of created instances of [tr]x-array-carriers is equivalent to number of CCs times number of arrays, where number of CCs is defined by operator. Appropriate values are configured to absolute-frequency-center, channel-bandwidth, gain, and so on. In provided configuration the value of leaf "active" is set as 'INACTIVE' (or leaf is omitted in configuration and O-RU uses its default value 'INACTIVE') for all just created [tr]x-array-carriers. O-RU replies `<rpc-reply><ok/></rpc-reply>`.
4. O-DU sends `<rpc><edit-config>` to create processing-elements related to interfaces offering access to desired endpoints. The key information such as MAC address is configured according to selected transport flow, eg, o-ru-mac-address, o-du-mac-address and vlan-id for C/U-Plane in case of Ethernet flow. O-RU replies `<rpc-reply><ok/></rpc-reply>`.

5. O-DU sends <rpc><edit-config> to create low-level-[tr]x-link(s) to make relationship between low-level-[tr]x-endpoint(s), [tr]x-array-carriers and processing elements belonging to transport. The number of instance(s) for low-level-[tr]x-links is equivalent to that of instance(s) of low-level-[tr]x-endpoints.

6.2.6.12 Step: Fault Management activation

1. O-DU sends <rpc><get><filter="subtree"> to get o-ran-fm.yang
2. O-RU responds with <rpc-reply><data> for <active-alarm-list><active-alarms>

Note: subscription to NETCONF default event stream fulfils the condition to signal notification alarm-notif when O-RU detects any alarm.

6.2.6.13 Step: Performance measurement activation (if required at start-up timing)

Note: Step for Retrieval of O-RU information may cover the configuration of o-ran-performance-management.yang if it is required at start-up installation timing.

6.2.6.14 Step: Retrieval of O-RU state, including synchronization information, from O-RU

If O-DU already knows at the moment this step is going to be performed (eg, from notification “synchronization-state-change”), that “sync-state” of O-RU is ‘LOCKED’, O-DU may skip this step.

S-Plane has to be operational.

1. O-DU sends <rpc><get><sync> to O-RU
2. S-Plane in O-RU has been locked, sends <rpc-reply><data> <sync-status><sync-state> LOCKED

6.2.6.15 Step: Configuring the O-RU operational parameters: carrier activation

S-Plane has to be operational (sync-state != FREERUN, preferable sync-state = LOCKED) prior to running this step

1. O-DU sends <rpc><edit-config> to perform activation by setting the value of the parameter “active” at [tr]x-array-carriers to “ACTIVE”. O-RU sends <rpc-reply><ok/>
2. O-RU sends notification [tr]x-array-carriers-state-change that indicates all [tr]x-array-carriers’ names with “state” = “READY”

Note: C/U-Plane service is available at this step.

6.2.7 Test Requirement (expected result)

Observe that both the O-DU and O-RU get in service successfully by monitoring correct transmission of synchronization signals and broadcast channel (ie, PSS/SSS and PBCH for LTE and SSB for NR).

Record downlink carrier frequency (EARFCN for LTE as specified in 3GPP TS 36.104 [12] and GSCN/NR-ARFCN for NR as specified in 3GPP TS 38.104 [13]), cell (PCI for LTE and for NR as specified in 3GPP TS 36.211 [14] and TS 38.211 [15], respectively) and system information (MIB for LTE and NR as specified in 3GPP TS 36.331 [16] and TS 38.331 [17], respectively). In case that beam sweeping is applied to SSB, system information should be recorded for each of the SSB indices detected.

The parameter values for downlink carrier frequency, cell, system information and SSB indices are part of the radio test setup configuration. This test is considered successful if the recorded measurements values match up with the values which are used for the radio test setup configuration.

6.2.8 Test Report (Failure)

If the test case fails, vendor specific methods will be relied on to assist with troubleshooting the root cause(s) which led to the failure. The steps outlined in this test case can be used to guide the troubleshooting process.

6.3 Start-up in hybrid mode

6.3.1 Test Description and Applicability

This test case is conditional MANDATORY. The condition is, the O-DU supports hybrid M-Plane deployment.

The DUT is composed of single O-DU, a single O-RU and a single NMS.

Test scenario refers to Chapter 5.1.2 “M-Plane Architectural Model” and Chapter 6 “Start-up” in hybrid architecture for M-Plane [3]. As per M-Plane Chapter 5.1.2, all compliant O-RUs shall be able to support both hierarchical and hybrid deployment. If the O-DU supports hybrid M-Plane deployment, the testing of hybrid use case shall therefore include being able to demonstrate that an O-RU is able to support simultaneous NETCONF sessions to the O-DU and NMS. In order to ensure limited impacts to the baseline hierarchical O-DU to NMS configuration management tasks, in this test the second NETCONF client associated with “nms” user privileges is limited to demonstrating that it is able to retrieve the configuration and operational state from an O-RU which has been configured by an O-DU.

6.3.2 Minimum Requirements (Prerequisites)

Single O-DU (DUT1(O-DU)), a single O-RU (DUT1(O-RU)) and a single NMS (DUT1(NMS)).

1. Are connected through the O-RAN FH

Assumptions which are required for this test scenario

1. DHCP server is configured for test purposes (either function served by O-DU or external DHCP server with O-DU or router as relay – that should have no impact on test scenario). In case of external DHCP server - Configuration of DHCP server and DHCP procedures are excluded from validation of IOT.
2. IPv4 and IPv6 are conditional mandatory from v07 of [3] and either IPv4 or IPv6 transport connection for M-Plane is used in this test.
3. A combination of VLAN identity and MAC address is used for C/U- Plane connectivity.
4. A common VLAN is used for M-Plane connectivity between the O-RU and O-DU and between the O-RU and the NMS.
5. Network between O-DU and O-RU and between NMS and O-RU allows for connectivity between actors.
6. An account user-name has been configured for group “nms” access control privileges in a previous O-DU/O-RU testing step as specified in sub-section 6.2.6.5.
7. O-RU has two accounts configured. One account has sudo access privilege with credentials used by the O-DU. The other account has nms access privilege and the credentials used by the NMS.
8. Appropriate software file for O-RU is pre-installed in O-RU and corresponding manifest.xml is located in O-DU. No software upgrade is required during startup test sequence of O-RU for this IOT test case.
9. An NMS is configured for supporting NETCONF Call Home functionality.

Testing tools which are required for this test scenario

- FH Protocol Analyzer monitoring traffic between O-DU and O-RU and between NMS and O-RU is used for validation of M-Plane procedural flows and contents of messages prior to the establishment of the SSH connection for M-Plane and when certain procedures do not use encryption (eg, loopback messages and responses of IEEE 802.1Q).

6.3.3 Test Purpose and Scope

Purpose of this test is to validate the startup sequence of O-RU and the interface to DHCP server and NETCONF clients in the O-DU and NMS for the start-up scenario. As described in [3], all O-RUs supporting the M-Plane specification shall support multiple NETCONF sessions, and hence all compliant O-RUs shall be able to support both hierarchical and hybrid deployment.

The detailed steps outlined in this test case are for informational purposes and can be useful for troubleshooting purposes in the event that the test case fails. Procedures which are not encrypted can be observed with the FH Protocol Analyzer for validation of the test progress.

Notwithstanding this being an M-Plane test, correct evaluation of the success of the M-Plane operation requires also that the fronthaul C-Plane and U-Plane also be operational, at least in the DL direction.

6.3.4 Testability requirements imposed on O-RU, O-DU and NMS

The appropriate software runs on the O-RU, O-DU and NMS for the test purpose.

6.3.5 Methodology and Initial Conditions

1. There is physical and or logical connectivity between O-RU and O-DU and between the O-RU and NMS
2. O-DU and NMS are powered up and in service or running
3. DHCP server is connected and available
4. O-RU is powered up for this start-up test scenario
5. The credential information (per the used M-Plane IOT profile) is configured on the O-RU according to one of test assumptions listed in in 6.3.2.

6.3.6 Procedure: M-Plane start up test

6.3.6.1 Step: Transport layer initialization

1. Depending on initial condition – either power-on to O-RU or physical or logical network connection is enabled between O-RU and O-DU/NMS.
2. FH Protocol Analyzer observation: DHCP Discovery coming from O-RU's MAC address

Note: message is sent as part of VLAN scan procedure, hence it can be in serial sequence "VID after VID with timer in between" or in parallel sequence "burst to subset of VIDs, timer, another burst to different subset of VIDs". FH Protocol Analyzer confirms that the O-RU includes an appropriate in the DHCP DISCOVER per the used IOT profile.
3. FH Protocol Analyzer observation: VLAN scan continues until response from DHCP server is received.
4. FH Protocol Analyzer observation: O-RU continues DHCP procedure using only the VID on which response from DHCP server has been received.
5. FH Protocol Analyzer observation: DHCP procedure is finished by DHCP Acknowledgement message sent towards O-RU. As a result of DHCP procedure, O-RU obtains its own IP details plus IP of corresponding NETCONF Clients in O-DU and in NMS per used IOT profile.

6.3.6.2 Step: RU calls home to discovered NETCONF client (TCP connection establishment)

1. FH Protocol Analyzer observes Call Home – a TCP session establishment initiated by O-RU towards O-DU per used IOT profile
2. FH Protocol Analyzer observes Call Home – a TCP session establishment initiated by O-RU towards NMS per used IOT profile

6.3.6.3 Step: SSH/TLS Secure connection establishment

1. SSH session/TLS connection establishment initiated by O-DU towards O-RU using sudo account privilege.
2. SSH session/TLS connection establishment initiated by NMS towards O-RU using nms account privilege.

6.3.6.4 Step: NETCONF Capability discovery

1. NETCONF Hello message sent by O-RU towards NETCONF clients (O-DU and NMS). The message exposes capability of ietf-yang-library
2. NETCONF Hello message sent by NETCONF clients (O-DU and NMS) towards O-RU.
3. `<rpc><get>` message sent by NETCONF clients (O-DU and NMS) to O-RU. The `<rpc><get>` has `<yang-library xmlns = "urn:ietf:params:xml:ns:yang:ietf-yang-library">` subtree filter imposed
4. `<rpc><reply>` message sent by O-RU to NETCONF Clients (O-DU and NMS). The message contains content of leaf "yang-library" of ietf-yang-library.yang module

6.3.6.5 Step: Optional provisioning of new management accounts

Note: this step is intentionally omitted as pre-provisioned O-RU is expected for IOT (meaning: no need to perform optional step and configure supplementary management accounts).

6.3.6.6 Step: Initial process of NETCONF Subscribe for each stream

1. NETCONF "create-subscription" RPC message(s) sent by O-DU towards O-RU. Number of NETCONF subscriptions is up to O-DU.

Note: O-DU can subscribe itself to default event stream or to specific streams of events. In case no subscription to default stream is performed, then O-DU shall subscribe itself at least to event streams "supervision-notification" and "alarm-notif". Subscription to other streams is optional, however not prohibited.

2. NETCONF "create-subscription" RPC Reply messages sent by O-RU for each RPC message

Note: subsequent steps use the NMS to retrieve the configuration and operational state of the O-RU and are performed after step 6.2.6.7 through 6.2.6.15 that complete O-RU start-up.

6.3.6.7 Step: NMS retrieval of O-RU information and Additional configuration

1. NMS sends `<rpc><get><filter="subtree">` to get each yang module listed in ietf-yang-library.yang in O-RU, with the exception of o-ran-usermgmt and o-ran-supervision, for example

ietf-hardware augmented by o-ran-hardware

ietf-interface augmented by o-ran-interfaces

o-ran-operations

o-ran-transceiver

o-ran-sync

o-ran-mplane-int

o-ran-lbm

o-ran-performance-management

o-ran-delay-management

o-ran-module-cap

o-ran-uplane-conf

o-ran-alarm-id

o-ran-fan

2. O-RU responds with `<rpc-reply><data>` for each yang module per `<rpc><get>`

6.3.7 Test Requirement (expected result)

Observe that the O-RU calls home to both the O-DU and NMS and then is able to establish simultaneous NETCONF sessions between the O-RU and O-DU and between the O-RU and NMS.

Observe that the NMS is able to retrieve configuration and operational state from models to which it does have NACM read privileges.

6.3.8 Test Report (Failure)

If the test case fails, vendor specific methods will be relied on to assist with troubleshooting the root cause(s) which led to the failure. The steps outlined in this test case can be used to guide the troubleshooting process.

6.3.9 Supplementary test: NETCONF Access Control (Failure)

This is a supplementary test to verify correct handling of NMS read attempt with incorrect privilege per 6.2.6.8.

1. The NETCONF client in the NMS attempts to read user accounts provisioned on the O-RU. The NMS sends `<rpc><get><filter>= subtree</filter></get>` to get the configuration and operational state for the o-ran-usermgmt yang module.
2. O-RU responds with `<rpc-error>` response with “access-denied”.

Expected result is to that an O-RU will deny access to an NMS that tries to read configuration and/or operational state from models to which it does not have NACM privileges.

If the test case fails, that is access is granted, vendor specific methods will be relied on to assist with troubleshooting the root cause(s) which led to the failure.

7 S-Plane IOT Test

7.1 Overview

This Section describes system level tests that are used to validate both S-Plane functionality and S-Plane performance between O-DU and O-RU from different vendors connected using the O-RAN WG4 FH interface specified in [2] and the O-RAN WG4 M-Plane interface specified in [3].

S-Plane functionality is determined by retrieval of the O-DU and O-RU synchronization state using the M-Plane or NMS as appropriate, whereas, S-Plane performance is determined by over-the-air (OTA) measurements or conductive (ie, a cabled electrical connection to the DUT's radio interface) measurements of O-RU synchronization signal.

This Section provides a common high-level list of the items that are expected to be covered in order to validate interoperability with the relevant standard, both functional and performance aspects are addressed:

- The tests apply to both 4G(LTE) and 5G(NR) and are applicable to all O-RAN IOT profiles listed in Annex A
- The O-RAN WG4 CUS Specification [2] Section validated is §11.3.2
- SyncE Master test case is optional and only valid when the SyncE clock (eg, Implemented in the O-RU) takes advantage of it. Therefore, the related SyncE Master test cases are optional
- Several test cases involve configuring the O-DU or O-RU
- M-Plane connection is established, meaning whole protocol stack ETH/IP/TCP/SSH/TLS is up and running, Capabilities are exchanged between NETCONF Client and Server, Subscriptions to notifications are created
 - Several test cases involve the configuration and collection of status of T-BC. However, as this is not specified in the Management Plane Specification [3], the parts of the tests below that require such functionality are not addressed in the present document.

- Further work needs to be done to align and refine the definition of the states in the O-RAN WG4 CUS Specification [2] and M-Plane [3] specifications for the next releases

At least one of the synchronization options (LLS-C1, LLS-C2, LLS-C3 or LLS-C4), depending on the relevant O-RAN deployment as described in Clause 4.5, shall be tested for functionality and performance.

The following bulleted notes apply specifically to S-Plane IOT functional tests in addition to the general notes above,

- Tests shall be done in the lab at constant temperature
- Test equipment is needed to generate a reference S-Plane signal
- For validation of synchronization information, use the FH M-Plane interface for the O-RU, and other interfaces such as NMS for O-DU
- Retrieving sync-state of O-DU using the NMS is not addressed in the present document
- O-RU reports over M-Plane the degraded received clock-class and clock-accuracy as well as locked state; details of how to do this are not addressed in the present document
- With the exception of lls-C1, that doesn't include any T-BC in the fronthaul link, the test configuration shall use from 1 up to the maximum number of T-BCs specified for performance test, as set out in the performance test specific notes set out below, according to the preference of the tester

The following bulleted notes apply specifically to S-Plane IOT performance tests in addition to the general notes above,

- All tests should be done in lab under constant and variable C/U-Plane profile. The variable C/U-Plane profile (for example similar to the test case 13 shown in figure VI.11 of ITU-T G.8261 [10]) should be defined for this test
- A previous version of the present document (v 1.00) contained a requirement to perform the IOT performance tests under variable temperature conditions. That requirement was removed, and instead replaced with the following text that was also used in the Conformance Test Specification [24]: The thermal profile is not defined. The tests are defined to be run at a constant temperature, but the thermal profile choice and range is left as a decision for the vendor
- Use the maximum number of T-BC devices as shown in O-RAN WG4 CUS Specification [2] in Annex H
 Note: Annex H includes example deployment cases that capture the influence of factors including, target end-to-end timing error requirement, clock type used in the network, and characteristics of the O-RU clock. Additional cases including, addressing O-RUs with different levels of clock performance, and different noise accumulation models etc are not addressed in the present document
- The acceptance criterion for the tests is to satisfy the 3GPP OTA TAE limits specified in 3GPP TS 36.141 [18] for 4G(LTE) and 38.141-2 [19] for 5G(NR), and summarized in eCPRI [11] and ITU-T G.8271 [7]. The particular conditions applicable to each test are set out in the relevant sections below

To perform the IOT S-Plane tests, following test set ups (Figure 7.1-1, 7.1-2 and 7.1-3) are proposed. The selection of the test setup is dependent on the configuration mode lls-C1, lls-C2, or lls-C3. This version of the document does not cover lls-C4:

- Each diagram covers both functional test and performance test of the respective synchronization configuration mode. The main difference between the setups for functional and performance test is the need for a measurement equipment at the radio interface for performance test as indicated in previous paragraphs.
- Each setup gives the option to select different types of S-Plane signals for the input of the O-DU. The selection of the option depends on the capability of the O-DU and operator's synchronization architecture. In lls-C3 mode, the O-DU has the option of selecting a master through its fronthaul interface or directly connecting to a T-GM.
- T-GM via a midhaul network (for lls-C1 or lls-C2) or a fronthaul network (for lls-C3) over an Ethernet interface with PTP and SyncE (if applicable)
- T-GM over an Ethernet interface with PTP and SyncE (if applicable)

- PRTC: 1pps according to ITU-T G.703, 10MHz, and ToD according to G.8271 Annex A or any other proprietary ToD decoding specification supported by the O-DU
- GNSS: RF input directly from an antenna used for GNSS signal reception
- The measurement equipment has to be synchronized with the DUT, thus the need for a line arriving at the measurement equipment.
- The arrows in these diagrams represent the direction of S-Plane messages (from master to subordinate), or from any source synchronization signal to the receiver.

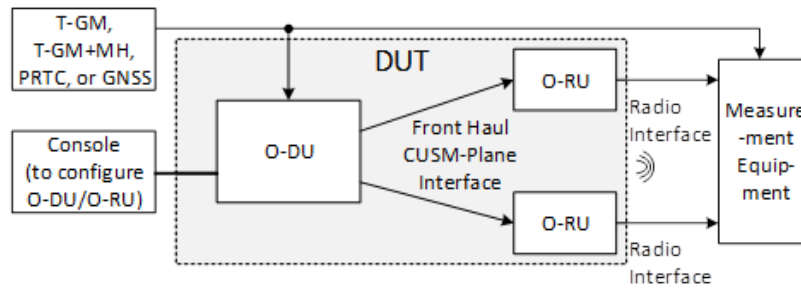


Figure 7.1-1: Test setup for IIs-C1

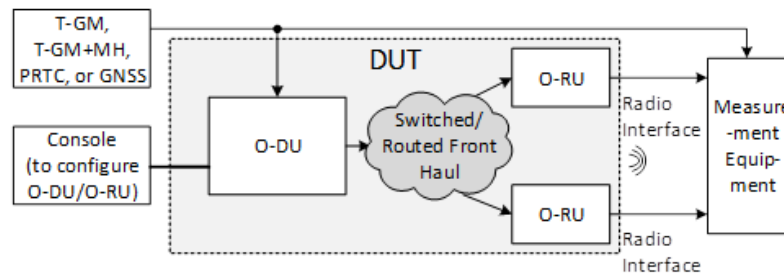


Figure 7.1-2: Test setup for IIs-C2

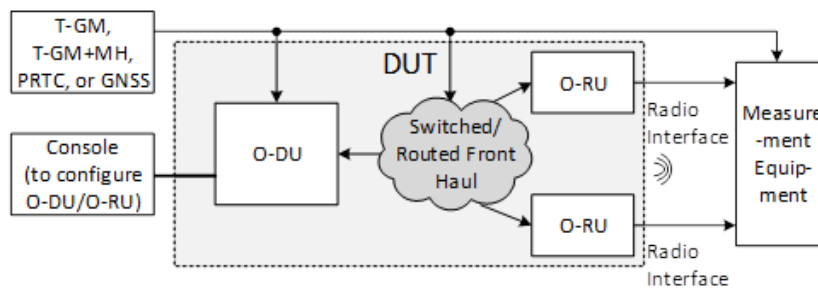


Figure 7.1-3: Test setup for IIs-C3

7.2 Functional test of O-DU + O-RU using ITU-T G.8275.1 profile (LLS-C1)

7.2.1 Test Description and Applicability

This test case is CONDITIONAL MANDATORY and shall be performed if the O-RU and O-DU declare support of LLS-C1 synchronization option (see Clause 4.4).

This test validates that O-RU is synchronizing from an O-DU that incorporates a PTP grand master and SyncE Master with ITU-T G.8275.1 [5] profile and is traceable to a PRTC.

This test involves one O-DU and one O-RU.

7.2.2 Minimum Requirements (Prerequisites)

1. O-RU is connected to O-DU via direct fiber O-RAN links.
2. The O-DU is connected to a PRTC traceable time source, directly (GNSS Receiver either embedded or connected to the O-DU) or via PTP, as described in Conformance Test Specification [24].

7.2.3 Purpose and Scope

The O-RU is synchronizing from the O-DU with the ITU-T G.8275.1 [5] profile. This test case validates the correct synchronization status of the O-RU.

7.2.4 Testability Requirements imposed on O-RU and O-DU

Requirements for M-Plane: is properly operating (as specified in 6.2).

Synchronization requirement: O-DU is connected to a local or remote PRTC.

7.2.5 Test Methodology

These tests use the O-RAN M-Plane and O-DU NMS features.

The O-DU acts as a PTP master compliant with the ITU-T G.8275.1 [5] profile.

Three conditions shall be covered:

- startup
- nominal
- degraded

7.2.5.1 Procedure: Startup conditions

1. Not yet configured.

External frequency and time source are available to the O-DU and deliver nominal status

O-DU is not yet configured to select the time source and align its frequency and time to it

O-DU not yet configured to act as PTP master on the FH ports

2. Configured.

O-DU is configured to align to the selected frequency and time source

3. Until disciplining

Until O-DU disciplining of the frequency and time to the selected source has completed, Startup conditions persists

The O-RUs are configured to synchronize from PTP in ITU-T G.8275.1 [5] profile and report their status

7.2.5.2 Procedure: Nominal conditions

1. O-DU is configured to start acting as a PTP master compliant with the ITU-T G.8275.1 [5] profile on selected FH ports.
2. O-DU acts as a PTP grand master or as a boundary clock with ports towards the FH interface in Master state, compliant with the ITU-T G.8275.1 [5] profile advertising “nominal” status.
3. O-DU reports status, acting as PTP master clock towards the FH interface.

7.2.5.3 Procedure: Degraded conditions

1. O-DU is configured to enter HOLDOVER based on local oscillator frequency.
2. O-DU acts as configured clock, with PTP ports in master state, compliant to ITU-T G.8275.1 profile advertising HOLDOVER status with degraded clockClass and clockAccuracy as specified by ITU-T G.8275.1 [5].
3. O-DU is configured to exit HOLDOVER and resumes normal frequency and phase disciplining using the source.
4. O-DU acts as configured clock, with PTP ports towards the FH interface in master state compliant to ITU-T G.8275.1 [5] in “nominal” status.

7.2.6 Test Requirements (expected results)

7.2.6.1 Startup conditions

The acceptance criterion is that the following status is observed for steps 1 to 3 (7.2.5.1 above):

- the FREERUN sync-state of the O-RU using the M-Plane
- the UNLOCKED PTP lock-state of the O-RU using the M-Plane
- the UNLOCKED SyncE lock-state of the O-RU using the M-Plane (optional)
- the FREERUN sync-state of the O-DU using the NMS

7.2.6.2 Nominal conditions

The acceptance criterion is that the following status is observed for steps 1 to 3 (7.2.5.2 above):

- the LOCKED sync-state of the O-DU using the NMS
- the received PTP clockClass level of the O-RU using the M-Plane
- the received SyncE SSM level of the O-RU using the M-Plane (optional)
- the LOCKED sync-state of the O-RU using the M-Plane
- the LOCKED PTP lock-state and PARENT PTP state of the O-RU using the M-Plane
- the LOCKED SyncE lock-state and OK or PARENT SyncE state of the O-RU using the M-Plane (optional)

7.2.6.3 Degraded conditions

The acceptance criterion is that the following status is observed for all steps 1 to 2 (7.2.5.3 above) (for steps 3 and 4, same criterion as “Nominal conditions” apply):

- the HOLDOVER sync-state of the O-DU using the NMS
- the received PTP clockClass level of the O-RU using the M-Plane
- the received SyncE SSM level of the O-RU using the M-Plane (optional). Note that if SyncE state is NOK, it is also permitted that the local default SSM level is reported using the M-Plane.
- the sync-state of the O-RU using the M-Plane is LOCKED if the received clockClass (or optional SSM QL) value matches the configured list of accepted values, otherwise HOLDOVER or FREERUN
- the PTP lock-state of the O-RU using the M-Plane is LOCKED, and the PTP state of the O-RU using the M-Plane is PARENT if the received clockClass matches the configured list of accepted values, otherwise NOK.
- the SyncE lock-state (optional) of the O-RU using the M-Plane is LOCKED or OK, and the SyncE state of the O-RU using the M-Plane is PARENT or OK if the received SSM matches the configured list of accepted values, otherwise NOK.

Note: If the NETCONF client triggers a reset procedure of the O-RU by FREERUN of the sync-state at these degraded conditions, the status observation using M-plane is not available for NETCONF client. It is observed that the alarm-notification is sent from the O-RU using M-Plane and the regular start-up procedures is performed.

7.3 Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.1 profile (LLS-C2)

7.3.1 Test Description and Applicability

This test case is **CONDITIONAL MANDATORY** and shall be performed if the O-RU and O-DU declare support of LLS-C2 synchronization option (see Clause 4.5).

This test validates that O-RU is synchronizing from an O-DU via a chain of T-BC using ITU-T G.8275.1 [5] profile.

This test involves one O-DU, one O-RU and multiple T-BCs.

The configuration, management, and retrieval of status of the T-BC are not addressed in the present document.

7.3.2 Minimum Requirements (Prerequisites)

1. O-RU is connected to O-DU via a chain of T-BCs.
2. The T-BCs are class B. The subordinate port of the first T-BC is connected to O-DU. The master clock of the last T-BC is connected to the O-RU.
3. The O-DU is connected to a PRTC traceable time source, directly (GNSS Receiver connected to the O-DU) or via PTP.

7.3.3 Purpose and Scope

The O-RU is synchronizing from the O-DU with the ITU-T G.8275.1 [5] profile over a bridged network that can deploy several T-BCs. This test case validates the correct synchronization status of the O-RU.

7.3.4 Testability Requirements imposed on O-RU and O-DU

Requirements for M-Plane: is properly operating (as specified in 6.2).

Synchronization requirement: O-DU is connected to a local PRTC/source traceable to PRTC.

7.3.5 Test Methodology

These tests use the O-RAN M-Plane and O-DU NMS features.

Three conditions shall be covered:

- startup
- nominal
- degraded

7.3.5.1 Procedure: Startup conditions

1. Not yet configured.

External frequency and time source are available to the O-DU and deliver nominal status.

O-DU is not yet configured to select the time source and align its frequency and time to it.

O-DU not yet configured to act as PTP master on the FH ports.

2. Configured.

O-DU is configured to align to the selected frequency and time source.

3. Until disciplining

Until O-DU disciplining of the frequency and time to the selected source has completed, Startup conditions persists.

The O-RUs are configured to synchronize from PTP in ITU-T G.8275.1 [5] profile and report their status.

7.3.5.2 Procedure: Nominal conditions

1. O-DU is configured to start acting as a PTP master compliant with the ITU-T G.8275.1 [5] profile on selected FH ports.
2. O-DU acts as a PTP grand master or as a boundary clock with ports towards the FH interface in Master state, compliant with the ITU-T G.8275.1 [5] profile advertising “nominal” status.
3. O-DU reports status, acting as PTP master clock towards the FH interface.

7.3.5.3 Procedure: Degraded conditions

1. O-DU is configured to enter HOLDOVER based on local oscillator frequency.
2. O-DU acts as configured clock, with PTP ports in master state, compliant to ITU-T G.8275.1 profile advertising HOLDOVER status with degraded clockClass and clockAccuracy as specified by ITU-T G.8275.1 [5].
3. O-DU is configured to exit HOLDOVER and resumes normal frequency and phase disciplining using the source.
4. O-DU acts as configured clock, with PTP ports towards the FH interface in master state compliant to ITU-T G.8275.1 [5] in “nominal” status.

7.3.6 Test Requirements (expected results)

7.3.6.1 Startup conditions

The acceptance criterion is that the following status is observed for steps 1 to 3 (7.4.5.1 above):

- the FREERUN sync-state of the O-DU using the NMS
- the FREERUN sync-state of the O-RU using the M-Plane
- the UNLOCKED PTP lock-state of the O-RU using the M-Plane
- the UNLOCKED SyncE lock-state of the O-RU using the M-Plane (optional)

7.3.6.2 Nominal conditions

The acceptance criterion is that the following status is observed for steps 1 to 3 (7.4.5.2 above):

- the LOCKED sync state of the O-DU using the NMS
- the LOCKED PTP lock-state of the O-DU using the NMS
- the “Master Enabled” SyncE status of the O-DU using the NMS (optional)
- the received PTP clockClass level of the O-RU using the M-Plane
- the received SyncE SSM level of the O-RU using the M-Plane (optional)
- the LOCKED sync-state of the O-RU using the M-Plane
- the LOCKED PTP lock-state and PARENT PTP state of the O-RU using the M-Plane
- the LOCKED SyncE lock-state and OK or PARENT SyncE state of the O-RU using the M-Plane (optional)
- the synchronization status of the deployed T-BC using the respective NMS

7.3.6.3 Degraded conditions

The acceptance criterion is that the following status is observed for all steps 1 to 2 (7.4.5.3 above) (for steps 3 and 4, same acceptance criterion as “Nominal conditions” apply)

- the HOLDOVER sync-state of the O-DU using the NMS
- the received PTP clockClass level of the O-RU using the M-Plane
- the received SyncE SSM level of the O-RU using the M-Plane (optional). Note that if SyncE state is NOK, it is also permitted that the local default SSM level is reported using the M-Plane.
- the sync-state of the O-RU using the M-Plane is LOCKED if the received clockClass (or optional SSM QL) value matches the configured list of accepted values, otherwise HOLDOVER or FREERUN
- the PTP lock-state of the O-RU using the M-Plane is LOCKED, and the PTP state of the O-RU using the M-Plane is PARENT or OK if the received SSM matches the configured list of accepted values, otherwise NOK
- the SyncE lock-state (optional) of the O-RU using the M-Plane is LOCKED or OK, and the SyncE state of the O-RU using the M-Plane is PARENT or OK if the received SSM matches the configured list of accepted values, otherwise NOK

Note: If the NETCONF client triggers a reset procedure of the O-RU by FREERUN of the sync-state at these degraded conditions, the status observation using M-plane is not available for NETCONF client. It is observed that the alarm-notification is sent from the O-RU using M-Plane and the regular start-up procedures is performed.

7.4 Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.1 profile (LLS-C3)

7.4.1 Test Description and Applicability

This test case is CONDITIONAL MANDATORY and shall be performed if the O-RU and O-DU declare support of LLS-C3 synchronization option (see Clause 4.5).

This test validates that both the O-DU and O-RU are synchronizing from a common PRTC via a chain of T-BCs using ITU-T G.8275.1 [5] profile.

This test involves one O-DU, one O-RU, a PRTC/T-GM and multiple T-BCs.

The configuration, management, and retrieval of status of the T-BC are not addressed in the present document.

7.4.2 Minimum Requirements (Prerequisites)

1. Both O-RU and O-DU are connected to a common PRTC via a chain of T-BCs that are either directly connected to a PRTC/T-GM in the FH network or are connected to another T-BC that is traceable to a PRTC.
2. The T-BCs are class B. The subordinate port of the first T-BC is connected to a PRTC/T-GM. The O-DU and O-RU are connected to master ports of either the same T-BC or different ones.

7.4.3 Purpose and Scope

Both O-DU and O-RU are synchronized via a chain of T-BC from a common PRTC/T-GM located in the FH networks using ITU-T G.8275.1 [5] profile. This test case validates the correct synchronization status of the O-RU and O-DU.

7.4.4 Testability Requirements imposed on O-RU, O-DU and bridged network

Requirements for M-Plane: is properly operating (as specified in 6.2)

Synchronization requirement: The T-BC is connected to a local PRTC or to another T-BC that is traceable to a PRTC.

7.4.5 Test Methodology

These tests use the O-RAN M-Plane and O-DU NMS features.

Three conditions shall be covered:

- startup
- nominal
- degraded

7.4.5.1 Procedure: Startup conditions

1. Not yet configured.
 - External frequency and time source are available to the PRTC/T-GM and deliver nominal status.
 - PRTC/T-GM is not yet configured to select the time source and align its frequency and time to it.
 - PRTC/T-GM not yet configured to act as PTP master on the FH ports.
2. Configured.
 - PRTC/T-GM is configured (eg, via proprietary) interface to align to the selected frequency and time source.

3. Until disciplining

Until PRTC/T-GM disciplining of the frequency and time to the selected source has completed, Startup conditions persists.

The O-RUs are configured to synchronize from PTP in ITU-T G.8275.1 [5] profile and report their status.

7.4.5.2 Procedure: Nominal conditions

1. PRTC/T-GM is configured to start acting as a PTP master compliant with the ITU-T G.8275.1 [5] profile on selected FH ports.
2. PRTC/T-GM acts as a PTP grand master towards the FH interface in Master state, compliant with the ITU-T G.8275.1 [5] profile advertising “nominal” status.
3. PRTC/T-GM reports status, acting as PTP master clock towards the FH interface.

7.4.5.3 Procedure: Degraded conditions

1. PRTC/T-GM is configured to enter HOLDOVER based on local oscillator frequency.
2. PRTC/T-GM acts as configured clock, with PTP ports in master state, compliant to ITU-T G.8275.1 profile advertising HOLDOVER status with degraded clockClass and clockAccuracy as specified by ITU-T G.8275.1 [5].
3. PRTC/T-GM is configured to exit HOLDOVER and resumes normal frequency and phase disciplining using the source.
4. PRTC/T-GM acts as configured clock, with PTP ports towards the FH interface in master state compliant to ITU-T G.8275.1 [5] in “nominal” status.

7.4.6 Test Requirements (expected results)

7.4.6.1 Startup conditions

The validation is done by checking the correct synchronization state is observed for steps 1 to 3 (7.4.6.1 above):

- the FREERUN sync-state of the O-DU using the NMS
- the UNLOCKED PTP lock-state of the O-DU using the NMS
- the UNLOCKED SyncE lock-state of the O-DU using the NMS (optional)
- the FREERUN sync-state of the O-RU using the M-Plane
- the UNLOCKED PTP lock-state of the O-RU using the M-Plane
- the UNLOCKED SyncE lock-state of the O-RU using the M-Plane (optional)
- the synchronization status of the deployed T-BC using the respective NMS

7.4.6.2 Nominal conditions

The validation is done by checking the correct synchronization state is observed for steps 1 to 3 (7.4.6.2 above):

- the LOCKED sync-state of the O-DU using the NMS
- the LOCKED PTP lock-state of the O-DU using the NMS
- the “Master Enabled” SyncE status of the O-DU using the NMS (optional)
- the received PTP clockClass level of the O-RU using the M-Plane

- the received SyncE SSM level of the O-RU using the M-Plane (optional).
- the LOCKED sync-state of the O-RU using the M-Plane
- the LOCKED PTP lock-state and PARENT PTP state of the O-RU using the M-Plane
- the LOCKED SyncE lock-state and OK or PARENT SyncE state of the O-RU using the M-Plane (optional)
- the synchronization status of the deployed T-BC using the respective NMS

7.4.6.3 Degraded conditions

The validation is done by checking the correct synchronization state is observed for all steps 1 to 2 (7.4.5.3 above) (for steps 3 and 4, same acceptance criterion as “Nominal conditions” apply):

- the HOLDOVER sync-state of the O-DU using the NMS
- the received PTP clockClass level of the O-DU using the NMS
- the received SyncE SSM level of the O-DU using the NMS (optional)
- the received PTP clockClass level of the O-RU using the M-Plane
- the received SyncE SSM level of the O-RU using the M-Plane (optional). Note that if SyncE state is NOK, it is also permitted that the local default SSM level is reported using the M-Plane
- the synchronization status of the deployed T-BC using the respective NMS - the sync-state of the O-RU using the M-Plane is LOCKED if the received clockClass (or optional SSM QL) value matches the configured list of accepted values, otherwise HOLDOVER or FREERUN
- the PTP lock-state of the O-RU using the M-Plane is LOCKED, and the PTP state of the O-RU using the M-Plane is PARENT if the received clockClass matches the configured list of accepted values, otherwise NOK
- the SyncE lock-state (optional) of the O-RU using the M-Plane is LOCKED or OK, and the SyncE state of the O-RU using the M-Plane is PARENT or OK if the received SSM matches the configured list of accepted values, otherwise NOK

Note: If the NETCONF client triggers a reset procedure of the O-RU by FREERUN of the sync-state at these degraded conditions, the status observation using M-plane is not available for NETCONF client. It is observed that the alarm-notification is sent from the O-RU using M-Plane and the regular start-up procedures is performed.

7.5 Functional test of O-DU + O-RU using ITU-T G.8275.2 profile (LLS-C1)

This test is not addressed in the present document.

7.6 Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.2 profile (LLS-C2)

This test is not addressed in the present document.

7.7 Functional test of O-DU + bridged network + O-RU using ITU-T G.8275.2 profile (LLS-C3)

This test is not addressed in the present document.

7.8 Functional test of O-DU + bridged network + O-RU (LLS-C4)

This test case is not addressed in the present document.

7.9 Performance test of O-DU + Two O-RUs using ITU-T G.8275.1 profile (LLS-C1)

7.9.1 Test Description and applicability

This test case is **CONDITIONAL MANDATORY** and shall be performed if the O-RU and O-DU declare support of LLS-C1 synchronization option (see Clause 4.5).

This test validates that the two O-RUs meet the 3GPP limits at their air interface, when

- the O-DU gets its synchronization from a PRTC using either a local or remote PRTC as specified in the Conformance Test Specification [24], under either ideal or normal conditions
- and is connected to two O-RUs via direct FH links
- and distributes frequency and time to these O-RUs using the ITU-T G.8275.1 (SyncE + PTP) profile

7.9.2 Minimum Requirements (Prerequisites)

1. IOT Functional test 7.2 is successfully passed and all O-DU, and O-RUs report LOCKED status
2. O-RUs are connected to the O-DU via direct fiber O-RAN links
3. O-RUs are suitable for Case 1.1 or 1.2 as specified in eCPRI [11]

7.9.3 Purpose and scope

The O-DU synchronizes the O-RUs with the ITU-T G.8275.1 profile [5]. This test validates that the frequency and time error on the O-RU air interfaces are within the limits of the 3GPP, in both constant and variable temperature and traffic load conditions.

Only LOCKED state is tested; HOLDOVER state test is not addressed in the present document.

7.9.4 Testability Requirements imposed on O-RU and O-DU

Both O-DU and O-RU are running nominal software.

Requirements for M-Plane: is “up and running.”

Synchronization requirement: O-DU and test equipment is connected to a local PRTC or source traceable to PRTC.

Test 7.2.6.2 is successfully passed and both O-DU and O-RUs report LOCKED status.

7.9.5 Test Methodology

After O-DU and O-RUs are frequency and phase locked to their PRTC synchronization source using the LLS-C1 configuration, the frequency and phase errors are measured on the O-RUs air interface using a test equipment referenced to the same PRTC.

The O-DU input time error on the O-DU is as specified in the Conformance Test Specification [24],

- Either “ideal” with zero value
- Or “normal”, with configured values with standard masks

7.9.6 Test Requirement (expected result)

For both constant and variable conditions tests, the acceptance criterion is to measure with the test equipment:

- ± 50 ppb maximum frequency error at the air interface
- ITU-T Level 4, eCPRI Cat C (mandatory): ± 1500 ns maximum absolute time error at the O-RU air interface
- ITU-T Level 6A, eCPRI Cat B (optional): ± 260 ns maximum relative time error between the two O-RUs air interfaces
Note: This level of accuracy assumes an O-RU implementation suitable for Case 1.1 and case 1.2 as specified in eCPRI [11].
- ITU-T Level 6B, eCPRI Cat A (optional) ± 130 ns maximum relative time error between the two O-RUs air interfaces
Note: This level of accuracy assumes an O-RU implementation suitable for Case 1.2 as specified in eCPRI [11]. It also assumes co-location of the O-RUs and O-DU.

7.10 Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.1 profile (LLS-C2)

7.10.1 Test Description and applicability

This test case is **CONDITIONAL MANDATORY** and shall be performed if the O-RU and O-DU declare support of LLS-C2 synchronization option (see Clause 4.5).

This test validates that the two O-RUs are meeting the 3GPP limits at their air interface, when:

- the O-DU gets its synchronization from a PRTC using a local or remote PRTC as specified in the Conformance Test Specification [24], under either ideal or normal conditions
- and is connected to two O-RUs via eCPRI FH links via bridged network elements acting as class B T-BCs
- and distributes frequency and time to these O-RUs using the ITU-T G.8275.1 (SyncE + PTP) profile [5]

7.10.2 Minimum Requirements (Prerequisites)

1. IOT Functional Test 7.3 is successfully passed and all O-DU, bridged network elements and O-RUs report LOCKED status
2. O-RUs are connected to O-DU via bridged network elements acting as ITU-T G.8275.1 class B T-BCs using O-RAN links [5]
3. O-RUs are suitable for Case 1.1 or 1.2 as specified in eCPRI [11]

7.10.3 Purpose and Scope

The O-DU synchronizes the O-RUs with the ITU-T G.8275.1 [5] profile via the bridged network elements.

This test validates that the frequency and time error on the O-RU air interfaces are within the limits of the 3GPP, in both constant and variable temperature and traffic load conditions.

Only LOCKED state is tested; HOLDOVER state test is not addressed in the present document.

7.10.4 Testability Requirements imposed on O-RU and O-DU

Both O-DU and O-RU are running nominal software.

Requirements for M-Plane: is “up and running”

Synchronization requirement: O-DU and test equipment is connected to a local PRTC or source traceable to PRTC.

Test 7.3.6.2 is successfully passed and both O-DU and O-RUs report LOCKED status.

7.10.5 Test Methodology

After O-DU and O-RUs are frequency and phase locked to their PRTC synchronization source using the LLS-C2 configuration, the frequency and phase errors are measured on the O-RUs air interface using a test equipment referenced to the same PRTC.

The O-DU input time error on the O-DU is as specified in the Conformance Test Specification [24],

- Either “ideal” with zero value
- Or “normal”, with configured values with standard masks

7.10.6 Test Requirement (expected result)

For both constant and variable conditions tests, the acceptance criterion is to measure with the test equipment:

- ± 50 ppb maximum frequency error
- ITU-T Level 4, eCPRI Cat C (mandatory): ± 1500 ns maximum absolute time error at the O-RU air interface
- ITU-T Level 6A, eCPRI Cat B (optional): ± 260 ns maximum relative time error between the two O-RUs air interfaces
Note: This level of accuracy assumes up to 1 (respectively 2) class B T-BC on the path between the branching one and each O-RU implementation suitable for case 1.1 (respectively case 1.2) as specified in eCPRI [11].
- ITU-T Level 6B, eCPRI Cat A (optional) ± 130 ns maximum relative time error between the two O-RUs air interfaces
Note: This level of accuracy assumes direct fiber link between the branching class B T-BC and each O-RU implementation suitable case 1.2 as specified in eCPRI [11] (not supported by case 1.1). It also assumes co-location of the O-RUs and O-DU.

7.11 Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.1 profile (LLS-C3)

7.11.1 Test Description and applicability

This test case is CONDITIONAL MANDATORY and shall be performed if the O-RU and O-DU declare support of LLS-C3 synchronization option (see Clause 4.5).

This test validates that the two O-RUs are meeting the 3GPP limits at their air interface, and the O-DU is meeting time error limits specified in O-RAN WG4 CUS Specification [2] when

- both O-DU and O-RUs get their synchronization from a PRTC using SyncE + PTP over a Full Timing Support network (ITU-T G.8275.1 profile [5], ITU-T G.8271.1 network limits [7])

7.11.2 Minimum Requirements (Prerequisites)

1. IOT Functional Test 7.4 is successfully passed and all O-DU, bridged network elements and O-RUs report LOCKED status
2. O-RUs and O-DU are connected to bridged network elements acting as ITU-T G.8275.1 class B T-BCs using O-RAN links [5]. This FH network has a local PRTC distributing SyncE and PTP to both O-DU and O-RUs.
3. O-RUs are suitable for Case 1.1 or 1.2 as specified in eCPRI [11]

7.11.3 Purpose and scope

The FH network synchronizes all O-DU and O-RUs with the ITU-T G.8275.1 [5] profile via the bridged network elements.

This test validates that the frequency and time error on the O-RU air interfaces and O-DU are within the limits of the 3GPP, in both constant and variable temperature and traffic load conditions.

Only LOCKED state is tested; HOLDOVER state test is not addressed in the present document.

7.11.4 Testability Requirements imposed on O-RU and O-DU

Both O-DU and O-RU are running nominal software.

Requirements for M-Plane: is “up and running”

Synchronization requirement: FH network and test equipment is connected to a local PRTC or source traceable to PRTC.

Test 7.4.6.2 is successfully passed and both O-DU and O-RUs report LOCKED status.

7.11.5 Test Methodology

After O-DU and O-RUs are frequency and phase locked to their PRTC synchronization source using the LLS-C3 configuration, the frequency and phase errors are measured at the O-RUs air interface using a test equipment referenced to the same PRTC.

The PRTC output time error on the O-DU is as specified in the Conformance Test Specification [24],

- Either “ideal” with zero value
- Or “normal”, with configured values with standard masks

7.11.6 Test Requirement (expected result)

For both constant and variable conditions tests, the acceptance criterion is to measure with the test equipment:

- ± 50 ppb maximum frequency error (mandatory): at the O-RU air interface
- ITU-T Level 4, eCPRI Cat C (mandatory): ± 1500 ns maximum absolute time error at the O-RU air interface and at the O-DU output test signal (for example 1PPS)
- ITU-T Level 6A, eCPRI Cat B (optional): ± 260 ns maximum relative time error between the two O-RUs air interfaces
Note: This level of accuracy assumes up to 1 (respectively 2) class B T-BC on the path between the branching one and each O-RU implementation suitable for case 1.1 (respectively case 1.2) as specified in eCPRI [11].
- ITU-T Level 6B, eCPRI Cat A (optional) ± 130 ns maximum relative time error between the two O-RUs air interfaces
Note: This level of accuracy assumes direct fiber link between the branching class B T-BC and each O-RU implementation suitable case 1.2 as specified in eCPRI [11] (not supported by case 1.1). It also assumes co-location of the O-RUs and O-DU.

7.12 Performance test of O-DU + Two O-RUs using ITU-T G.8275.2 profile (LLS-C1)

This test is not addressed in the present document.

7.13 Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.2 profile ((LLS-C2)

This test is not addressed in the present document.

7.14 Performance test of O-DU + bridged network + Two O-RUs using ITU-T G.8275.2 profile (LLS-C3)

This test is not addressed in the present document.

7.15 Performance test of O-DU + bridged network + Two O-RUs (LLS-C4)

This test is not addressed in the present document.

8 C/U-Plane IOT Test

8.1 Overview

C/U-Plane IOT tests are used to validate the performance of DUT and interoperability of O-DU and O-RU from different vendors using O-RAN specified or customized IOT profile test configuration(s). IOT profiles and associated PTCs to be tested are selected by the participating vendors. Not all PTCs within a profile are required to be tested. See Annex A for details on the IOT profiles and PTCs.

The test environment should be arranged to match the expectations of the test eg, provide a noise-free channel and a traffic demand matching the intent of the test.

An O-DU and O-RU are considered interoperable under a profile test configuration (PTC) if the DUT can pass all mandatory and applicable conditional mandatory tests using that PTC, such passing being subject to the DUT using “to be validated” capabilities that may be listed in that PTC. The usage of such capabilities need not be continuous, eg, neither in every section description nor in every slot. The exact means to validate the usage of such capabilities is out of scope of the present document. Exemplary means include,

- a. Proprietary capability of the DUT, eg, a special test port or test log readout
- b. O-RAN FH protocol analysis, including that data packets are sent to appropriately configured endpoints

The PTC may also contain a list of “to be reported” capabilities. The usage, or absence of usage, of these capabilities shall be reported in the test report. The usage of these capabilities is not required to pass the test. The means of determination of the usage of these capabilities is out of the scope of the present document

It is expected that the DUT is capable of meeting each test and PTC requirement. The exact means to achieve this is out of scope of the present document. Exemplary means include,

- a. By design of the DUT
- b. Proprietary capability of the DUT such as a special configuration mechanism

8.2 Radio Layer 3 C-Plane establishment and Initial Radio U-Plane data transfer

8.2.1 Test Description and Applicability

This test case is MANDATORY.

This is a Radio system level test which is used to validate the radio system functionalities, performance and multi-vendor interoperability of the O-DU and O-RU from different vendors connected using the O-RAN WG4 specified FH interface [2], [3].

This test validates if a UE can perform Radio Layer 3 C-Plane establishment and initial Radio U-Plane data transfer procedures with the network which includes the O-DU and O-RU as an integrated system under test in this test setup.

Although there is no FH focused testing for C/U-Plane in this test, it is still possible to observe successful interoperability via positive test outcomes for this test. ie, if radio system level test passes, it can be inferred that O-RAN WG4 specified FH interface C/U-Plane is successfully working.

The DUT shall comprise a single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU)).

8.2.2 Minimum Requirements (Prerequisites)

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU))

1. Are connected through the O-RAN FH
2. Are synchronized with the common S-Plane configuration

Testing tools which are required for this test scenario

- Single Test UE or UE emulator: used to perform Radio Layer 3 C-Plane establishment and Radio U-Plane data transfers with the network
- O-CU or O-CU emulator either as a disaggregated node or as an aggregated node with the O-DU(s) (DUT): used to provide Layer 2 and Layer 3 radio processing on the network side. In case of a disaggregated node, terminates the 3GPP 5G F1 interface with the O-DU(s) (DUT)
- 4G Core network or 4G Core network emulator: used to terminate UEs (emulator) NAS protocol in NSA mode
- 4G MeNB or 4G MeNB emulator: used to terminate the 3GPP EN-DC X2 interface with 5G CU in NSA mode
- 5G Core Network or 5G Core Network emulator: used to terminate UEs (emulator) NAS protocol in SA mode
- Application test server: used to generate and terminate application layer traffic (eg, UDP, TWAMP, etc) and provide application layer processing on the network side

Testing tools which can be useful for this test scenario particularly for validating that the DUTs are configured and operating correctly during the test, troubleshooting and detailed validation purposes

- FH Protocol Analyzer: used for protocol analysis of O-RAN FH protocols in this specific test scenario, C/U-Plane procedural flows and contents
- RF Spectrum and Beam Signal Analyzer: used for RF and Beam power and quality analysis ensuring that the O-RU (DUT1(O-RU)) is transmitting correctly on the configured broadcast and synchronization signals on the downlink

8.2.3 Test Purpose and Scope

Purpose of this test is to validate key radio operation after M-Plane startup, ie, Radio Layer 3 C-Plane establishment and initial Radio U-Plane data transfer on system level with integration of O-DU and O-RU from different vendors.

Note that this test requires both Downlink and Uplink Radio Layer 3 C-Plane message. This means that this test also validates both transfer of Downlink and Uplink FH U-Plane message and related Downlink FH C-Plane messages.

8.2.4 Testability requirements imposed on O-RU and O-DU

Nominal software runs on the O-RU and O-DU.

8.2.5 Test Methodology

8.2.5.1 Initial Conditions

1. O-RU and O-DU are both in service, ie, M-Plane start-up procedure is completed, and broadcast channels are being transmitted
2. Test UE or UE emulator has not yet been registered with the network

8.2.5.2 Procedure: Nominal test

Performs Radio Layer 3 C-Plane establishment procedure using the Test UE or UE emulator. Note that Radio Layer 3 C-Plane establishment procedure depends on the 5G NR operation mode. In the case when the DUTs are operating in NSA mode, Radio Layer 3 C-Plane establishment procedure includes Attach procedure and/or service request procedure specified in 3GPP TS 23.401 [20] and EN-DC setup procedure specified in 3GPP TS 37.340 [22]. In the case when the DUTs are operating in SA mode, Radio Layer 3 C-Plane establishment procedure includes Registration procedure and service request procedure specified in 3GPP TS 23.502 [21].

Performs data transfer from the application test server to the Test UE or UE emulator. The application test server generates and transmits 10 IP packets with each packet 32 bytes in size.

Note that data transfer depends on the operation mode. In the case when the DUTs are operating in NSA mode, data transfers can be performed over Default EPS bearer using SN terminated split bearer specified in 3GPP TS 37.340 [22]. In case when the DUTs are operating in SA mode, data transfers can be performed over PDU Session and QoS flow specified in 3GPP TS 23.502 [21].

This test case does not specify the test data pattern generated by the application test server, but it is recommended that the test data pattern should include some level of randomness (ie, avoiding all zeros).

8.2.6 Test Requirement (expected result)

Observe the Test UE or emulated UE can perform Radio Layer 3 C-Plane establishment successfully and can perform data transfers over the network particularly through the O-DU and O-RU.

Record Test UE or UE emulator logs that the Radio Layer 3 (eg, RRC/NAS) message flows are per 3GPP TS 23.401 [20] Sections 5.3.4, 5.4 and 3GPP TS 37.340 [22] in NSA mode and 3GPP TS 23.502 [21] Sections 4.2.3, 4.3 in SA mode.

Record Test UE or UE emulator Radio U-Plane logs that the data packets transferred by application test server (ie, 10 IP packets of 32 bytes) are received correctly.

8.3 Radio U-Plane downlink data transfer (Downlink throughput performance)

8.3.1 Radio U-Plane downlink data transfer performance with one UE

8.3.1.1 Test Description and Applicability

This test case is MANDATORY.

The DUT shall comprise a single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU)). This scenario allows to test if a UE can perform Radio U-Plane data transfers with the network through O-DU and O-RU from different vendors.

8.3.1.2 Minimum Requirements (Prerequisites)

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU)):

1. Are connected through the O-RAN FH

2. Are synchronized with the common S-Plane configuration

Testing tools which are required for this test scenario:

- Single Test UE or UE emulator: used to perform Radio U-Plane data transfers with the network
- BFN when beamforming IOT profiles are to be tested
- O-CU or O-CU emulator either as a disaggregated node or as an aggregated node with the O-DU(s) (DUT): used to provide Layer 2 and Layer 3 radio processing on the network side. In case of disaggregated node, terminates the 3GPP 5G F1 interface with the O-DU(s) (DUT)
- 4G Core network or 4G Core network emulator: used to terminate UEs (emulator) NAS protocol in NSA mode
- 4G MeNB or 4G MeNB emulator: used to terminate the 3GPP EN-DC X2 interface with 5G CU in NSA mode
- 5G Core Network or 5G Core Network emulator: used to terminate UEs (emulator) NAS protocol in SA mode
- Application test server: used to generate and terminate application layer traffic (eg UDP, TWAMP, etc) and provide application layer processing on the network side

Testing tools which can be useful for this test scenario particularly for validating that the DUTs are configured and operating correctly during the test, troubleshooting and detailed validation purposes:

- FH Protocol Analyzer: used for protocol analysis of O-RAN FH protocols in this specific test scenario, FH C/U-Plane procedural flows and contents
- RF Spectrum and Beam Signal Analyzer: used for RF and Beam power and quality analysis ensuring that the O-RU (DUT1(O-RU)) is transmitting correctly on the configured broadcast and synchronization signals on the downlink

8.3.1.3 Test Purpose and Scope

Purpose of this test is to validate key radio operation after Radio Layer 3 C-Plane establishment and initial Radio U-Plane data transfer, the Radio U-Plane data transfer including throughput performance on system level with integration of O-DU and O-RU from different vendors.

Note that this test requires Maximum Layer 1 Radio data rate (with some margin). This means that this test also validates transfer of Downlink FH C/U-Plane message with higher MIMO layers and higher order modulation schemes.

8.3.1.4 Testability requirements imposed on O-RU and O-DU

Nominal software runs on the O-RU and O-DU.

8.3.1.5 Test Methodology

8.3.1.5.1 Initial Conditions

1. O-RU and O-DU are both in service, ie, M-Plane start-up procedure has been completed and broadcast channels are being transmitted
2. Test UE or UE emulator has registered to the network, ie, Radio Layer 3 C-Plane establishment procedure is completed. Note that Radio Layer 3 C-Plane establishment procedure depends on operation mode. In case when the of DUTs are operating in NSA mode, Radio Layer 3 C-Plane establishment procedure includes means Attach procedure and/or service request procedure specified in 3GPP TS 23.401 [20] and EN-DC setup procedure specified in 3GPP TS 37.340 [22]. In case when the of DUTs are operating in SA mode, Radio Layer 3 C-Plane establishment procedure includes means Registration procedure and service request procedure specified in 3GPP TS 23.502 [21].

8.3.1.5.2 Procedure: Nominal test

Performs downlink data transfer from application test server to the Test UE or UE emulator. The application test server generates and transmits downlink data with data size large enough to achieve the maximum Layer 1 Radio data rate, which is specified in 8.2.6, for the duration of the test. The duration of the test is 20 seconds.

Note that data transfer depends on the operation mode. In case when the DUTs are operating in NSA mode, data transfers can be performed over Default EPS bearer using SN terminated split bearer specified in 3GPP TS 37.340 [22]. In case when the DUTs are operating in SA mode, data transfers can be performed over PDU Session and QoS flow specified in 3GPP TS 23.502 [21].

This test case does not specify the test data pattern generated by the application test server, but it is recommended that the test data pattern should include some level of randomness (ie, avoiding all zeros).

8.3.1.6 Test Requirement (expected result)

Observe that the Test UE or emulated UE can perform Radio U-Plane data transfers over the network particularly through the O-DU and O-RU at the target data rate.

Record the Test UE or UE emulator Radio U-Plane logs and determine that they contain measurements such as the measured Radio U-Plane data rates during the test. The acceptance criterion is that Radio U-Plane data rate on average during the test duration achieves the performance level defined as follows

Performance level

Target data rate = Maximum Layer 1 Radio data rate * (1 - Ratio of Radio resource overhead) * Margin,

where

Maximum Layer 1 Radio data rate is calculated based on 3GPP TS 38.211 [15] and TS 38.214 [23] by using RAN parameters, ie, Total channel bandwidth, TDD configuration (when TDD), number of spatial/antenna streams, modulation order and coding rate. Note that these RAN parameters are defined in the IOT profiles (see Annex A for details).

Ratio of Radio resource overhead is calculated based on 3GPP TS 38.211 [15] by using Radio physical layer channel configurations, eg, PDCCH resource mapping, Reference signal resource mapping, and so on, assumed in the IOT setup. Note that the physical layer channel configurations are part of the radio test setup configuration.

Margin is assumed in order to account for issues other than the FH. The value of *Margin* is 0.8.

8.3.2 Radio U-Plane downlink data transfer performance with two UEs

8.3.2.1 Test Description and Applicability

This test is CONDITIONAL MANDATORY; the condition being the DUT supports at least two simultaneous UE data layers. Any PTC not stating a UE data layer configuration is not required to execute the test.

The DUT shall comprise a single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU)).

This scenario allows to test that two UEs can simultaneously perform Radio U-Plane data transfer with the network through O-DU and O-RU from different vendors as set out below.

8.3.2.2 Minimum Requirements (Prerequisites)

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU))

1. Are connected through the O-RAN FH
2. Are synchronized with the common S-Plane configuration

Testing tools which are required for this test scenario:

1. Two UEs, either test UEs or UE emulator capable to support two UEs, where each UE supports a single layer: used to perform Radio U-Plane data transfer with the network
2. BFN able to simultaneously provide two downlink channels with different effective angle of departure for the UEs. The angle of departure and channel conditioning to provide sufficient rank of channel matrix such that each UE can respectively receive the UE layer directed to it. The BFN shall provide symmetrical uplink and downlink channels
3. O-CU or O-CU emulator either as a disaggregated node or as an aggregated node with the (DUT1) O-DU. In case of disaggregated node, O-CU or O-CU emulator terminates the 3GPP 5G F1 interface with the DUT1 (O-DU)
4. 5G Core Network or 5G Core Network emulator: used to terminate UEs (emulator) NAS protocol
5. Application test server: used to generate and terminate application layer traffic (eg UDP, TWAMP, etc) and provide application layer processing on the network side

Testing tools which can be useful for this test scenario particularly for validating that the DUTs are configured and operating correctly during the test, troubleshooting and detailed validation purposes:

1. RF Spectrum and Beam Signal Analyzer: used for RF and Beam power and quality analysis ensuring that the O-RU (DUT1(O-RU)) is correctly transmitting the configured broadcast and synchronization signals on the downlink
2. FH Protocol Analyzer: For protocol analysis of O-RAN FH protocols and message flows (for example to verify or report used capabilities)

Additional requirements for this test scenario:

1. A means to determine that the DUT has used the capabilities declared in the “to be validated” and “to be reported” tables associated with the PTC. The exact means of determination is out of scope of the present document

8.3.2.3 Test Purpose and Scope

Purpose of this test is to validate, after Radio Layer 3 C-Plane establishment and initial Radio U-Plane data transfer when 2 UEs are connected to the network according to the relevant constraints in the PTC, the Radio U-Plane data transfer including throughput performance on system level with integration of O-DU and O-RU from different vendors.

This test requires Maximum Layer 1 Radio data rate (with some margin) for successful completion. This means that this test also validates transfer of Downlink FH C/U-Plane messages with two layers, one per mobile, and higher order modulation schemes.

8.3.2.4 Testability requirements imposed on O-RU and O-DU

Nominal software runs on the O-RU and O-DU.

8.3.2.5 Test Methodology

8.3.2.5.1 Initial Conditions

1. O-RU and O-DU are both in service, ie, M-Plane start-up procedure has been completed and broadcast channels are being transmitted

The two UEs, either test UE or UE emulator, have registered to the network, ie, Radio Layer 3 C-Plane establishment procedures are completed. Radio Layer 3 C-Plane establishment procedure includes Registration procedure and service request procedure specified in 3GPP TS 23.502 [21]

8.3.2.5.2 Procedure: Nominal test

Perform downlink data transfer from the application test server on the network side to the UEs. The application test server connected to the UEs generates and transmits data with data size that, when aggregated across both UEs, is large

enough to achieve the maximum Layer 1 Radio data rate, which is specified in 8.3.2.6, for the duration of the test. The duration of the test is 20 seconds.

Data transfers shall be performed over PDU Session and QoS flow specified in 3GPP TS 23.502 [21].

This test case does not specify the test data pattern generated by the application test server connected to the UEs, but it is recommended that the test data pattern should include some level of randomness (ie, avoid all zeros).

8.3.2.6 Test Requirement (expected result)

Observe that both UEs perform Radio U-Plane data reception over the network through the O-DU and O-RU, with aggregated UE transfer rate achieving the target data rate.

Collect the Radio U-Plane logs of both UEs and analyze to determine each contains measurements such as the measured Radio U-Plane data rates during the test

Observe, for example by analyzing message flows over FH, that “to be validated” capabilities listed in the PTC are used and such usage is recorded in the test report.

Observe that the usage or absence of usage of the “to be reported” capabilities listed in the PTC is recorded in the test report.

The performance acceptance criterion is that for the duration of the test both UEs simultaneously receive downlink data such that the aggregated Radio U-Plane rate achieves the performance level defined as follows,

Performance level

Target data rate = Maximum Layer 1 Radio data rate * (1 – Margin),

where

Maximum Layer 1 Radio data rate is calculated based on TS 38.306, clause 4.1.2 [26] with $v=2$ (for two layers, one per UE). The data rate shall be pro-rated with the TDD configuration specified in IOT profile. Parameter values for number of carriers (J), numerology (μ), and total channel bandwidth (BW) are also specified in the IOT profiles. Maximum supported modulation order (Q_m) is DUT’s maximum supported modulation order in DL for the PTC being tested. Scaling factor (f) shall be set to one. Any parameters in Equation 8.3.2.6-1 not defined in the IOT profiles are defined in TS 38.306 [26]

$$\text{data rate (in Mbps)} = 10^{-6} \cdot \sum_{j=1}^J \left(v_{\text{Layers}}^{(j)} \cdot Q_m^{(j)} \cdot f^{(j)} \cdot R_{\text{max}} \cdot \frac{N_{\text{PRB}}^{BW^{(j)}, \mu} \cdot 12}{T_s^{\mu}} \cdot (1 - OH^{(j)}) \right) \quad [\text{Equation 8.3.2.6-1}]$$

Margin is assumed in order to account for impairments other than the FH. The value of Margin is 0.2

Multi-layer operation is implicitly tested as achievement of target performance requires that each UEs nominally consumes all available resources on its respective layer

If the data throughput performance achieves the necessary performance level but use of “to be validated” capabilities listed in the PTC is NOT confirmed, the inconclusive outcome of the test is marked as failure.

8.4 Radio U-Plane uplink data transfer (Uplink throughput performance)

8.4.1 Radio U-Plane uplink data transfer performance with one UE

8.4.1.1 Test Description and Applicability

This test case is MANDATORY.

The DUT shall comprise a single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU)).

This scenario allows to test if a UE can perform Radio U-Plane data transfers with the network through O-DU and O-RU from different vendors.

8.4.1.2 Minimum Requirements (Prerequisites)

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU))

1. Are connected through the O-RAN FH
2. Are synchronized with the common S-Plane configuration

Testing tools which are required for this test scenario

- Single Test UE or UE emulator: used to perform Radio U-Plane data transfers with the network
- BFN when beamforming IOT profiles are to be tested
- O-CU or O-CU emulator either as a disaggregated node or as an aggregated node with the O-DU(s) (DUT): used to provide Layer 2 and Layer 3 radio processing on the network side. In case of a disaggregated node, terminates the 3GPP 5G F1 interface with the O-DU(s) (DUT).
- 4G Core network or 4G Core network emulator: used to terminate UEs (emulator) NAS protocol in NSA mode
- 4G MeNB or 4G MeNB emulator: used to terminate the 3GPP EN-DC X2 interface with 5G CU in NSA mode
- 5G Core Network or 5G Core Network emulator: used to terminate UEs (emulator) NAS protocol in SA mode
- Application test server: used to generate and terminate application layer traffic (eg UDP, TWAMP, etc) and provide application layer processing on the network side

Testing tools which can be useful for this test scenario particularly for validating that the DUTs are configured and operating correctly during the test, troubleshooting and detailed validation purposes

- FH Protocol Analyzer: used for protocol analysis of O-RAN FH protocols in this specific test scenario, FH C/U-Plane procedural flows and contents
- RF Spectrum and Beam Signal Analyzer: used for RF and Beam power and quality analysis ensuring that the O-RU (DUT1(O-RU)) is transmitting correctly on the configured broadcast and synchronization signals on the downlink

8.4.1.3 Test Purpose and Scope

Purpose of this test is to validate key radio operation after Radio Layer 3 C-Plane establishment and initial Radio U-Plane data transfer, the Radio U-Plane data transfer including throughput performance on system level with integration of O-DU and O-RU from different vendors.

Note that this test requires Maximum Layer 1 Radio data rate (with some margin). This means that this test also validates transfer of Uplink FH C/U-Plane message with higher MIMO layers and higher order modulation schemes.

8.4.1.4 Testability requirements imposed on O-RU and O-DU

Nominal software runs on the O-RU and O-DU.

8.4.1.5 Test Methodology

8.4.1.5.1 Initial Conditions

1. O-RU and O-DU are both in service, ie, M-Plane start-up procedure has been completed and broadcast channels are being transmitted
2. Test UE or UE emulator has registered to the network, ie, Radio Layer 3 C-Plane establishment procedure is completed. Note that Radio Layer 3 C-Plane establishment procedure depends on operation mode. In case when

the of DUTs are operating in NSA mode, Radio Layer 3 C-Plane establishment procedure includes means Attach procedure and/or service request procedure specified in 3GPP TS 23.401 [20] and EN-DC setup procedure specified in 3GPP TS 37.340 [22]. In case when the of DUTs are operating in SA mode, Radio Layer 3 C-Plane establishment procedure includes means Registration procedure and service request procedure specified in 3GPP TS 23.502 [21].

8.4.1.5.2 Procedure: Nominal test

Performs uplink data transfer from the Test UE or UE emulator to application test server on the network side. Either the Test UE or UE emulator, or application test server connected with UE generates and transmits uplink data with data size large enough to achieve the maximum Layer 1 Radio data rate, which is specified in 8.3.6, for the duration of the test. The Test UE or UE emulator transmits the uplink data. The duration of the test is 20 seconds.

Note that data transfer depends on the operation mode. In case when the DUTs are operating in NSA mode, data transfers can be performed over Default EPS bearer using SN terminated split bearer specified in 3GPP TS 37.340 [22]. In case when the DUTs are operating in SA mode, data transfers can be performed over PDU Session and QoS flow specified in 3GPP TS 23.502 [21].

This test case does not specify the test data pattern generated by either the Test UE or UE emulator, or application test server connected with UE, but it is recommended that the test data pattern should include some level of randomness (ie, avoiding all zeros).

8.4.1.6 Test Requirement (expected result)

Observe that the Test UE or emulated UE can perform Radio U-Plane data transfers over the network particularly through the O-DU and O-RU at the target data rate.

Record the Test UE or UE emulator Radio U-Plane logs and determine that they contain measurements such as the measured Radio U-Plane data rates during the test. The acceptance criterion is that Radio U-Plane data rate on average during the test duration achieves the performance level defined as follows

Performance level

Target data rate = Maximum Layer 1 Radio data rate * (1 - Ratio of Radio resource overhead) * Margin,

where

Maximum Layer 1 Radio data rate is calculated based on 3GPP TS 38.211 [15] and TS 38.214 [23] by using RAN parameters, ie, Total channel bandwidth, TDD configuration (when TDD), number of spatial/antenna streams, modulation order and coding rate. Note that these RAN parameters are defined in the IOT profiles (see Annex A for details).

Ratio of Radio resource overhead is calculated based on 3GPP TS 38.211 [15] by using Radio physical layer channel configurations, eg, PUCCH resource mapping, Reference signal resource mapping, and so on, assumed in the IOT setup. Note that the physical layer channel configurations are part of the radio test setup configuration.

Margin is assumed in order to account for issues other than the FH. The value of *Margin* is 0.8.

8.4.2 Radio U-Plane uplink data transfer performance with two UEs

8.4.2.1 Test Description and Applicability

This test is CONDITIONAL MANDATORY; the condition being the DUT supports at least two simultaneous UE data layers. Any PTC not stating a UE data layer configuration is not required to execute the test.

The DUT shall comprise a single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU)).

This scenario allows to test that two UEs can simultaneously perform Radio U-Plane data transfer with the network through O-DU and O-RU from different vendors as set out below.

8.4.2.2 Minimum Requirements (Prerequisites)

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU))

1. Are connected through the O-RAN FH
2. Are synchronized with the common S-Plane configuration

Testing tools which are required for this test scenario:

1. Two UEs, either test UEs or UE emulator capable to support two UEs, where each UE supports a single layer: used to perform Radio U-Plane data transfer with the network
2. BFN able to simultaneously provide two uplink channels with different effective angle of arrival for the UEs. The angle of arrival and channel conditioning to provide sufficient rank of the channel matrix, such that both UE layers can be properly detected by the DUT during channel estimation and equalisation. The BFN shall provide symmetrical uplink and downlink channels
3. O-CU or O-CU emulator either as a disaggregated node or as an aggregated node with the (DUT1) O-DU. In case of disaggregated node, O-CU or O-CU emulator terminates the 3GPP 5G F1 interface with the DUT1 (O-DU)
4. 5G Core Network or 5G Core Network emulator: used to terminate UEs (emulator) NAS protocol
5. Application test server: used to generate and terminate application layer traffic (eg UDP, TWAMP, etc) and provide application layer processing on the network side

Testing tools which can be useful for this test scenario particularly for validating that the DUTs are configured and operating correctly during the test, troubleshooting and detailed validation purposes:

1. RF Spectrum and Beam Signal Analyzer: used for RF and Beam power and quality analysis ensuring that the O-RU (DUT1(O-RU)) is correctly transmitting the configured broadcast and synchronization signals on the downlink
2. FH Protocol Analyzer: For protocol analysis of O-RAN FH protocols and message flows (for example to verify or report used capabilities)

Additional requirements for this test scenario:

1. A means to determine that the DUT has used the capabilities declared in the “to be validated” and “to be reported” tables associated with the PTC. The exact means of determination is out of scope of the present document

8.4.2.3 Test Purpose and Scope

Purpose of this test is to validate, after Radio Layer 3 C-Plane establishment and initial Radio U-Plane data transfer when 2 UEs are connected to the network according to the relevant constraints in the PTC, the Radio U-Plane data transfer including throughput performance on system level with integration of O-DU and O-RU from different vendors.

This test requires Maximum Layer 1 Radio data rate (with some margin) for successful completion. This means that this test also validates transfer of Uplink FH C/U-Plane messages with two layers, one per mobile, and higher order modulation schemes.

8.4.2.4 Testability requirements imposed on O-RU and O-DU

Nominal software runs on the O-RU and O-DU.

8.4.2.5 Test Methodology

8.4.2.5.1 Initial Conditions

1. O-RU and O-DU are both in service, ie, M-Plane start-up procedure has been completed and broadcast channels are being transmitted

The two UEs, either test UE or UE emulator, have registered to the network, ie, Radio Layer 3 C-Plane establishment procedures are completed. Radio Layer 3 C-Plane establishment procedure includes Registration procedure and service request procedure specified in 3GPP TS 23.502 [21]

8.4.2.5.2 Procedure: Nominal test

Perform uplink data transfer from the UEs to the application test server on the network side. Both UEs, or application test server connected to the UEs, generate and transmit uplink data with data size, when aggregated across both UEs, is large enough to achieve the maximum Layer 1 Radio data rate, which is specified in 8.4.2.6, for the duration of the test. The duration of the test is 20 seconds.

Data transfers shall be performed over PDU Session and QoS flow specified in 3GPP TS 23.502 [21].

This test case does not specify the test data pattern generated by the UEs, or application test server connected to the UEs, but it is recommended that the test data pattern should include some level of randomness (ie, avoid all zeros).

8.4.2.6 Test Requirement (expected result)

Observe that both UEs perform Radio U-Plane data transfer over the network through the O-DU and O-RU, with aggregated UE transfer rate achieving the target data rate.

Collect the Radio U-Plane logs of both UEs and analyze to determine each contains measurements such as the measured Radio U-Plane data rates during the test

Observe, for example by analyzing message flows over FH, that “to be validated” capabilities listed in the PTC are used and such usage is recorded in the test report.

Observe that the usage or absence of usage of the “to be reported” capabilities listed in the PTC is recorded in the test report.

The performance acceptance criterion is that for the duration of the test both UEs simultaneously transmit uplink data such that the aggregated Radio U-Plane rate achieves the performance level defined as follows,

Performance level

Target data rate = Maximum Layer 1 Radio data rate * (1 – Margin),

where

Maximum Layer 1 Radio data rate is calculated based on TS 38.306, clause 4.1.2 [26], Equation 8.3.2.6-1 above defines the relationship, with $v=2$ (for two layers, one per UE). The data rate shall be pro-rated with the TDD configuration specified in IOT profile. Parameter values for number of carriers (J), numerology (μ), and total channel bandwidth (BW) are also specified in the IOT profiles. Maximum supported modulation order (Q_m) is DUT’s maximum supported modulation order in UL for the PTC being tested. Scaling factor (f) shall be set to one. Any parameters in Equation 8.3.2.6-1 not defined in the IOT profiles are defined in TS 38.306 [26]

Margin is assumed in order to account for impairments other than the FH. The value of Margin is 0.2

Multi-layer operation is implicitly tested as achievement of target performance requires that each UEs nominally consumes all available resources on its respective layer

If the performance acceptance criterion has been met, but use of “to be validated” capabilities listed in the PTC is NOT confirmed, the inconclusive outcome of the test is marked as failure.

9 C/U-Plane Delay Management IOT Test

9.1 General

The test cases defined in this section will validate that the transmitted Control- and User-data packets sent from the O-DU are received within the reception windows at the O-RU by checking that the frame timing on the air interface is correct.

The same is applicable for uplink direction, ie, that the User-data packets sent from the O-RU are received within the reception window on the O-DU, this is done by checking that it is possible for a test UE to attach and send data.

For all currently defined IOT Profiles the O-DU timing advance type is ‘Fixed Timing Advance’. The Figure below is copied from the O-RAN WG4 CUS Specification [2] section 4.4.6 “Latency Categories for O-DU with fixed timing advance”.

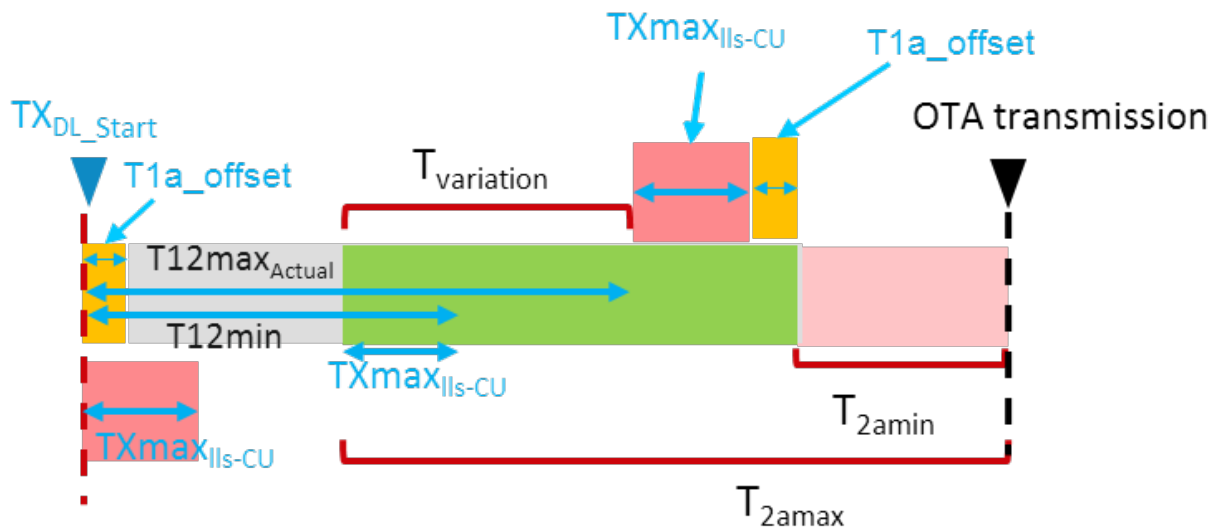


Figure 9.1-1: O-RAN WG4 CUS Specification [2] section 4.4.6 “Latency Categories for O-DU with fixed timing advance”.

The Figure shows the downlink timing when fixed timing is used, the point TX_{DL_Start} is a fixed point in relation to the “OTA transmission” point.

The O-DU needs only to be configured with T_{12max}, T_{12min} and T_{cp_adv_dl} the values are taken from the profile that is used when running these test cases. The T_{cp_adv_dl} value is the time in advance the O-RU needs the C-Plane messages before the U-Plane information is received by the O-RU. These test cases assume a symmetric latency in the Fronthaul Network, ie, T₁₂=T₃₄.

These test cases assume that the difference between “fixed timing advance” and “dynamic timing advance” is that for “fixed timing advance” the O-DU will advance the transmission of downlink data the same amount of time for all connected O-RUs regardless of the individual fronthaul latency, for “dynamic timing advance” the advancement could be different per O-RU and thus eg, T_{12max} shall be configured per O-RU.

An O-DU running in “Fixed Timing Advance”-mode is an O-DU that has the ability to handle a specific value for T_{1a} (sum of fronthaul latency T₁₂ and O-RU processing delay T_{2a}). The O-DU will set the TX_{DL_Start} to this fixed value independent on actual T_{12_max} value.

TXmax_{II_S-CU} will vary due to different line-rates of the fronthaul interface and amount of data (control+user) that is sent for a specific symbol.

9.2 Test environment

Testing tools which are required for the delay management test cases.

- Single Test UE or UE emulator: used to perform Radio Layer 3 C-Plane establishment and Radio U-Plane data transfers with the network
- O-CU or O-CU emulator either as a disaggregated node or as an aggregated node with the O-DU(s) (DUT): used to provide Layer 2 and Layer 3 radio processing on the network side. In case of a disaggregated node, terminates the 3GPP 5G F1 interface with the O-DU(s) (DUT)
- 4G Core network or 4G Core network emulator: used to terminate UEs (emulator) NAS protocol in NSA mode
- 4G MeNB or 4G MeNB emulator: used to terminate the 3GPP EN-DC X2 interface with 5G CU in NSA mode
- 5G Core Network or 5G Core Network emulator: used to terminate UEs (emulator) NAS protocol in SA mode
- Application test server: used to generate and terminate application layer traffic (eg, UDP, TWAMP, etc) and provide application layer processing on the network side

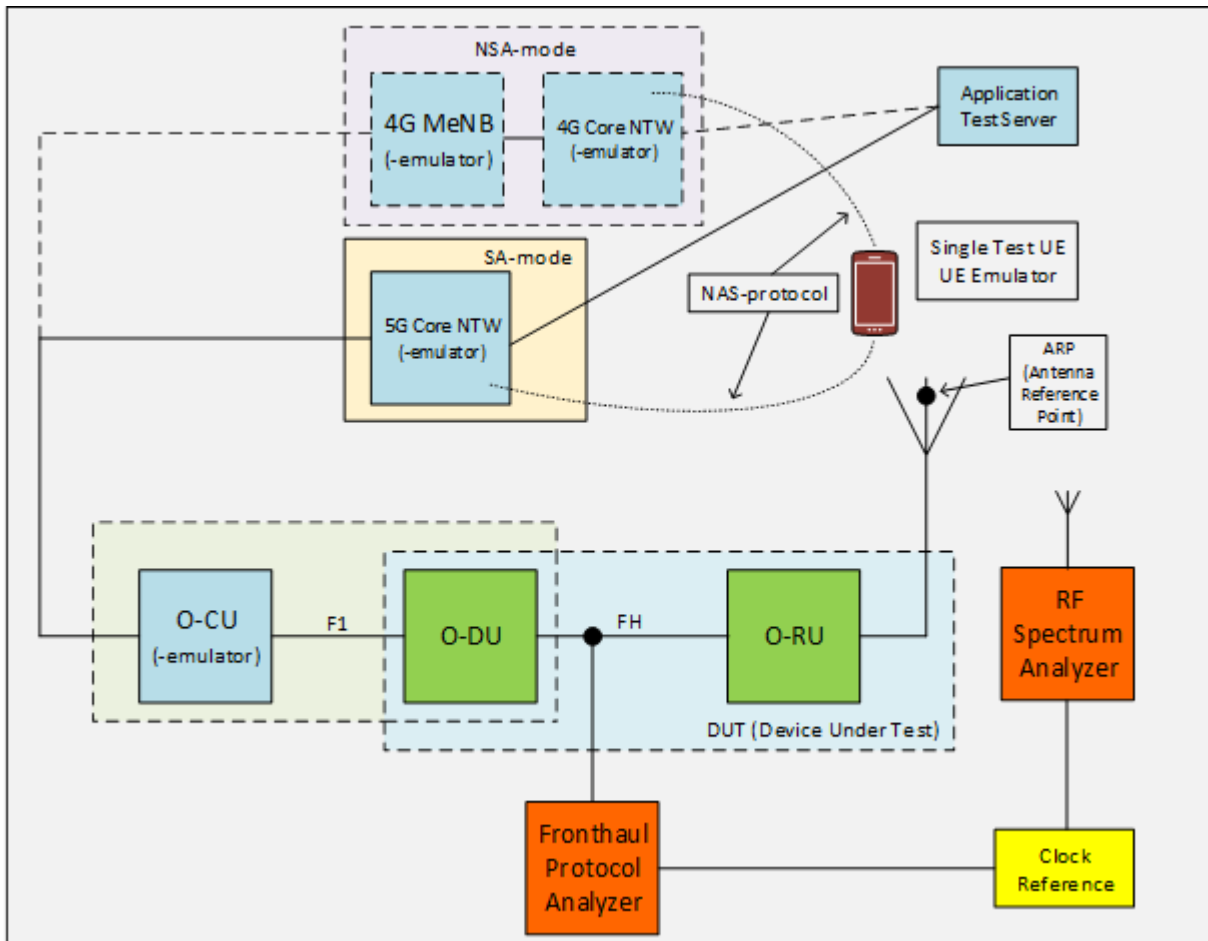


Figure 9.2-1: Delay-management test environment

Testing tools which can be useful for this test scenario particularly for validating that the DUTs are configured and operating correctly during the test, troubleshooting and detailed validation purposes

- FH Protocol Analyzer: used for protocol analysis of O-RAN FH protocols in this specific test scenario, C/U-Plane procedural flows and contents

- RF Spectrum and Beam Signal Analyzer: used for RF and Beam power and quality analysis ensuring that the O-RU (DUT1(O-RU)) is transmitting correctly on the configured broadcast and synchronization signals on the downlink

9.3 Timing accuracy definition

When in “Test Requirement (expected result)” it is stated “... *timing accuracy at the antenna reference point is according to requirement*” the following is applicable.

In ORAN WG4 CUS Specification [2] section 11.2.5.5 “Air interface maximum time error”, tables 11.2.5.5-1 and 11.2.5.5-2 list the 3GPP air interface timing accuracy for different LTE and 5G features. The timing accuracy requirements are divided into 2 alternatives, relative or absolute time errors.

Depending on what 3GPP features that are to be supported by a specific IOT-profile different timing accuracy requirement is thus applicable.

9.4 Delay Management #1, minimum fronthaul latency

9.4.1 Test Description and Applicability

This test case is OPTIONAL.

This is a Radio system level test which is used to validate the radio system functionalities, performance and multi-vendor interoperability of the O-DU and O-RU from different vendors connected using the O-RAN WG4 specified FH interface [2], [3].

This test validates that the O-RU will transmit the user data at correct point in time and with correct content during different conditions related to the fronthaul interface latency between the O-DU and the O-RU.

9.4.2 Minimum Requirements

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU))

1. Are connected through the O-RAN FH
2. Are synchronized with the common S-Plane configuration

9.4.3 Test Purpose

Purpose of this test is to validate that correct timing is achieved on the air interface regardless of the O-RU processing time and using the minimum feasible fronthaul latency.

9.3.4 Testability requirements imposed on O-RU and O-DU

Nominal software runs on the O-RU and O-DU.

9.4.5 Test Methodology

9.4.5.1 Initial Conditions

1. The fronthaul latency should be at a minimum for this test case. The T12_min in current IOT-Profiles is 0µs. 0µs is not possible to achieve in reality, instead the shortest possible fronthaul latency with existing test equipment is sufficient for this initial condition.
2. O-RU and O-DU are both in service, ie, M-Plane start-up procedure is completed and broadcast channels are being transmitted

3. Test UE or UE emulator has registered to the network, ie, Radio Layer 3 C-Plane establishment procedure is completed

9.4.5.2 Procedure: Nominal test

Performs downlink and uplink data transfer to/from the test UE.

9.4.6 Test Requirement (expected result)

Observe that:

1. With Fronthaul Protocol Analyzer (connected at the O-DU-end of fronthaul interface), observe that the fronthaul C- and U-Plane messages transmitted by the O-DU are transmitted within respective plane's transmission windows.
2. With Fronthaul Protocol Analyzer (connected at the O-RU-end of fronthaul interface), observe that the fronthaul C- and U-Plane messages in downlink direction are received within respective plane's reception window by the O-RU.
3. With the RF Spectrum Analyzer, observe that the air interface transmission timing accuracy at the antenna reference point is according to requirements as specified in section 9.3.
4. With Fronthaul Protocol Analyzer (connected at the O-RU-end of fronthaul interface), observe that the U-Plane messages transmitted by the O-RU (uplink direction) are transmitted within the O-RU's transmission window

9.5 Delay Management #2, maximum fronthaul latency

9.5.1 Test Description and Applicability

This test case is OPTIONAL.

This is a Radio system level test which is used to validate the radio system functionalities, performance and multi-vendor interoperability of the O-DU and O-RU from different vendors connected using the O-RAN WG4 specified FH interface [2], [3].

This test validates that the O-RU will transmit the user data at correct point in time and with correct content during different conditions related to the fronthaul interface between the O-DU and the O-RU.

9.5.2 Minimum Requirements

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU))

1. Are connected through the O-RAN FH
2. Are synchronized with the common S-Plane configuration

9.5.3 Test Purpose

Purpose of this test is to validate that correct timing is achieved on the air interface regardless of the O-RU processing time and using the maximum supported fronthaul latency as per the corresponding IOT profile.

9.5.4 Testability requirements imposed on O-RU and O-DU

Nominal software runs on the O-RU and O-DU.

9.5.5 Test Methodology

9.5.5.1 Initial Conditions

1. The fronthaul latency should be at a maximum for this test case. The T12_max in current IOT-Profiles is 160μs.
2. O-RU and O-DU are both in service, ie, M-Plane start-up procedure is completed and broadcast channels are being transmitted
3. Test UE or UE emulator has registered to the network, ie, Radio Layer 3 C-Plane establishment procedure is completed.

9.5.5.2 Procedure: Nominal test

Performs downlink and uplink data transfer to/from the test UE.

9.5.6 Test Requirement (expected result)

Observe that:

1. With Fronthaul Protocol Analyzer (connected at the O-DU-end of fronthaul interface), observe that the fronthaul C- and U-Plane messages transmitted by the O-DU are transmitted within respective plane's transmission windows.
2. With Fronthaul Protocol Analyzer (connected at the O-RU-end of fronthaul interface), observe that the fronthaul C- and U-Plane messages in downlink direction are received within respective plane's reception window by the O-RU.
3. With the RF Spectrum Analyzer, observe that the air interface transmission timing accuracy at the antenna reference point is according to requirements as specified in section 9.3.
4. With Fronthaul Protocol Analyzer (connected at the O-RU-end of fronthaul interface), observe that the U-Plane messages transmitted by the O-RU (uplink direction) are transmitted within the O-RU's transmission window.

9.6 Delay Management #3, normal fronthaul latency

9.6.1 Test Description and Applicability

This test case is OPTIONAL.

This is a Radio system level test which is used to validate the radio system functionalities, performance and multi-vendor interoperability of the O-DU and O-RU from different vendors connected using the O-RAN WG4 specified FH interface [2], [3].

This test validates that the O-RU will transmit the user data at correct point in time and with correct content during different conditions related to the fronthaul interface between the O-DU and the O-RU.

9.6.2 Minimum Requirements

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU))

1. Are connected through the O-RAN FH
2. Are synchronized with the common S-Plane configuration

9.6.3 Test Purpose

Purpose of this test is to validate that correct timing is achieved on the air interface regardless of the O-RU processing time and using a normal fronthaul latency as per the corresponding IOT profile.

9.6.4 Testability requirements imposed on O-RU and O-DU

Nominal software runs on the O-RU and O-DU.

9.6.5 Test Methodology

9.6.5.1 Initial Conditions

1. The fronthaul latency should be set to an arbitrary value between 0 and 160 μ s eg, 75 μ s
2. O-RU and O-DU are both in service, ie, M-Plane start-up procedure is completed and broadcast channels are being transmitted
3. Test UE or UE emulator has registered to the network, ie, Radio Layer 3 C-Plane establishment procedure is completed

9.6.5.2 Procedure: Nominal test

Performs downlink and uplink data transfer to/from the test UE.

9.6.6 Test Requirement (expected result)

Observe that:

1. With Fronthaul Protocol Analyzer (connected at the O-DU-end of fronthaul interface), observe that the fronthaul C- and U-Plane messages transmitted by the O-DU are transmitted within respective plane's transmission windows.
2. With Fronthaul Protocol Analyzer (connected at the O-RU-end of fronthaul interface), observe that the fronthaul C- and U-Plane messages in downlink direction are received within respective plane's reception window by the O-RU.
3. With the RF Spectrum Analyzer, observe that the air interface transmission timing accuracy at the antenna reference point is according to requirements as specified in section 9.3.
4. With Fronthaul Protocol Analyzer (connected at the O-RU-end of fronthaul interface), observe that the U-Plane messages transmitted by the O-RU (uplink direction) are transmitted within the O-RU's transmission window.

9.7 Delay Management #4, larger fronthaul latency then supported

9.7.1 Test Description and Applicability

This test case is OPTIONAL.

This is a Radio system level test which is used to validate the radio system functionalities, performance and multi-vendor interoperability of the O-DU and O-RU from different vendors connected using the O-RAN WG4 specified FH interface [2], [3].

This test validates that the O-RU will **not** transmit anything on the air interface when the latency in the fronthaul network is larger than is supported by the O-RU, ie, the C- and U-Plane messages will be received too late on the O-RU (outside its reception window) and thus be dropped by the O-RU.

9.7.2 Minimum Requirements

Single O-DU (DUT1(O-DU)) and a single O-RU (DUT1(O-RU))

1. Are connected through the O-RAN FH

2. Are synchronized with the common S-Plane configuration

9.7.3 Test Purpose

Validate that the O-RU will not transmit anything OTA when the configured fronthaul latency is larger than the maximum supported value as per the corresponding IOT profile.

9.7.4 Testability requirements imposed on O-RU and O-DU

Nominal software runs on the O-RU and O-DU.

9.7.5 Test Methodology

9.7.5.1 Initial Conditions

1. The fronthaul latency should be set to a value that is larger than the O-DU's category specifies

9.7.5.2 Procedure: Nominal test

-

9.7.6 Test Requirement (expected result)

Observe that the no cell is being transmitted by the O-RU over the air interface.

Annex A (normative): Profiles used for Interoperability Testing

A.1 General

This Annex contains the IOT Profiles and the IOT Profile Test Configurations (PTC) within each IOT Profile.

Sometimes the value of existing IOT Profile Test Configuration parameter will change from one version of the specification to the next. Therefore, it is necessary to refer to both the IOT Profile Test Configuration name and the particular version of the WG4 IOT specification to ensure that the appropriate set of profile parameters are referenced.

For IOT testing, the Profile Test Configuration serves as entry criteria for O-DU and O-RU. The terminology used in the IOT profiles is listed in Table A.1-1.

Table A.1-1: Meaning of terminology used in the IOT profiles.

Terminology	O-DU entry criterion	O-RU entry criterion	Additional information
REQ	Capability is supported	Capability is supported	Capability is expected to be used in the IOT test but actual usage is not guaranteed
O-RU REQ	Capability may be supported (not a test entry criterion)	Capability is supported	The capability is supported by the O-RU and thus depending on O-DU support, O-DU may use the capability. In the IOT testing, the capability need not be invoked. This allows many O-DU designs to match an O-RU capabilities
NOT REQ	Capability may be supported (not a test entry criterion)	Capability may be supported (not a test entry criterion)	There is no support requirement on O-DU and O-RU and O-DU is allowed to use the capability in IOT testing if O-RU supports it. This enables a wide variety of equipment to be tested without increasing the number of PTCs
N/A			The capabilities that are not directly relevant to a test configuration or are not expected to be invoked during an IOT test are categorized under N/A

The intent of an IOT Profile Test Configuration is to characterize the parameters for an interoperability test such that, for a given DUT, the same result will be obtained irrespective of where the test is conducted and who conducts the test; future releases of the present document may amend an IOT Profile Test Configuration description as required to achieve this intent. In particular, test definitions including parameter lists as defined by O-RAN shall be complete, and the test creator shall have no discretion when creating the test but shall follow the test definitions in the present document exactly. However, in cases where no IOT Profile Test Configuration matches the needs of the test creator, a Customized IOT Configuration may be used; see clause 5.1.8 for more information on the use of Customized IOT Configurations.

The IOT Profiles refer to the specific versions of the CUS-Plane and M-Plane specifications referenced in clause 2.1 of the present document.

In some cases deployment/implementation specific constraints make it desirable to customize the parameters in the IOT profiles in this specification. M-Plane-parameters are excluded from customization. To generate a customized profile, an existing CUS IOT profile is chosen and Table A.1-2 provides a list of CUS-Plane IOT Profile parameters that may be modified for the Custom IOT Configuration.

Table A.1-2: Modifiable and Forbidden CUS-Plane Parameters

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	Forbidden
	TDD configuration	-	Modifiable
	Nominal sub-carrier spacing	-	Modifiable
	SSB sub-carrier spacing	-	Modifiable
	Nominal FFT size	-	Modifiable
	Total channel bandwidth	-	Modifiable
	Number of spatial/antenna streams	-	Modifiable
	Fronthaul Ethernet link	-	Modifiable
	PRACH preamble format	-	Modifiable
	O-RU category	4.2.1	Forbidden
	LAA	-	Forbidden
Delay management	Network delay determination	4.4.4.2	Forbidden
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	Forbidden
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Forbidden
	T1a_max_up	4.4, Annex B	Forbidden
	T1a_min_up	4.4, Annex B	Forbidden
	T2a_max_up	4.4, Annex B	Forbidden

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	T2a_min_up	4.4, Annex B	Forbidden
	Tcp_adv_dl	4.3.2, Annex B	Forbidden
	Ta3_max_up	4.4, Annex B	Forbidden
	Ta3_min_up	4.4, Annex B	Forbidden
	Ta4_max_up	4.4, Annex B	Forbidden
	Ta4_min_up	4.4, Annex B	Forbidden
	Ta3_max_up (scs=1.25kHz)	4.4, Annex B	Forbidden
	Ta3_min_up (scs=1.25kHz)	4.4, Annex B	Forbidden
	Ta4_max_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	Forbidden
	Ta4_min_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	Forbidden
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Forbidden
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	Forbidden
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	Forbidden
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Forbidden
	T12_max	4.4, Annex B	Forbidden
	T12_min	4.4, Annex B	Forbidden
	T34_max	4.4, Annex B	Forbidden
	T34_min	4.4, Annex B	Forbidden
	Non-delay managed U-Plane traffic	4.4.7	Forbidden
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	Modifiable

Category	Item	Related O-RAN CUS-Plane specification section(s)	
C/U-Plane Transport	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	Modifiable
	eAxC ID CC_ID bitwidth	5.1.3.2.7	Modifiable
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	Modifiable
	All other C/U-plane transport items as defined in baseline IOT profile		All other C/U-plane transport parameters shall align with base IOT profile selected (ie Forbidden)
Digital Power Scaling	UL gain_correction	8.1.3.2	Forbidden
Beamforming	All items as defined in baseline IOT profile		All Beamforming parameters shall align with base IOT profile selected (ie Forbidden)
IQ compression	All items as defined in baseline IOT profile		All IQ compression parameters shall align with base IOT profile selected (ie Forbidden)
C-Plane	All items as defined in baseline IOT profile		All C-plane parameters shall align with base IOT profile selected (ie Forbidden)
S-Plane	All items as defined in baseline IOT profile		All S-plane parameters shall align with base IOT profile selected (ie Forbidden)

A.2 M-Plane IOT Profile

A.2.1 M-Plane IOT Profile 1 Hierarchical-sudo

Table A.2.1-1: Hierarchical-sudo

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
High Level Description	Architectural models	5.1.2 M-Plane architecture model	Hierarchical model
	IP version	5.1.3 Transport network	IPv4
	Hash algorithm for data integrity	5.4 Security	HMAC-SHA2-256

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
	Cyphering algorithm	5.4 Security	AES128-CTR
“Start up” installation	O-RU identification by DHCP option	6.2.2 O-RU identification in DHCP	DHCPv4(Option: 60)
	VLAN Discovery	6.2.3 Management plane VLAN discovery aspects	support VLAN SCAN
	IP address assignment	6.2.4 O-RU management plane IP address assignment	IPv4 configuration using DHCPv4
	O-RU controller discovery	6.2.5 O-RU controller discovery	DHCPv4 option 43
	DHCP format of O-RU controller discovery	6.2.5 O-RU controller discovery	O-RU Controller IP Address
	NETCONF Call Home	6.3 NETCONF call home to O-RU controller(s)	call home(port4334)
	SSH/TLS Connection Establishment	6.4 NETCONF connection establishment	SSH: password-based authentication
	TCP port for SSH/TLS establishment (for test purpose)	6.4.1 NETCONF security YANG (4) o-ran-mplane-int.yang	SSH: Default (port 830)
	NETCONF Authentication	6.4.2 NETCONF authentication	password-based authentication
	User Account Provisioning	6.4.3 User account provisioning	default sudo
	sudo	6.5 NETCONF access control	used
	nms	6.5 NETCONF access control	not used
	fm-pm	6.5 NETCONF access control	not used
	swm	6.5 NETCONF access control	not used
	NETCONF capability	6.6 NETCONF capability discovery	yang-library, Writable-running Capability, XPath capability, Notifications, Interleave capability
	Watchdog timer	6.7 Monitoring NETCONF connectivity	used
O-RU to O-DU Interface Management	VLAN tagging for C/U/M-Plane	6.2.3 Management plane VLAN discovery aspects 7.3 C/U-Plane VLAN configuration	used for C/U/M-Plane

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
	C/U-Plane IP Address Assignment	7.4 O-RU C/U-Plane IP address assignment	not used
	Definition of processing elements	7.5 Definition of processing elements	a combination of VLAN identity and MAC address
	C/U-Plane Transport Connectivity	7.6 O-DU verification of C/U-Plane transport connectivity	Loop-back Protocol (LBM/LBR)
	O-RU Monitoring of C/U-Plane Connectivity	7.10 O-RU Monitoring of C/U-Plane connectivity	not used
Configuration Management	Baseline configuration	9.1 Baseline configuration	1 phase
Fault Management	subscribe notification	11.3 Manage alarm subscription to NETCONF clients	default stream
Synchronization Aspects	Sync Capability Object	13.3 Sync capability object	CLASS_B
Details of O-RU Operations	Activation, deactivation and sleep	15.3.2 Activation, deactivation and sleep	used

A.2.2 M-Plane IOT Profile 2 Hybrid-sudo+nms

Profile Test Configurations:

- Hybrid-sudo+nms-[DHCP]
- Hybrid-sudo+nms-[o-ran-mplane-int.yang]

Table A.2.2-1: Hybrid-sudo+nms

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
High Level Description	Architectural models	5.1.2 M-Plane architecture model	Hybrid model
	IP version	5.1.3 Transport network	IPv4

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
	Hash algorithm for data integrity	5.4 Security	HMAC-SHA2-256
	Cyphering algorithm	5.4 Security	AES128-CTR
“Start up” installation	O-RU identification by DHCP option	6.2.2 O-RU identification in DHCP	DHCPv4(Option: 60)
	VLAN Discovery	6.2.3 Management plane VLAN discovery aspects	support VLAN SCAN
	IP address assignment	6.2.4 O-RU management plane IP address assignment	IPv4 configuration using DHCPv4
	O-RU controller discovery	6.2.5 O-RU controller discovery	Entry 1) DHCPv4 option 43 Entry 2) Configured by o-ran-mplane-int.yang
	DHCP format of O-RU controller discovery	6.2.5 O-RU controller discovery	O-RU Controller IP Address
	NETCONF Call Home	6.3 NETCONF call home to O-RU controller(s)	call home(port4334)
	SSH/TLS Connection Establishment	6.4 NETCONF connection establishment	SSH: password-based authentication
	TCP port for SSH/TLS establishment (for test purpose)	6.4.1 NETCONF security YANG (4) o-ran-mplane-int.yang	SSH: Default (port 830)
	NETCONF Authentication	6.4.2 NETCONF authentication	password-based authentication
	User Account Provisioning	6.4.3 User account provisioning	default sudo
	sudo	6.5 NETCONF access control	used
	nms	6.5 NETCONF access control	used
	fm-pm	6.5 NETCONF access control	not used
	swm	6.5 NETCONF access control	not used
	NETCONF capability	6.6 NETCONF capability discovery	yang-library, Writable-running Capability, XPath capability, Notifications, Interleave capability
	Watchdog timer	6.7 Monitoring NETCONF connectivity	used

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
O-RU to O-DU Interface Management	VLAN tagging for C/U/M-Plane	6.2.3 Management plane VLAN discovery aspects 7.3 C/U-Plane VLAN configuration	used for C/U/M-Plane
	C/U-Plane IP Address Assignment	7.4 O-RU C/U-Plane IP address assignment	not used
	Definition of processing elements	7.5 Definition of processing elements	a combination of VLAN identity and MAC address
	C/U-Plane Transport Connectivity	7.6 O-DU verification of C/U-Plane transport connectivity	Loop-back Protocol (LBM/LBR)
	O-RU Monitoring of C/U-Plane Connectivity	7.10 O-RU monitoring of C/U-Plane connectivity	not used
Configuration Management	Baseline configuration	9.1 Baseline configuration	1 phase
Fault Management	subscribe notification	11.3 Manage alarm subscription to NETCONF clients	default stream
Synchronization Aspects	Sync Capability Object	13.3 Sync capability object	CLASS_B
Details of O-RU Operations	Activation, deactivation and sleep	15.3.2 Activation, deactivation and sleep	not used

A.2.3 M-Plane IOT Profile 3 Hierarchical-sudo-IPv6

Profile Test Configurations:

- Hierarchical-sudo-IPv6_[SSH-IPaddStateFull-SyncENHANCED]
- Hierarchical-sudo-IPv6_[SSH- IPaddStateless-SyncENHANCED]
- Hierarchical-sudo-IPv6_[TLS- IPaddStateFull-SyncCLASS_B]
- Hierarchical-sudo-IPv6_[TLS- IPaddStateless-SyncCLASS_B]

Table A.2.3-1: Hierarchical-sudo-IPv6

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
High Level Description	Architectural models	5.1.2 M-Plane architecture model	Hierarchical model
	IP version	5.1.3 Transport network	IPv6
	Hash algorithm for data integrity	5.4 Security	Entry 1) SSHv2 -HMAC-SHA2-256 Entry 2) TLS1.2 - SHA256
	Cyphering algorithm	5.4 Security	Entry 1) SSHv2 - AES128-CTR Entry 2) TLS1.2 -AES128-GCM
“Start up” installation	O-RU identification by DHCP option	6.2.2 O-RU identification in DHCP	DHCPv6 (Option: 16)
	VLAN Discovery	6.2.3 Management plane VLAN discovery aspects	support VLAN SCAN
	IP address assignment	6.2.4 O-RU management plane IP address assignment	Entry 1) State-full address configuration Entry 2) Stateless Address Auto-Configuration (SLAAC)
	O-RU controller discovery	6.2.5 O-RU controller discovery	DHCPv6 (Option: 17)
	DHCP format of O-RU controller discovery	6.2.5 O-RU controller discovery	O-RU Controller IP Address
	NETCONF Call Home	6.3 NETCONF call home to O-RU controller(s)	Entry 1) SSH - call home port:4334 Entry 2) TLS - call home port 4335
	SSH/TLS Connection Establishment	6.4 NETCONF connection establishment	Entry 1) SSH - password-based authentication Entry 2) TLS - X.509 Certificate
	TCP port for SSH/TLS establishment (for test purpose)	6.4.1 NETCONF security YANG (4) o-ran-mplane-int.yang	Entry 1) SSH Default (port 830) Entry 2) TLS Default (port 6513)
	NETCONF Authentication	6.4.2 NETCONF authentication	Entry 1) SSH password-based authentication Entry 2) TLS - X.509 Certificate
	User Account Provisioning	6.4.3 User account provisioning	default sudo
	sudo	6.5 NETCONF access control	used
	nms	6.5 NETCONF access control	not used
	fm-pm	6.5 NETCONF access control	not used
	swm	6.5 NETCONF access control	not used

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
	NETCONF capability	6.6 NETCONF capability discovery	yang-library, Writable-running Capability, rollback on error, XPATH capability, Notifications, Interleave capability
	Watchdog timer	6.7 Monitoring NETCONF connectivity	used
O-RU to O-DU Interface Management	VLAN tagging for C/U/M-Plane	6.2.3 Management plane VLAN discovery aspects 7.3 C/U-Plane VLAN configuration	used for C/U/M-Plane
	C/U-Plane IP Address Assignment	7.4 O-RU C/U-Plane IP address assignment	not used
	Definition of processing elements	7.5 Definition of processing elements	a combination of VLAN identity and MAC address
	C/U-Plane Transport Connectivity	7.6 O-DU verification of C/U-Plane transport connectivity	Loop-back Protocol (LBM/LBR)
	O-RU Monitoring of C/U-Plane Connectivity	7.10 O-RU monitoring of C/U-Plane connectivity	not used
Configuration Management	Baseline configuration	9.1 Baseline configuration	1 phase
Fault Management	subscribe notification	11.3 Manage alarm subscription to NETCONF clients	default stream
Synchronization Aspects	Sync Capability Object	13.3 Sync capability object	Entry 1: CLASS_B Entry 2: ENHANCED
Details of O-RU Operations	Activation, deactivation and sleep	15.3.2 Activation, deactivation and sleep	used

A.2.4 M-Plane IOT Profile 4 Hybrid-sudo+nms-IPv6

Profile Test Configurations:

- Hybrid-sudo+nms-IPv6_[SSH-IPaddStateFull-SyncENHANCED-DHCP]

- Hybrid-sudo+nms-IPv6_[SSH- IPaddStateless-SyncENHANCED-DHCP]
- Hybrid-sudo+nms-IPv6_[TLS- IPaddStateFull-SyncCLASS_B-DHCP]
- Hybrid-sudo+nms-IPv6_[TLS- IPaddStateless-SyncCLASS_B-DHCP]
- Hybrid-sudo+nms-IPv6_[TLS- IPaddStateFull-SyncCLASS_B-int.yang]

Table A.2.4-1: Hybrid-sudo+nms-IPv6

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
High Level Description	Architectural models	5.1.2 M-Plane architecture model	Hybrid model
	IP version	5.1.3 Transport network	IPv6
	Hash algorithm for data integrity	5.4 Security	Entry 1) SSHv2 -HMAC-SHA2-256 Entry 2) TLS1.2 - SHA256
	Cyphering algorithm	5.4 Security	Entry 1) SSHv2 - AES128-CTR Entry 2) TLS1.2 -AES128-GCM
“Start up” installation	O-RU identification by DHCP option	6.2.2 O-RU identification in DHCP	DHCPv6 (Option: 16)
	VLAN Discovery	6.2.3 Management plane VLAN discovery aspects	support VLAN SCAN
	IP address assignment	6.2.4 O-RU Management plane IP address assignment	Entry 1) State-full address configuration Entry 2) Stateless Address Auto-Configuration (SLAAC)
	O-RU controller discovery	6.2.5 O-RU controller discovery	Entry 1) DHCPv6 (Option: 17) Entry 2) Configured by o-ran-mplane-int.yang
	DHCP format of O-RU controller discovery	6.2.5 O-RU controller discovery	O-RU Controller IP Address
	NETCONF Call Home	6.3 NETCONF call home to O-RU controller(s)	Entry 1) SSH - call home port:4334 Entry 2) TLS - call home port 4335
	SSH/TLS Connection Establishment	6.4 NETCONF connection establishment	Entry 1) SSH - password-based authentication Entry 2) TLS - X.509 Certificate
	TCP port for SSH/TLS establishment (for test purpose)	6.4.1 NETCONF security YANG (4) o-ran-mplane-int.yang	Entry 1) SSH Default (port 830) Entry 2) TLS Default (port 6513)

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
	NETCONF Authentication	6.4.2 NETCONF authentication	Entry 1) SSH password-based authentication Entry 2) TLS - X.509 Certificate
	User Account Provisioning	6.4.3 User account provisioning	default sudo
	sudo	6.5 NETCONF access control	used
	nms	6.5 NETCONF access control	used
	fm-pm	6.5 NETCONF access control	not used
	swm	6.5 NETCONF access control	not used
	NETCONF capability	6.6 NETCONF capability discovery	yang-library, Writable-running Capability, rollback on error, XPATH capability, Notifications, Interleave capability
	Watchdog timer	6.7 Monitoring NETCONF connectivity	used
O-RU to O-DU Interface Management	VLAN tagging for C/U/M-Plane	6.2.3 Management plane VLAN discovery aspects 7.3 C/U-Plane VLAN configuration	used for C/U/M-Plane
	C/U-Plane IP Address Assignment	7.4 O-RU C/U-Plane IP address assignment	not used
	Definition of processing elements	7.5 Definition of processing elements	a combination of VLAN identity and MAC address
	C/U-Plane Transport Connectivity	7.6 O-DU verification of C/U-Plane transport connectivity	Loop-back Protocol (LBM/LBR)
	O-RU Monitoring of C/U-Plane Connectivity	7.10 O-RU monitoring of C/U-Plane connectivity	not used
Configuration Management	Baseline configuration	9.1 Baseline configuration	1 phase
Fault Management	subscribe notification	11.3 Manage alarm subscription to NETCONF clients	default stream
Synchronization Aspects	Sync Capability Object	13.3 Sync capability object	Entry 1: CLASS_B Entry 2: ENHANCED

Category	Item	Related O-RAN M-Plane/ YANG specification section(s)	
Details of O-RU Operations	Activation, deactivation and sleep	15.3.2 Activation, deactivation and sleep	not used

A.3 CUS-Plane IOT Profiles

A.3.1 NR TDD

A.3.1.1 NR TDD IOT Profile 1 - NR-TDD-FR1-CAT-A-NoBF

Profile Test Configurations:

- NR-TDD-FR1-CAT-A-NoBF_[ConfigDDDSUUDDDD-4SS-14bitIQ-25Gbpsx1lane-PRACHB4-eAxCID2644-llsC1C2]
- NR-TDD-FR1-CAT-A-NoBF_[ConfigDDDSUUDDDD-4SS-14bitIQ-25Gbpsx1lane-PRACHF0-eAxCID2644-llsC1C2]
- NR-TDD-FR1-CAT-A-NoBF_[ConfigDDDSUUDSUU-4SS-14bitIQ-25Gbpsx1lane-PRACHB4-eAxCID2644-llsC1C2]
- NR-TDD-FR1-CAT-A-NoBF_[ConfigDDDSUUDSUU-4SS-14bitIQ-25Gbpsx1lane-PRACHF0-eAxCID2644-llsC1C2]
- NR-TDD-FR1-CAT-A-NoBF_[ConfigDDDSUUDDDD-4SS-14bitIQ-10Gbpsx2lane-PRACHC2-eAxCID4246-llsC1C2]
- NR-TDD-FR1-CAT-A-NoBF_[ConfigDDDDDDDSUU-2SS-8bitIQ-10Gbpsx1lane-PRACHF0-eAxCID2644-llsC1C2]
- NR-TDD-FR1-CAT-A-NoBF_[ConfigDDDDDDDSUU-4SS-8bitIQ-10Gbpsx1lane-PRACHB4-eAxCID2644-llsC1C2]
- NR-TDD-FR1-CAT-A-NoBF_[ConfigDDDSUDDDSU-4SS-9bitIQ-10Gbpsx1lane-PRACHC2-eAxCID2644-llsC3]

Table A.3.1.1-1: NR TDD IOT Profile 1 - NR-TDD-FR1-CAT-A-NoBF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	TDD configuration	-	Entry1: pattern1 { dl-UL-TransmissionPeriodicity ms3 nrofDownlinkSlots 3 nrofDownlinkSymbols 6 nrofUplinkSlots 2 nrofUplinkSymbols 4} pattern2 { dl-UL-TransmissionPeriodicity ms2 nrofDownlinkSlots 4 nrofDownlinkSymbols 0 nrofUplinkSlots 0 nrofUplinkSymbols 0} Entry2: dl-UL-TransmissionPeriodicity ms5 nrofDownlinkSlots 7 nrofDownlinkSymbols 6 nrofUplinkSlots 2 nrofUplinkSymbols 4 Entry3: dl-UL-TransmissionPeriodicity ms2p5 nrofDownlinkSlots 3 nrofDownlinkSymbols 10 nrofUplinkSlots 1 nrofUplinkSymbols 2 Entry4: pattern1 { dl-UL-TransmissionPeriodicity ms3 nrofDownlinkSlots 3 nrofDownlinkSymbols 6 nrofUplinkSlots 2 nrofUplinkSymbols 4}

Category	Item	Related O-RAN CUS-Plane specification section(s)	
			pattern2{ dl-UL-TransmissionPeriodicity ms2 nrofDownlinkSlots 1 nrofDownlinkSymbols 10 nrofUplinkSlots 2 nrofUplinkSymbols 0}
	Nominal sub-carrier spacing	-	30 kHz
	SSB sub-carrier spacing	-	30 kHz
	Nominal FFT size	-	4096
	Total channel bandwidth	-	100MHz x 1CC
	Number of spatial/antenna streams	-	Entry1: 2 Entry2: 4
	Fronthaul Ethernet link	-	Entry1: 25Gbps x 1lane Entry2: 10Gbps x 2lane Entry3: 10Gbps x 1lane
	PRACH preamble format	-	Entry1: B4 Entry2: C2 Entry3: Long preamble F0
	RU category	4.2.1	Category A
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 345us
	T1a_min_up	4.4, Annex B	More than or equal to 294us
	T2a_max_up	4.4, Annex B	More than or equal to 345us
	T2a_min_up	4.4, Annex B	Less than or equal to 134us
	Tcp_adv_dl	4.3.2, Annex B	125 us
	Ta3_max_up	4.4, Annex B	Less than or equal to 171us
	Ta3_min_up	4.4, Annex B	More than or equal to 50us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Ta4_max_up	4.4, Annex B	More than or equal to 331us
	Ta4_min_up	4.4, Annex B	Less than or equal to 50us
	Ta3_max_up (scs=1.25kHz)	4.4, Annex B	Less than or equal to 1650us
	Ta3_min_up (scs=1.25kHz)	4.4, Annex B	More than or equal to 827us
	Ta4_max_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	More than or equal to 1810us
	Ta4_min_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	Less than or equal to 827us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 336us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 336us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	Entry1: 2 Entry2: 4
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	Entry1: 6 Entry2: 2
	eAxC ID CC_ID bitwidth	5.1.3.2.7	4
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	Entry1: 4 Entry2: 6
	Fragmentation	5.5	Application layer fragmentation

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0dB
Beamforming	RU beamforming type	4.2.1..(10.1)	No beamforming
	Beamforming control method	7.3, Annex J	Beamforming ID based (always "0")
IQ compression	U-Plane data compression method	8, Annex A	Block floating point
	U-Plane data IQ bitwidth	8, Annex D	Entry1: 8 Entry2: 14 Entry3: 9
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C1 (can also apply lls-C2) Entry2: lls-C3

A.3.1.2 NR TDD IOT Profile 2 - NR-TDD-FR1-CAT-A-DBF

Profile Test Configurations:

- NR-TDD-FR1-CAT-A-DBF_[ConfigDDDSUUDDDD-100MHz-PRACHB4-llsC1C2]
- NR-TDD-FR1-CAT-A-DBF_[ConfigDDDSUUDDDD-100MHz- PRACHF0-llsC1C2]
- NR-TDD-FR1-CAT-A-DBF_[ConfigDDDSUUDSUU-100MHz-PRACHB4-llsC1C2]
- NR-TDD-FR1-CAT-A-DBF_[ConfigDDDSUUDSUU-100MHz- PRACHF0-llsC1C2]
- NR-TDD-FR1-CAT-A-DBF_[ConfigDDDSUUDDDD- 80MHz-PRACHB4-llsC3]

Table A.3.1.2-1: NR TDD IOT Profile 2 - NR-TDD-FR1-CAT-A-DBF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	Entry1: pattern1 { dl-UL-TransmissionPeriodicity ms3 nrofDownlinkSlots 3 nrofDownlinkSymbols 6 nrofUplinkSlots 2 nrofUplinkSymbols 4}

Category	Item	Related O-RAN CUS-Plane specification section(s)	
			pattern2 { dl-UL-TransmissionPeriodicity ms2 nrofDownlinkSlots 4 nrofDownlinkSymbols 0 nrofUplinkSlots 0 nrofUplinkSymbols 0} Entry2: pattern1 { dl-UL-TransmissionPeriodicity ms3 nrofDownlinkSlots 3 nrofDownlinkSymbols 6 nrofUplinkSlots 2 nrofUplinkSymbols 4} pattern2 { dl-UL-TransmissionPeriodicity ms2 nrofDownlinkSlots 1 nrofDownlinkSymbols 10 nrofUplinkSlots 2 nrofUplinkSymbols 0}
	Nominal sub-carrier spacing	-	30 kHz
	SSB sub-carrier spacing	-	30 kHz
	Nominal FFT size	-	4096
	Total channel bandwidth	-	Entry1: 100MHz x 1CC Entry2: 80MHz x 1CC
	Number of spatial/antenna streams	-	4
	Fronthaul Ethernet link	-	25Gbps x 1lane
	PRACH preamble format	-	Entry1: B4 Entry2: Long preamble F0
	RU category	4.2.1	Category A
	LAA	-	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 345us
	T1a_min_up	4.4, Annex B	More than or equal to 294us
	T2a_max_up	4.4, Annex B	More than or equal to 345us
	T2a_min_up	4.4, Annex B	Less than or equal to 134us
	Tcp_adv_dl	4.3.2, Annex B	125 us
	Ta3_max_up	4.4, Annex B	Less than or equal to 171us
	Ta3_min_up	4.4, Annex B	More than or equal to 50us
	Ta4_max_up	4.4, Annex B	More than or equal to 331us
	Ta4_min_up	4.4, Annex B	Less than or equal to 50us
	Ta3_max_up (scs=1.25kHz)	4.4, Annex B	Less than or equal to 1650us
	Ta3_min_up (scs=1.25kHz)	4.4, Annex B	More than or equal to 827us
	Ta4_max_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	More than or equal to 1810us
	Ta4_min_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	Less than or equal to 827us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 336us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 336us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE
	Transport encapsulation	5.1.1-5.1.2	Ethernet

Category	Item	Related O-RAN CUS-Plane specification section(s)	
C/U-Plane transport	Jumbo frames	5.1.2	FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	6
	eAxC ID CC_ID bitwidth	5.1.3.2.7	4
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	4
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0dB
Beamforming	RU beamforming type	4.2.1..(10.1)	Digital beamforming
	Beamforming control method	7.3, Annex J	Beamforming ID based
IQ compression	U-Plane data compression method	8, Annex A	Block floating point
	U-Plane data IQ bitwidth	8, Annex D	14
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	FALSE
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C1 (can also apply lls-C2) Entry2: lls-C3

A.3.1.3 NR TDD IOT Profile 3 - NR-TDD-FR2-CAT-A-ABF

Profile Test Configurations:

- NR-TDD-FR2-CAT-A-ABF_[100MHz-SSB240kHz&PRACHC0-eAxCID2644-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[200MHz-SSB240kHz&PRACHC0-eAxCID2644-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[300MHz-SSB240kHz&PRACHC0-eAxCID2644-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[400MHz-SSB240kHz&PRACHC0-eAxCID2644-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[100MHz-SSB120kHz&PRACHA3-eAxCID2644-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[200MHz-SSB120kHz&PRACHA3-eAxCID2644-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[300MHz-SSB120kHz&PRACHA3-eAxCID2644-llsC1C2]

- NR-TDD-FR2-CAT-A-ABF_[400MHz-SSB120kHz&PRACHA3-eAxCID2644-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[800MHz-SSB120kHz&PRACHA3-eAxCID2644-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[100MHz-SSB240kHz&PRACHC2-eAxCID4246-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[200MHz-SSB240kHz&PRACHC2-eAxCID4246-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[400MHz-SSB240kHz&PRACHC2-eAxCID4246-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[600MHz-SSB240kHz&PRACHC2-eAxCID4246-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[800MHz-SSB240kHz&PRACHC2-eAxCID4246-llsC1C2]
- NR-TDD-FR2-CAT-A-ABF_[200MHz-SSB120kHz&PRACHA3-eAxCID2644-llsC3]
- NR-TDD-FR2-CAT-A-ABF_[400MHz-SSB120kHz&PRACHA3-eAxCID2644-llsC3]
- NR-TDD-FR2-CAT-A-ABF_[600MHz-SSB120kHz&PRACHA3-eAxCID2644-llsC3]
- NR-TDD-FR2-CAT-A-ABF_[800MHz-SSB120kHz&PRACHA3-eAxCID2644-llsC3]

Table A.3.1.3-1: NR TDD IOT Profile 3 - NR-TDD-FR2-CAT-A-ABF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	dl-UL-TransmissionPeriodicity: ms0p625 nrofDownlinkSlots: 3 nrofDownlinkSymbols: 10 nrofUplinkSlots: 1 nrofUplinkSymbols: 2
	Nominal sub-carrier spacing	-	120 kHz
	SSB sub-carrier spacing	-	Entry1: 240kHz Entry2: 120KHz

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Nominal FFT size	-	1024
	Total channel bandwidth	-	Entry1: 100MHz x 1CC Entry2: 100MHz x 2CC Entry3: 100MHz x 3CC Entry4: 100MHz x 4CC Entry5: 100MHz x 6CC Entry6: 100MHz x 8CC
	Number of spatial/antenna streams	-	2
	Fronthaul Ethernet link	-	25Gbps x 1lane for 100MHz x 1,2,3,4CC, 25Gbps x 2lane for 100MHz x 6,8CC
	PRACH preamble format	-	Entry1: C0 Entry2: A3 Entry3: C2
	RU category	4.2.1	Category A
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 264us
	T1a_min_up	4.4, Annex B	More than or equal to 213us
	T2a_max_up	4.4, Annex B	More than or equal to 264us
	T2a_min_up	4.4, Annex B	Less than or equal to 53us
	Tcp_adv_dl	4.3.2, Annex B	63 us
	Ta3_max_up	4.4, Annex B	Less than or equal to 90us
	Ta3_min_up	4.4, Annex B	More than or equal to 20us
	Ta4_max_up	4.4, Annex B	More than or equal to 250us
	Ta4_min_up	4.4, Annex B	Less than or equal to 20us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 274us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 223us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 274us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 63us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	Entry1: 2 Entry2: 4
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	Entry1: 6 Entry2: 2
	eAxC ID CC_ID bitwidth	5.1.3.2.7	4
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	Entry1: 4 Entry2: 6
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0dB

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Beamforming	RU beamforming type	4.2.1..(10.1)	Analog beamforming
	Beamforming control method	7.3, Annex J	Beamforming ID based
IQ compression	U-Plane data compression method	8, Annex A	Block floating point
	U-Plane data IQ bitwidth	8, Annex D	14
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	FALSE
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	7.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C1 (can also apply lls-C2) Entry2: lls-C3

A.3.1.4 NR TDD IOT M-MIMO Profile 1 - NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP

Profile Test Configurations:

- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFUDDDD-100MHz1CC-8SS-25Gbpsx2lane-PRACHF0-noJumbo-BFW.12bitFP-lsC1-SE1_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFUDDDD-100MHz1CC-8SS-25Gbpsx2lane-PRACHC2-Jumbo-BFW.12bitFP-lsC1noPLFS-SE11_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFUDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-noJumbo-BFW.9bitBFP-lsC1noPLFS-SE11_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFUDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHB4-noJumbo-BFW.12bitFP-lsC3-SE1_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFU-100MHz1CC-8SS-25Gbpsx2lane-PRACHF0-noJumbo-BFW.12bitFP-lsC1noPLFS-SE1]_DM1
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFU-100MHz1CC-8SS-25Gbpsx2lane-PRACHB4-noJumbo-BFW.9bitBFP-lsC1-SE11]_DM1
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFU-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-noJumbo-BFW.12bitFP-lsC1-SE1_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFU-100MHz1CC-16SS-25Gbpsx2lane-PRACHC2-Jumbo-BFW.12bitFP-lsC3-SE11_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFUDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-noJumbo-BFW.9bitBFP-lsC1noPLFS-SE11_DM2]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP_[ConfigDDDFU-100MHz1CC-8SS-25Gbpsx2lane-PRACHB4-noJumbo-BFW.9bitBFP-lsC1-SE11_DM2]

Table A.3.1.4-1: NR TDD IOT M-MIMO Profile 1 - NR-TDD-FR1-CAT-B-mMIMO-RTWeights-BFP

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	Entry1: DDDFUDDDD Entry2: DDDFU
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size	-	4096(100MHz_BW)

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Total channel bandwidth	-	100MHz x 1 CC
	Number of spatial/antenna streams	-	Entry1: 8L (DL), 8 streams (UL) Entry2: 16L (DL), 16 streams (UL)
	Fronthaul Ethernet link	-	25Gbps x 2 lane (16L, 1CC)
	PRACH preamble format	-	Entry1: Long preamble F0 Entry2: B4 Entry3: C2
	RU category	4.2.1	Category B
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to Entry1: 345us Entry2: 475us (Note 2)
	T1a_min_up	4.4, Annex B	More than or equal to Entry1: 294us Entry2: 400us
	T1a_max_cp_dl	4.4, Annex B	Less than or equal to Entry1: 669us Entry2: 971us
	T1a_min_cp_dl	4.4, Annex B	More than or equal t Entry1: 419us Entry2: 550us
	T2a_max_up	4.4, Annex B	More than or equal to Entry1: 345us Entry2: 475us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	T2a_min_up	4.4, Annex B	Less than or equal to Entry1: 134us Entry2: 240us
	T2a_max_cp_dl	4.3.2, Annex B	More than or equal to Entry1: 669us Entry2: 971us
	T2a_min_cp_dl	4.4, Annex B	Less than or equal to Entry1: 259us Entry2: 390us
	Tcp_adv_dl	4.4, Annex B	Entry1: 125us Entry2: 150us
	Ta3_max_up	4.4, Annex B	Less than or equal to Entry1: 171us Entry2: 280us
	Ta3_min_up	4.4, Annex B	More than or equal to 50us
	Ta4_max_up	4.4, Annex B	More than or equal to Entry1: 331us Entry2: 440us
	Ta4_min_up	4.4, Annex B	Less than or equal to 50us
	Ta3_max_up (scs=1.25kHz)	4.4, Annex B	Less than or equal to 1650us
	Ta3_min_up (scs=1.25kHz)	4.4, Annex B	More than or equal to 827us
	Ta4_max_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	More than or equal to 1810us
	Ta4_min_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	Less than or equal to 827us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to Entry1: 535us Entry2: 840us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to Entry1: 285us Entry2: 419us
	T2a_max_cp_ul	4.4, Annex B	More than or equal to Entry1: 535us Entry2: 840us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	T2a_min_cp_ul	4.4, Annex B	Less than or equal to Entry1: 125us Entry2: 259us
	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
	Non-delay managed U-Plane traffic	4.4.7	Support
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	Entry1: TRUE Entry2: FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	5
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	1
	eAxC ID CC_ID bitwidth	5.1.3.2.7	2
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	8
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority
	Transport prioritization within U-Plane	5.3	TRUE
	Seperation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0 dB
Beamforming	O-RU beamforming type	4.2.1..(10.1)	Digital beamforming

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Beamforming control method	7.3, Annex J	RT Weights (Frequency Domain)
	BFW IQ	7.4, Annex J	Entry 1: Fixed Point, 12-bit Entry 2: BFP, 9-bit
IQ compression	U-Plane data compression method	8, Annex A	BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	9
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to "0")
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	Entry 1: FALSE Entry 2: TRUE (Note 1)
	Section extension 2 (beamforming attributes)	7.7.2	FALSE
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	FALSE
	Section extension 4 (modulation compr. params)	7.7.4	FALSE
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	FALSE
	Section extension 7 (Multiple-eAxC designation)	7.7.7	FALSE
	Section extension 8 (regularization factor)	7.7.8	FALSE
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	FALSE
	Section extension 10 (Multiple ports grouping)	7.7.10	FALSE
	Section extension 11 (Flexible BF weights)	7.7.11	Entry 1: TRUE Entry 2: FALSE (Note 1)
	Section extension 12	7.7.12	FALSE
	Section extension 13	7.7.13	FALSE
	Section extension 14	7.7.14	FALSE
	Section extension 15	7.7.15	FALSE
	Section extension 16	7.7.16	FALSE
	Section extension 17	7.7.17	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry 1) lls-C1 (can also apply lls-C2) Entry 2) lls-C1 (PLFS not required by O-RU) Entry 3) lls-C3

Note 1: Either Section extension 1 or Section extension 11 should be used for IOT (ie, (FALSE, TRUE) or (TRUE, FALSE)).

Note 2: When two entries are present for parameters in section ‘Delay Management’, DM1 refers to Entry1 and DM2 to Entry2. DM1/2 are used as identifiers in list “Profile Test Configurations”.

A.3.1.5 NR TDD IOT M-MIMO Profile 2 - NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp

Profile Test Configurations:

- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz1CC-8SS-25Gbpsx2lane-PRACHF0-noJumbo-12bitFP.BFW-llsC1-SE1-SE4_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz1CC-8SS-25Gbpsx2lane-PRACHC2-Jumbo-12bitFP.BFW-llsC1noPLFS-SE11-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-noJumbo-9bitBFP.BFW-llsC1noPLFS-SE11-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHB4-noJumbo-12bitFP.BFW-llsC3-SE1-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-Jumbo-9bitBFP.BFW-llsC1-SE11-SE4_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz2CC-8SS-25Gbpsx3lane-PRACHF0-noJumbo-12bitFP.BFW-llsC1-SE1-SE4_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz2CC-8SS-25Gbpsx3lane-PRACHC2-Jumbo-12bitFP.BFW-llsC1noPLFS-SE11-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz2CC-16SS-25Gbpsx3lane-PRACHF0-noJumbo-9bitBFP.BFW-llsC1noPLFS-SE11-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz2CC-16SS-25Gbpsx3lane-PRACHB4-noJumbo-12bitFP.BFW-llsC3-SE1-SE4_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz1CC-8SS-25Gbpsx2lane-PRACHF0-noJumbo-12bitFP.BFW-llsC1noPLFS-SE1-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz1CC-8SS-25Gbpsx2lane-PRACHB4-noJumbo-9bitBFP.BFW-llsC1-SE11-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-noJumbo-12bitFP.BFW-llsC1-SE1-SE4_DM1]

- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz1CC-16SS-25Gbpsx2lane-PRACHC2-Jumbo-12bitFP.BFW-llsC3-SE11-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-Jumbo-9bitBFP.BFW-llsC1noPLFS-SE11-SE4_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz2CC-8SS-25Gbpsx3lane-PRACHF0-noJumbo-12bitFP.BFW-llsC1noPLFS-SE1-SE4_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz2CC-8SS-25Gbpsx3lane-PRACHB4-noJumbo-9bitBFP.BFW-llsC1-SE11-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz2CC-16SS-25Gbpsx3lane-PRACHF0-noJumbo-12bitFP.BFW-llsC1-SE1-SE4_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz2CC-16SS-25Gbpsx3lane-PRACHC2-Jumbo-12bitFP.BFW-llsC3-SE11-SE5_DM1]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFUDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-noJumbo-9bitBFP.BFW-llsC1noPLFS-SE11-SE5_DM2]
- NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp_[ConfigDDDFU-100MHz1CC-8SS-25Gbpsx2lane-PRACHB4-noJumbo-9bitBFP.BFW-llsC1-SE11-SE5_DM2]

Table A.3.1.5-1: NR TDD IOT M-MIMO Profile 2 - NR-TDD-FR1-CAT-B-mMIMO-RTWeights-ModComp

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	Entry1: DDDFUDDDD Entry2: DDDFU
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size	-	4096(100MHz_BW)
	Total channel bandwidth	-	Entry1: 100MHz x 1 CC Entry2: 100 MHz x 2 CC
	Number of spatial/antenna streams	-	Entry1: 8L (DL), 8 streams (UL) Entry2: 16L (DL), 16 streams (UL)

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Fronthaul Ethernet link	-	Entry1: 25Gbps x 2 lane (16L, 1CC) Entry2: 25Gbps x 3 lane (16L, 2CC)
	PRACH preamble format	-	Entry1: Long preamble F0 Entry2: B4 Entry3: C2
	O-RU category	4.2.1	Category B
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	O-RU adaptation of delay profile information (based on IIs-CU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to Entry1: 345us Entry2: 475us (Note 3)
	T1a_min_up	4.4, Annex B	More than or equal to Entry1: 294us Entry2: 400us
	T1a_max_cp_dl	4.4, Annex B	Less than or equal to Entry1: 669us Entry2: 971us
	T1a_min_cp_dl	4.4, Annex B	More than or equal to Entry1: 419us Entry2: 550us
	T2a_max_up	4.4, Annex B	More than or equal to Entry1: 345us Entry2: 475us
	T2a_min_up	4.4, Annex B	Less than or equal to Entry1: 134us Entry2: 240us
	T2a_max_cp_dl	4.3.2, Annex B	More than or equal to Entry1: 669us Entry2: 971us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	T2a_min_cp_dl	4.4, Annex B	Less than or equal to Entry1: 259us Entry2: 390us
	Tcp_adv_dl	4.4, Annex B	Entry1: 125us Entry2: 150us
	Ta3_max_up	4.4, Annex B	Less than or equal to Entry1: 171us Entry2: 280us
	Ta3_min_up	4.4, Annex B	More than or equal to 50us
	Ta4_max_up	4.4, Annex B	More than or equal to Entry1: 331us Entry2: 440us
	Ta4_min_up	4.4, Annex B	Less than or equal to 50us
	Ta3_max_up (scs=1.25kHz)	4.4, Annex B	Less than or equal to 1650us
	Ta3_min_up (scs=1.25kHz)	4.4, Annex B	More than or equal to 827us
	Ta4_max_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	More than or equal to 1810us
	Ta4_min_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	Less than or equal to 827us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to Entry1: 535us Entry2: 840us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to Entry1: 285us Entry2: 419us
	T2a_max_cp_ul	4.4, Annex B	More than or equal to Entry1: 535us Entry2: 840us
	T2a_min_cp_ul	4.4, Annex B	Less than or equal to Entry1: 125us Entry2: 259us
	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	T34_min	4.4, Annex B	0us
	Non-delay managed U-Plane traffic	4.4.7	Support
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	Entry1: TRUE Entry2: FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	5
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	1
	eAxC ID CC_ID bitwidth	5.1.3.2.7	2
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	8
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority
	Transport prioritization within U-Plane	5.3	TRUE
	Seperation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0 dB
Beamforming	O-RU beamforming type	4.2.1..(10.1)	Digital beamforming
	Beamforming control method	7.4, Annex J	RT Weights (Frequency Domain)
	BFW IQ	7.4, Annex J	Entry 1: Fixed Point, 12-bit Entry 2: BFP, 9-bit
IQ compression	U-Plane data compression method	8, Annex A	DL: Mod COMP, UL: BFP

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	DL: Mod COMP, 4 bits (max) UL: BFP, 9 bits
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to "0")
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	Entry 1: FALSE Entry 2: TRUE (Note 1)
	Section extension 2 (beamforming attributes)	7.7.2	FALSE
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	FALSE
	Section extension 4 (modulation compr. params)	7.7.4	Entry 1: TRUE Entry 2: FALSE (Note 2)
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	Entry 1: FALSE Entry 2: TRUE (Note 2)
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 7 (Multiple-eAxC designation)	7.7.7	FALSE
	Section extension 8 (regularization factor)	7.7.8	FALSE
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	FALSE
	Section extension 10 (Multiple ports grouping)	7.7.10	FALSE
	Section extension 11 (Flexible BF weights)	7.7.11	Entry 1: TRUE Entry 2: FALSE (Note 1)
	Section extension 12	7.7.12	FALSE
	Section extension 13	7.7.13	FALSE
	Section extension 14	7.7.14	FALSE
	Section extension 15	7.7.15	FALSE
	Section extension 16	7.7.16	FALSE
	Section extension 17	7.7.17	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry 1) lls-C1 (can also apply lls-C2) Entry 2) lls-C1 (PLFS not required by O-RU) Entry 3) lls-C3

Note 1: Either Section extension 1 or Section extension 11 should be used for IOT (ie, (FALSE, TRUE) or (TRUE, FALSE)).

Note 2: Either Section extension 4 or Section extension 5 should be used for IOT (ie, (FALSE, TRUE) or (TRUE, FALSE)).

Note 3: When two entries are present for parameters in section ‘Delay Management’, DM1 refers to Entry1 and DM2 to Entry2. DM1/2 are used as identifiers in list “Profile Test Configurations”

A.3.1.6 NR TDD IOT M-MIMO Profile 3 - NR-TDD-FR1-CAT-B-mMIMO-ChInfo-BFP

Profile Test Configurations:

- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-BFP_[ConfigDDDFUDDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-Jumbo-llsC1noPLFS]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-BFP_[ConfigDDDFUDDDDD-100MHz1CC-8SS-25Gbpsx2lane-PRACHC2-noJumbo-llsC1]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-BFP_[ConfigDDDFUDDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHB4-Jumbo-llsC3]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-BFP_[ConfigDDDFU-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-noJumbo-llsC1]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-BFP_[ConfigDDDFU-100MHz1CC-8SS-25Gbpsx2lane-PRACHB4-noJumbo-llsC1noPLFS]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-BFP_[ConfigDDDFU-100MHz1CC-16SS-25Gbpsx2lane-PRACHC2-Jumbo-llsC3]

Table A.3.1.6-1: NR TDD IOT M-MIMO Profile 3 - NR-TDD-FR1(30kHzSRS)-CAT-B-mMIMO-ChInfo-BFP

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	Entry1: DDDFUDDDDD Entry2: DDDFU
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size	-	4096(100MHz_BW)
	Total channel bandwidth	-	100MHz x 1 CC
	Number of spatial/antenna streams	-	Entry1: 8L (DL), 8 streams (UL) Entry2: 16L (DL), 16 streams (UL)
	Fronthaul Ethernet link	-	25Gbps x 2 lane (16L, 1CC)

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	PRACH preamble format	-	Entry1: Long preamble F0 Entry2: B4 Entry3: C2
	O-RU category	4.2.1	Category B
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	O-RU adaptation of delay profile information (based on lls-CU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 345us
	T1a_min_up	4.4, Annex B	More than or equal to 294us
	T1a_max_cp_dl	4.4, Annex B	Less than or equal to 820us
	T1a_min_cp_dl	4.4, Annex B	More than or equal to 769us
	T2a_max_up	4.4, Annex B	More than or equal to 345us
	T2a_min_up	4.4, Annex B	Less than or equal to 134us
	T2a_max_cp_dl	4.3.2, Annex B	More than or equal to 820us
	T2a_min_cp_dl	4.4, Annex B	Less than or equal to 609us
	Tcp_adv_dl	4.4, Annex B	475us
	Ta3_max_up	4.4, Annex B	Less than or equal to 171us
	Ta3_min_up	4.4, Annex B	More than or equal to 50us
	Ta4_max_up	4.4, Annex B	More than or equal to 331us
	Ta4_min_up	4.4, Annex B	Less than or equal to 50us
	Ta3_max_up (scs=1.25kHz)	4.4, Annex B	Less than or equal to 1650us
	Ta3_min_up (scs=1.25kHz)	4.4, Annex B	More than or equal to 827us
	Ta4_max_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	More than or equal to 1810us
	Ta4_min_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	Less than or equal to 827us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 336us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4, Annex B	More than or equal to 336us
	T2a_min_cp_ul	4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
	Non-delay managed U-Plane traffic	4.4.7	Support
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	Entry1: TRUE Entry2: FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	5
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	1
	eAxC ID CC_ID bitwidth	5.1.3.2.7	2
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	8
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority
	Transport prioritization within U-Plane	5.3	TRUE
	Seperation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Digital Power Scaling	UL gain_correction	8.1.3.2	0 dB
Beamforming	O-RU beamforming type	4.2.1..(10.1)	Digital beamforming
	Beamforming control method	7.4, Annex J	Chan Info (Frequency Domain)
	BFW IQ	7.4, Annex J	NA
IQ compression	U-Plane data compression method	8, Annex A	BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	9
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	TRUE
	Section Type 6	7.4.8	TRUE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to "0")
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	FALSE
	Section extension 2 (beamforming attributes)	7.7.2	FALSE
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	FALSE
	Section extension 4 (modulation compr. params)	7.7.4	FALSE
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	FALSE
	Section extension 7 (Multiple-eAxC designation)	7.7.7	FALSE
	Section extension 8 (regularization factor)	7.7.8	FALSE
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	FALSE
	Section extension 10 (Multiple ports grouping)	7.7.10	TRUE
	Section extension 11 (Flexible BF weights)	7.7.11	FALSE
	Section extension 12	7.7.12	FALSE
	Section extension 13	7.7.13	FALSE
	Section extension 14	7.7.14	FALSE
	Section extension 15	7.7.15	FALSE
	Section extension 16	7.7.16	TRUE
	Section extension 17	7.7.17	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry 1) IIs-C1 (can also apply IIs-C2) Entry 2) IIs-C1 (PLFS not required by O-RU) Entry 3) IIs-C3

A.3.1.7 NR TDD IOT M-MIMO Profile 4 - NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp

Profile Test Configurations:

- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFUDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-Jumbo-IIsC1noPLFS]

- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFUUDDDD-100MHz1CC-8SS-25Gbpsx2lane-PRACHC2-noJumbo-llsC1]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFUUDDDD-100MHz1CC-16SS-25Gbpsx2lane-PRACHB4-Jumbo-llsC3]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFUUDDDD-100MHz2CC-16SS-25Gbpsx3lane-PRACHF0-Jumbo-llsC1noPLFS]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFUUDDDD-100MHz2CC-8SS-25Gbpsx3lane-PRACHC2-noJumbo-llsC1]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFUUDDDD-100MHz2CC-16SS-25Gbpsx3lane-PRACHB4-Jumbo-llsC3]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFU-100MHz1CC-16SS-25Gbpsx2lane-PRACHF0-noJumbo-llsC1]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFU-100MHz1CC-8SS-25Gbpsx2lane-PRACHB4-noJumbo-llsC1noPLFS]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFU-100MHz1CC-16SS-25Gbpsx2lane-PRACHC2-Jumbo-llsC3]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFU-100MHz2CC-16SS-25Gbpsx3lane-PRACHF0-noJumbo-llsC1]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFU-100MHz2CC-8SS-25Gbpsx3lane-PRACHB4-noJumbo-llsC1noPLFS]
- NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp_[ConfigDDDFU-100MHz2CC-16SS-25Gbpsx3lane-PRACHC2-Jumbo-llsC3]

Table A.3.1.7-1: NR TDD IOT M-MIMO Profile 4 - NR-TDD-FR1-CAT-B-mMIMO-ChInfo-ModComp

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	Entry1: DDDFUUDDDD Entry2: DDDFU
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size	-	4096(100MHz_BW)
	Total channel bandwidth	-	Entry1: 100MHz x 1 CC Entry2: 100 MHz x 2 CC
	Number of spatial/antenna streams	-	Entry1: 8L (DL), 8 streams (UL) Entry2: 16L (DL), 16 streams (UL)

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Fronthaul Ethernet link	-	Entry1: 25Gbps x 2 lane (16L, 1CC) Entry2: 25Gbps x 3 lane (16L, 2CC)
	PRACH preamble format	-	Entry1: Long preamble F0 Entry2: B4 Entry3: C2
	O-RU category	4.2.1	Category B
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	O-RU adaptation of delay profile information (based on lls-CU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 345us
	T1a_min_up	4.4, Annex B	More than or equal to 294us
	T1a_max_cp_dl	4.4, Annex B	Less than or equal to 820us
	T1a_min_cp_dl	4.4, Annex B	More than or equal to 769us
	T2a_max_up	4.4, Annex B	More than or equal to 345us
	T2a_min_up	4.4, Annex B	Less than or equal to 134us
	T2a_max_cp_dl	4.3.2, Annex B	More than or equal to 820us
	T2a_min_cp_dl	4.4, Annex B	Less than or equal to 609us
	Tcp_adv_dl	4.4, Annex B	475us
	Ta3_max_up	4.4, Annex B	Less than or equal to 171us
	Ta3_min_up	4.4, Annex B	More than or equal to 50us
	Ta4_max_up	4.4, Annex B	More than or equal to 331us
	Ta4_min_up	4.4, Annex B	Less than or equal to 50us
	Ta3_max_up (scs=1.25kHz)	4.4, Annex B	Less than or equal to 1650us
	Ta3_min_up (scs=1.25kHz)	4.4, Annex B	More than or equal to 827us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Ta4_max_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	More than or equal to 1810us
	Ta4_min_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	Less than or equal to 827us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 336us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4, Annex B	More than or equal to 336us
	T2a_min_cp_ul	4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
	Non-delay managed U-Plane traffic	4.4.7	Support
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	Entry1: TRUE Entry2: FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	5
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	1
	eAxC ID CC_ID bitwidth	5.1.3.2.7	2
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	8
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority
	Transport prioritization within U-Plane	5.3	TRUE
	Seperation of C/U-Plane and M-Plane traffic	5.4	VLAN ID

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0 dB
Beamforming	O-RU beamforming type	4.2.1..(10.1)	Digital beamforming
	Beamforming control method	7.4, Annex J	Chan Info (Frequency Domain)
	BFW IQ	7.4, Annex J	NA
IQ compression	U-Plane data compression method	8, Annex A	DL: Mod COMP, UL: BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	DL: Mod COMP, 4 bits (max) UL: BFP, 9 bits
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	TRUE
	Section Type 6	7.4.8	TRUE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to "0")
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	FALSE
	Section extension 2 (beamforming attributes)	7.7.2	FALSE
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 4 (modulation compr. params)	7.7.4	FALSE
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	TRUE
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	FALSE
	Section extension 7 (Multiple-eAxC designation)	7.7.7	FALSE
	Section extension 8 (regularization factor)	7.7.8	FALSE
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	FALSE
	Section extension 10 (Multiple ports grouping)	7.7.10	TRUE
	Section extension 11 (Flexible BF weights)	7.7.11	FALSE
	Section extension 12	7.7.12	FALSE
	Section extension 13	7.7.13	FALSE
	Section extension 14	7.7.14	FALSE
	Section extension 15	7.7.15	FALSE
	Section extension 16	7.7.16	TRUE
	Section extension 17	7.7.17	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry 1) lls-C1 (can also apply lls-C2) Entry 2) lls-C1 (PLFS not required by O-RU) Entry 3) lls-C3

A.3.1.8 NR TDD IOT M-MIMO Profile 5 - NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Base

Table A.3.1.8-1 describes the base profile. The following PTCs are defined.

- 1) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Base_PTC1_[12bitFP.BFW]
- 2) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Base_PTC2_[8bitBFP.BFW]

Table A.3.1.8-1: NR TDD IOT M-MIMO Profile 5: NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Base

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	DDDSUUDDDD
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size		4096
	Total channel bandwidth	-	100MHz x 1 CC
	Number of user layers	-	DL: 2 – 16 UL: 2 – 8
	Fronthaul Ethernet link	-	25Gbps x 1 lane
	PRACH preamble format	-	O-RU REQ: F0
	O-RU category	4.2.1	Category B
	LAA	7.2.5	N/A
Delay management, general	Network delay determination	4.4.4	Defined Transport Method
	O-RU adaptation of delay profile information (based on IIs-CU delay profile and transport delay)	4.4.4.3	NOT REQ
	O-DU timing advance type	4.4.5-4.4.6, Annex B	REQ: Fixed timing Advance
	SRS transfer	4.4.7	REQ: Non-delay managed U-Plane
	Delay Sets for timing parameters		REQ: Error! Reference source not found.

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Delay management, default	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
Delay management, non default	non-default beamforming methods #1	12.6.3	REQ: DMRS-BF-EQ
	Delay Sets for timing parameters		REQ: Error! Reference source not found.
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	NOT REQ
	Transport header	5.1.3	REQ: eCPRI
	eCPRI concatenation	5.1.3.2	NOT REQ
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 3 and 2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	O-RU REQ: 2 and 3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	O-RU REQ: 3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 8
	Fragmentation	5.5	Application layer fragmentation REQ
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority REQ
	Transport prioritization within U-Plane	5.3	NOT REQ
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID REQ
	Transport-based separation within C/U-Plane traffic	5.4	NOT REQ
Digital Power Scaling	UL_gain_correction (not applicable to DMRS-BF-EQ)	8.1.3.2	0 dB
	equalization-data-scaling	12.6.3.5	REQ: Scaling-function1 or Scaling-function2
Beamforming	O-RU beamforming type	4.2.1	REQ: Digital beamforming
	Beamforming control method	7.2.1, 12.6, Annex J	DL: REQ: WDBF and PDBF UL: REQ: DMRS-BF for PUSCH, WDBF or PDBF for non-PUSCH
	BFW IQ	7.7.1, Annex J	REQ PTC1: Fixed point, 12 bits

Category	Item	Related O-RAN CUS-Plane specification section(s)	
			PTC2: BFP, 8 bits
IQ compression	U-Plane data compression method	8, Annex A	REQ: BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	DL: REQ: 9 bits UL: REQ: 8 and 9 bits (PUSCH), 9 bits for other UL channels
	IQ data frame format not including udCompHdr field	8.3.3.13	REQ
C-Plane	Section Type 0	7.4.2	O-RU REQ
	Section Type 1	7.4.3	REQ
	Section Type 3	7.4.5	REQ (for PRACH)
	Section Type 4	7.4.6	NOT REQ
	Section Type 5	7.4.7	REQ (for DMRS-BF)
	Section Type 6	7.4.8	NOT REQ
	Section Type 7	7.4.9	N/A
	Section Type 8	7.4.10	NOT REQ
	Section Type 9	7.4.11	REQ
	Section Type 10	7.4.12	REQ
	Section Type 11	7.4.13	O-RU REQ
	"symInc" flag	7.5.3.3	NOT REQ
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	REQ: Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	REQ
	Section extension 2 (beamforming attributes)	7.7.2	NOT REQ
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	NOT REQ
	Section extension 4 (modulation compr. params)	7.7.4	NOT REQ
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	NOT REQ
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 7 (Multiple-eAxC designation)	7.7.7	NOT REQ
	Section extension 8 (regularization factor)	7.7.8	NOT REQ
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	NOT REQ
	Section extension 10 (Multiple ports grouping)	7.7.10	REQ (for DMRS-BF)
	Section extension 11 (Flexible BF weights)	7.7.11	NOT REQ
	Section extension 12	7.7.12	NOT REQ
	Section extension 13	7.7.13	NOT REQ
	Section extension 14	7.7.14	NOT REQ
	Section extension 15	7.7.15	NOT REQ
	Section extension 16	7.7.16	NOT REQ
	Section extension 17	7.7.17	NOT REQ
	Section extension 18	7.7.18	NOT REQ
	Section extension 19	7.7.19	NOT REQ
	Section extension 20	7.7.20	NOT REQ
	Section extension 21	7.7.21	NOT REQ
	Section extension 22	7.7.22	NOT REQ
	Section extension 23	7.7.23	NOT REQ
	Section extension 24	7.7.24	REQ
	Section extension 25	7.7.25	NOT REQ
	Section extension 26	7.7.26	NOT REQ
	Section extension 27	7.7.27	NOT REQ
DMRS-BF	Port reduced DMRS data sending	12.6.3.1	O-RU REQ
	User group self assembly	7.7.24	NOT REQ
	Per-window symbol reordering with DMRS sending	7.7.25, 12.6.1.5	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Per section tx-window reassignment	7.7.25, 12.6.1.5	NOT REQ
	symbolId meaning when reordering (up-symbolId-type-supported)	7.7.25	N/A
	Transform precoding supported	7.7.24	O-RU REQ
	PUSCH and DMRS mux supported	7.7.24	NOT REQ
	DMRS configuration Type (dType)	7.7.24, 7.7.24.14	REQ: Type 1 (dType=0)
	dmrsSymbolMask	7.7.24	REQ: 1+1 (symbol location: 2, 11)
	different-transform-precoding-in-user-group-supported	7.7.24	NOT REQ
	different-cdm-without-data-in-user-group-supported	7.7.24	NOT REQ
Measurements	SINR reporting	7.2.11, 7.4.11	O-RU REQ
	MEAS-UE-TAE (UE timing advance error)	9.2.1	O-RU REQ
	MEAS-UE-LAYER-POWER	9.2.1	O-RU REQ
	MEAS-UE-FREQ-OFFSET	9.2.1	O-RU REQ
	MEAS-IPN-ALLOC	9.2.1	O-RU REQ
	MEAS-IPN-UNALLOC	9.2.1	O-RU REQ
	MEAS-ANT-DMRS-SNR	9.2.1	NOT REQ
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	REQ
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	NOT REQ
	Local PRTC	5.2.3, 10.1, 11	N/A
	Topology configuration	11.2.2, 11.2.3 (for PLFS)	REQ: LLS-C1 or LLS-C3

Table A.3.1.8-2 lists “to be reported” and “to be validated” capabilities in radio layer 3 C-plane establishment and initial radio U-plane data transfer test. Table A.3.1.8-3 lists “to be reported” and “to be validated” capabilities in U-plane DL data transfer test. Table A.3.1.8-4 lists “to be reported” and “to be validated” capabilities in U-plane UL data transfer test.

Table A.3.1.8-2: Capability to be reported or validated for NR TDD IOT M-MIMO profile 5 in radio layer 3 C-Plane establishment and initial radio U-Plane data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH
	Capabilities to be reported: PDBF for DL, BFW Fixed point 12 bits, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer

Table A.3.1.8-3: Capability to be reported or validated for NR TDD IOT M-MIMO profile 5 in U-Plane DL data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL
	Capabilities to be reported: PDBF for DL, BFW Fixed point 12 bits, DMRS-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits)
PTC2	Capabilities to be validated: WDBF for DL
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, DMRS-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits)

Table A.3.1.8-4: Capability to be reported or validated for NR TDD IOT M-MIMO profile 5 in U-Plane data UL transfer test

PTCs	
PTC1	Capabilities to be validated: DMRS-BF-EQ for PUSCH
	Capabilities to be reported: WDBF or PDBF for DL, BFW Fixed point 12 bits, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: DMRS-BF-EQ for PUSCH
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer

A.3.1.9 NR TDD IOT M-MIMO Profile 6 - NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Base

Table A.3.1.9-1 describes the base profile. The following PTCs are defined.

- 1) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Base_PTC1_[12bitFP.BFW]
- 2) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Base_PTC2_[8bitBFP.BFW]

Table A.3.1.9-1: NR TDD IOT M-MIMO Profile 6: NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Base

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	DDDSUDDDD
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size		4096
	Total channel bandwidth	-	100MHz x 1 CC
	Number of user layers	-	DL: 2 – 16 UL: 2 – 8
	Fronthaul Ethernet link	-	25Gbps x 1 lane
	PRACH preamble format	-	O-RU REQ: F0
	O-RU category	4.2.1	Category B
	LAA	7.2.5	N/A
Delay management, general	Network delay determination	4.4.4	Defined Transport Method
	O-RU adaptation of delay profile information (based on IIs-CU delay profile and transport delay)	4.4.4.3	NOT REQ
	O-DU timing advance type	4.4.5-4.4.6, Annex B	REQ: Fixed timing Advance

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	SRS transfer	4.4.7	REQ: Non-delay managed U-Plane
Delay management, default	Delay Sets for timing parameters		REQ: Error! Reference source not found.
	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
Delay management, non default	non-default beamforming methods #1	12.6.3	REQ: DMRS-BF-NEQ
	Delay Sets for timing parameters		REQ: Error! Reference source not found.
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	NOT REQ
	Transport header	5.1.3	REQ: eCPRI
	eCPRI concatenation	5.1.3.2	NOT REQ
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 3 and 2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	O-RU REQ: 2 and 3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	O-RU REQ: 3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 8
	Fragmentation	5.5	Application layer fragmentation REQ
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority REQ
	Transport prioritization within U-Plane	5.3	NOT REQ
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID REQ
	Transport-based separation within C/U-Plane traffic	5.4	NOT REQ
Digital Power Scaling	UL gain_correction (not applicable to DMRS-BF-EQ)	8.1.3.2	0 dB
	equalization-data-scaling	12.6.3.5	N/A

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Beamforming	O-RU beamforming type	4.2.1	REQ: Digital beamforming
	Beamforming control method	7.2.1, 12.6, Annex J	DL: REQ: WDBF and PDBF UL: REQ: DMRS-BF for PUSCH, WDBF or PDBF for non-PUSCH
	BFW IQ	7.7.1, Annex J	REQ PTC1: Fixed point, 12 bits PTC2: BFP, 8 bits
IQ compression	U-Plane data compression method	8, Annex A	REQ: BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	DL: REQ: 9 bits UL: REQ: 9 bits
	IQ data frame format not including udCompHdr field	8.3.3.13	REQ
C-Plane	Section Type 0	7.4.2	O-RU REQ
	Section Type 1	7.4.3	REQ
	Section Type 3	7.4.5	REQ (for PRACH)
	Section Type 4	7.4.6	NOT REQ
	Section Type 5	7.4.7	REQ (for DMRS-BF)
	Section Type 6	7.4.8	NOT REQ
	Section Type 7	7.4.9	N/A
	Section Type 8	7.4.10	NOT REQ
	Section Type 9	7.4.11	NOT REQ
	Section Type 10	7.4.12	NOT REQ
	Section Type 11	7.4.13	NOT REQ
	"symInc" flag	7.5.3.3	NOT REQ
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	REQ: Single C-Plane message

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 1 (beamforming weights)	7.7.1	REQ
	Section extension 2 (beamforming attributes)	7.7.2	NOT REQ
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	NOT REQ
	Section extension 4 (modulation compr. params)	7.7.4	NOT REQ
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	NOT REQ
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	NOT REQ
	Section extension 7 (Multiple-eAxC designation)	7.7.7	NOT REQ
	Section extension 8 (regularization factor)	7.7.8	NOT REQ
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	NOT REQ
	Section extension 10 (Multiple ports grouping)	7.7.10	REQ (for DMRS-BF)
	Section extension 11 (Flexible BF weights)	7.7.11	NOT REQ
	Section extension 12	7.7.12	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 13	7.7.13	NOT REQ
	Section extension 14	7.7.14	NOT REQ
	Section extension 15	7.7.15	NOT REQ
	Section extension 16	7.7.16	NOT REQ
	Section extension 17	7.7.17	NOT REQ
	Section extension 18	7.7.18	NOT REQ
	Section extension 19	7.7.19	NOT REQ
	Section extension 20	7.7.20	NOT REQ
	Section extension 21	7.7.21	NOT REQ
	Section extension 22	7.7.22	NOT REQ
	Section extension 23	7.7.23	NOT REQ
	Section extension 24	7.7.24	REQ
	Section extension 25	7.7.25	NOT REQ
	Section extension 26	7.7.26	NOT REQ
	Section extension 27	7.7.27	NOT REQ
DMRS-BF	Port reduced DMRS data sending	12.6.3.1	REQ
	User group self assembly	7.7.24	N/A
	Per-window symbol reordering with DMRS sending	7.7.25, 12.6.1.5	NOT REQ
	Per section tx-window reassignment	7.7.25, 12.6.1.5	NOT REQ
	symbolid meaning when reordering (up-symbolId-type-supported)	7.7.25	N/A
	Transform precoding supported	7.7.24	O-RU REQ
	PUSCH and DMRS mux supported	7.7.24	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	DMRS configuration Type (dType)	7.7.24, 7.7.24.14	REQ: Type 1 (dType=0)
	dmrsSymbolMask	7.7.24	REQ: 1+1 (symbol location: 2, 11)
	different-transform-precoding-in-user-group-supported	7.7.24	NOT REQ
	different-cdm-without-data-in-user-group-supported	7.7.24	NOT REQ
Measurements	SINR reporting	7.2.11, 7.4.11	N/A
	MEAS-UE-TAE (UE timing advance error)	9.2.1	NOT REQ. O-RU declares feature DMRS-BF-NEQ-UNALTERED-TAE.
	MEAS-UE-LAYER-POWER	9.2.1	NOT REQ
	MEAS-UE-FREQ-OFFSET	9.2.1	NOT REQ. O-RU declares feature DMRS-BF-NEQ-UNALTERED-FREQ-OFFSET.
	MEAS-IPN-ALLOC	9.2.1	NOT REQ
	MEAS-IPN-UNALLOC	9.2.1	NOT REQ
	MEAS-ANT-DMRS-SNR	9.2.1	NOT REQ
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	REQ
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	NOT REQ
	Local PRTC	5.2.3, 10.1, 11	N/A
	Topology configuration	11.2.2, 11.2.3 (for PLFS)	REQ: LLS-C1 or LLS-C3

Table A.3.1.9-2 lists “to be reported” and “to be validated” capabilities in radio layer 3 C-plane establishment and initial radio U-plane data transfer test. Table A.3.1.9-3 lists “to be reported” and “to be validated” capabilities in U-plane DL data transfer test. Table A.3.1.9-4 lists “to be reported” and “to be validated” capabilities in U-plane UL data transfer test.

Table A.3.1.9-2: Capability to be reported or validated for NR TDD IOT M-MIMO profile 6 in radio layer 3 C-Plane establishment and initial radio U-Plane data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH
	Capabilities to be reported: PDBF for DL, BFW Fixed point 12 bits, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer

Table A.3.1.9-3: Capability to be reported or validated for NR TDD IOT M-MIMO profile 6 in U-Plane DL data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL
	Capabilities to be reported: PDBF for DL, BFW Fixed point 12 bits, DMRS-BF-NEQ for PUSCH, Non-delay managed U-plane for SRS transfer, eAxC-ID format (3238 or 2338)
PTC2	Capabilities to be validated: WDBF for DL
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, DMRS-BF-NEQ for PUSCH, Non-delay managed U-plane for SRS transfer, eAxC-ID format (3238 or 2338)

Table A.3.1.9-4: Capability to be reported or validated for NR TDD IOT M-MIMO profile 5 in U-Plane data UL transfer test

PTCs	
PTC1	Capabilities to be validated: DMRS-BF-NEQ for PUSCH
	Capabilities to be reported: WDBF or PDBF for DL, BFW Fixed point 12 bits, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: DMRS-BF-NEQ for PUSCH
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer

A.3.1.10 NR TDD IOT M-MIMO Profile 7 - NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Enhanced

Table A.3.1.10-1 describes the base profile. The following PTCs are defined.

- 1) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Enh_PTC1_[12bitFP.BFW-nondelay.SRS-SE4.ModComp-noSE5-noSE6-noSE11-noSE12-noSE23-noSE27-noSelRE-noSA]
- 2) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Enh_PTC2_[8bitBFP.BFW-nondelay.SRS-SE4.ModComp-noSE5-noSE6-noSE11-noSE12-noSE23-noSE27-noSelRE-noSA]
- 3) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Enh_PTC3_[8bitBFP.BFW-nondelay.SRS-SE4.SE5.ModComp-SE6-SE11-SE12-noSE23-noSE27-noSelRE-noSA]
- 4) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Enh_PTC4_[8bitBFP.BFW-MplaneTxWindow.SRS-SE4.ModComp-noSE5-SE6-SE11-SE12-noSE23-noSE27-noSelRE-noSA]
- 5) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Enh_PTC5_[8bitBFP.BFW-MplaneTxWindow.SRS-SE4.ModComp-noSE5-SE6-SE11-SE12-noSE23-SE27-noSelRE-noSA]
- 6) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Enh_PTC6_[8bitBeamSpaceII.8bitBFP.BFW-MplaneTxWindow.SRS-SE4.SE5.ModComp-SE6-SE11-SE12-SE23-noSE27-SelRE-SAID]

Table A.3.1.10-1: NR TDD IOT M-MIMO Profile 7: NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Enhanced

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	DDDSUUDDDD
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size		4096
	Total channel bandwidth	-	100MHz x 1 CC
	Number of user layers	-	DL: 2 – 16 UL: 2 – 8
	Fronthaul Ethernet link	-	25Gbps x 1 lane

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	PRACH preamble format	-	O-RU REQ: F0
	O-RU category	4.2.1	Category B
	LAA	7.2.5	N/A
Delay management, general	Network delay determination	4.4.4	Defined Transport Method
	O-RU adaptation of delay profile information (based on IIs-CU delay profile and transport delay)	4.4.4.3	NOT REQ
	O-DU timing advance type	4.4.5-4.4.6, Annex B	REQ: Fixed timing Advance
	SRS transfer	4.4.7, 4.6.4	PTC4, PTC5, PTC6: M-Plane static window control (STATIC-TRANSMISSION-WINDOW-CONTROL) REQ PTC1, PTC2, PTC3: Non-delay managed U-Plane REQ
Delay management, default	Delay Sets for timing parameters		REQ: Error! Reference source not found.
	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
Delay management, non default	non-default beamforming methods #1	12.6.3	REQ: DMRS-BF-EQ
	Delay Sets for timing parameters		REQ: Error! Reference source not found.
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	O-RU REQ
	Transport header	5.1.3	REQ: eCPRI
	eCPRI concatenation	5.1.3.2	NOT REQ
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 3 and 2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	O-RU REQ: 2 and 3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	O-RU REQ: 3

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 8
	Fragmentation	5.5	Application layer fragmentation REQ
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority REQ
	Transport prioritization within U-Plane	5.3	NOT REQ
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID REQ
	Transport-based separation within C/U-Plane traffic	5.4	NOT REQ
Digital Power Scaling	UL gain_correction (not applicable to DMRS-BF-EQ)	8.1.3.2	0 dB
	equalization-data-scaling	12.6.3.5	REQ: Scaling-function1 or Scaling-function2
Beamforming	O-RU beamforming type	4.2.1	REQ: Digital beamforming
	Beamforming control method	7.2.1, 12.6, Annex J	DL: REQ: WDBF and PDBF UL: REQ: DMRS-BF for PUSCH, WDBF or PDBF for non-PUSCH
	BFW IQ	7.7.1, Annex J	REQ: PTC1: Fixed point, 12 bits PTC2, PTC3, PTC4, PTC5: BFP, 8 bits O-RU REQ: PTC6: Beamspace compression Type II (8 bits) and BFP (8 bits)
	U-Plane data compression method	8, Annex A	DL: O-RU REQ PTC1, PTC2, PTC3, PTC4, PTC5: Mod Comp, no sel RE PTC6: Mod Comp, with sel RE. UL: REQ BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	REQ for Mod Comp in DL: 4 bits for BFP in UL: 8 and 9 bits for DMRS-BF and 9 bits for other BF methods
	IQ data frame format not including udCompHdr field	8.3.3.13	REQ
C-Plane	Section Type 0	7.4.2	O-RU REQ
	Section Type 1	7.4.3	REQ
	Section Type 3	7.4.5	REQ (for PRACH)

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section Type 4	7.4.6	NOT REQ
	Section Type 5	7.4.7	REQ (for DMRS-BF)
	Section Type 6	7.4.8	NOT REQ
	Section Type 7	7.4.9	N/A
	Section Type 8	7.4.10	NOT REQ
	Section Type 9	7.4.11	REQ
	Section Type 10	7.4.12	REQ
	Section Type 11	7.4.13	O-RU REQ
	"symInc" flag	7.5.3.3	NOT REQ
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	REQ: Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	REQ
	Section extension 2 (beamforming attributes)	7.7.2	NOT REQ
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	NOT REQ
	Section extension 4 (modulation compr. params)	7.7.4	O-RU REQ
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	PTC1, PTC2, PTC4, PTC5: NOT REQ PTC3, PTC6: O-RU REQ
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	PTC1, PTC2: NOT REQ PTC3, PTC4, PTC5, PTC6: O-RU REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 7 (Multiple-eAxC designation)	7.7.7	NOT REQ
	Section extension 8 (regularization factor)	7.7.8	NOT REQ
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	NOT REQ
	Section extension 10 (Multiple ports grouping)	7.7.10	REQ (for DMRS-BF)
	Section extension 11 (Flexible BF weights)	7.7.11	PTC1, PTC2: NOT REQ PTC3, PTC4, PTC5, PTC6: REQ
	Section extension 12	7.7.12	PTC1, PTC2: NOT REQ PTC3, PTC4, PTC5, PTC6: O-RU REQ
	Section extension 13	7.7.13	NOT REQ
	Section extension 14	7.7.14	NOT REQ
	Section extension 15	7.7.15	NOT REQ
	Section extension 16	7.7.16	NOT REQ
	Section extension 17	7.7.17	NOT REQ
	Section extension 18	7.7.18	NOT REQ
	Section extension 19	7.7.19	NOT REQ
	Section extension 20	7.7.20	NOT REQ
	Section extension 21	7.7.21	NOT REQ
	Section extension 22	7.7.22	NOT REQ
	Section extension 23	7.7.23	PTC1, PTC2, PTC3, PTC4, PTC5: NOT REQ PTC6: O-RU REQ
	Section extension 24	7.7.24	REQ
	Section extension 25	7.7.25	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 26	7.7.26	O-RU REQ
	Section extension 27	7.7.27	PTC1, PTC2, PTC3, PTC4, PTC6: NOT REQ PTC5: REQ
DMRS-BF	Port reduced DMRS data sending	12.6.3.1	O-RU REQ
	User group self assembly	7.7.24	PTC1, PTC2, PTC3, PTC4, PTC5: NOT REQ (no self-assembly = noSA) PTC6: O-RU REQ: SAID (self assembly with id)
	Per-window symbol reordering with DMRS sending	7.7.25, 12.6.1.5	NOT REQ
	Per section tx-window reassignment	7.7.25, 12.6.1.5	NOT REQ
	symbolid meaning when reordering (up-symbolId-type-supporte)	7.7.25	N/A
	Transform precoding supported	7.7.24	O-RU REQ
	PUSCH and DMRS mux supported	7.7.24	O-RU REQ
	DMRS configuration Type (dType)	7.7.24, 7.7.24.14	REQ: Type 1 (dType=0)
	dmrsSymbolMask	7.7.24	REQ: 1+1 (symbol location: 2, 11)
	different-transform-precoding-in-user-group-supported	7.7.24	NOT REQ
	different-cdm-without-data-in-user-group-supported	7.7.24	NOT REQ
Measurements	SINR reporting	7.2.11, 7.4.11	O-RU REQ
	MEAS-UE-TAE (UE timing advance error)	9.2.1	O-RU REQ
	MEAS-UE-LAYER-POWER	9.2.1	O-RU REQ
	MEAS-UE-FREQ-OFFSET	9.2.1	O-RU REQ
	MEAS-IPN-ALLOC	9.2.1	O-RU REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	MEAS-IPN-UNALLOC	9.2.1	O-RU REQ
	MEAS-ANT-DMRS-SNR	9.2.1	NOT REQ
	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	REQ
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	NOT REQ
	Local PRTC	5.2.3, 10.1, 11	N/A
	Topology configuration	11.2.2, 11.2.3 (for PLFS)	REQ: LLS-C1 or LLS-C3

Table A.3.1.10-2 lists “to be reported” and “to be validated” capabilities in radio layer 3 C-plane establishment and initial radio U-plane data transfer test. Table A.3.1.10-3 lists “to be reported” and “to be validated” capabilities in U-plane DL data transfer test. Table A.3.1.10-4 lists “to be reported” and “to be validated” capabilities in U-plane UL data transfer test.

Table A.3.1.10-2: Capability to be reported or validated for NR TDD IOT M-MIMO profile 7 in radio layer 3 C-Plane establishment and initial radio U-Plane data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW Fixed point 12 bits, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC3	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, SE5, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC4	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), M-Plane static window control for SRS transfer
PTC5	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, SE6, SE12, SE11, SE27, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), M-Plane static window control for SRS transfer
PTC6	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BeamSpace II + BFP 8 bits, SE4, SE5, SE6, SE12, SE11, SE23, Selective RE sending, User group self assembly, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), M-Plane static window control for SRS transfer

Table A.3.1.10-3: Capability to be reported or validated for NR TDD IOT M-MIMO profile 7 in U-Plane DL data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, PUSCH-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW Fixed point 12 bits, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits)
PTC2	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, PUSCH-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits)
PTC3	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, PUSCH-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW BFP 8 bits, SE4, SE5, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits)
PTC4	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, PUSCH-BF-EQ for PUSCH, M-Plane static window control for SRS transfer, BFW BFP 8 bits, SE4, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits)
PTC5	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, PUSCH-BF-EQ for PUSCH, M-Plane static window control for SRS transfer, BFW BFP 8 bits, SE4, SE6, SE12, SE11, SE27, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits)
PTC6	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, PUSCH-BF-EQ for PUSCH, M-Plane static window control for SRS transfer, BFW BeamSpace II + BFP 8 bits, SE4, SE5, SE6, SE12, SE11, SE23, Selective RE sending, User group self assembly, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits)

Table A.3.1.10-4: Capability to be reported or validated for NR TDD IOT M-MIMO profile 7 in U-Plane data UL transfer test

PTCs	
PTC1	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW Fixed point 12 bits, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC3	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, SE5, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC4	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), M-Plane static window control for SRS transfer
PTC5	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, SE6, SE12, SE11, SE27, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), M-Plane static window control for SRS transfer
PTC6	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW BeamSpace II + BFP 8 bits, SE4, SE5, SE6, SE12, SE11, SE23, Selective RE sending, User group self assembly, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), M-Plane static window control for SRS transfer

A.3.1.11 NR TDD IOT M-MIMO Profile 8 - NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Enhanced

Table A.3.1.11-1 describes the base profile. The following PTCs are defined.

- 1) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Enh_PTC1_[12bitFP.BFW-nondelay.SRS-SE4.ModComp-noSE5-noSE6-noSE11-noSE12-noSE23-noSE27-noSelRE-noSA]
- 2) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Enh_PTC2_[8bitBFP.BFW-nondelay.SRS-SE4.ModComp-noSE5-noSE6-noSE11-noSE12-noSE23-noSE27-noSelRE-noSA]
- 3) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Enh_PTC3_[8bitBFP.BFW-nondelay.SRS-SE4.SE5.ModComp-SE6-SE11-SE12--noSE23-noSE27-noSelRE-noSA]

- 4) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Enh_PTC4_[8bitBFP.BFW-MplaneTxWindow.SRS-SE4.ModComp-noSE5-SE6-SE11-SE12-noSE23-noSE27-noSelRE-noSA]
- 5) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Enh_PTC5_[8bitBFP.BFW-MplaneTxWindow.SRS-SE4.ModComp-noSE5-SE6-SE11-SE12-noSE23-SE27-noSelRE-noSA]
- 6) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Enh_PTC6_[8bitBeamSpaceII.8bitBFP.BFW-MplaneTxWindow.SRS-SE4.SE5.ModComp-SE6-SE11-SE12-SE23-noSE27-SelRE-SAID]

Table A.3.1.11-1: NR TDD IOT M-MIMO Profile 8: NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Enhanced

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	DDDSUDDDD
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size		4096
	Total channel bandwidth	-	100MHz x 1 CC
	Number of user layers	-	DL: 2 – 16 UL: 2 – 8
	Fronthaul Ethernet link	-	25Gbps x 1 lane
	PRACH preamble format	-	O-RU REQ: F0
	O-RU category	4.2.1	Category B
	LAA	7.2.5	N/A
Delay management, general	Network delay determination	4.4.4	Defined Transport Method
	O-RU adaptation of delay profile information (based on IIs-CU delay profile and transport delay)	4.4.4.3	NOT REQ
	O-DU timing advance type	4.4.5-4.4.6, Annex B	REQ: Fixed timing Advance
	SRS transfer	4.4.7, 4.6.4	PTC4, PTC5, PTC6: M-Plane static window control (STATIC-TRANSMISSION-WINDOW-CONTROL) REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
			PTC1, PTC2, PTC3: Non-delay managed U-Plane REQ
Delay management, default	Delay Sets for timing parameters		REQ: Error! Reference source not found.
	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
Delay management, non default	non-default beamforming methods #1	12.6.3	REQ: DMRS-BF-NEQ
	Delay Sets for timing parameters		REQ: Error! Reference source not found.
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	O-RU REQ
	Transport header	5.1.3	REQ: eCPRI
	eCPRI concatenation	5.1.3.2	NOT REQ
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 3 and 2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	O-RU REQ: 2 and 3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	O-RU REQ: 3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 8
	Fragmentation	5.5	Application layer fragmentation REQ
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority REQ
	Transport prioritization within U-Plane	5.3	NOT REQ
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID REQ
	Transport-based separation within C/U-Plane traffic	5.4	NOT REQ
Digital Power Scaling	UL gain_correction (not applicable to DMRS-BF-EQ)	8.1.3.2	0 dB
	equalization-data-scaling	12.6.3.5	N/A
Beamforming	O-RU beamforming type	4.2.1	REQ: Digital beamforming

Category	Item	Related O-RAN CUS-Plane specification section(s)	
IQ compression	Beamforming control method	7.2.1, 12.6, Annex J	DL: REQ: WDBF and PDBF UL: REQ: DMRS-BF for PUSCH, WDBF or PDBF for non-PUSCH
	BFW IQ	7.7.1, Annex J	REQ PTC1: Fixed point, 12 bits PTC2, PTC3, PTC4, PTC5: BFP, 8 bits O-RU REQ PTC6: Beamspace compression Type II (8 bits) and BFP (8 bits)
	U-Plane data compression method	8, Annex A	DL: O-RU REQ PTC1, PTC2, PTC3, PTC4, PTC5: Mod Comp, no sel RE PTC6: Mod Comp, with sel RE. UL: REQ BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	DL: REQ: Mod Comp with 4 bits UL: REQ: BFP with 9 bits
	IQ data frame format not including udCompHdr field	8.3.3.13	REQ
C-Plane	Section Type 0	7.4.2	O-RU REQ
	Section Type 1	7.4.3	REQ
	Section Type 3	7.4.5	REQ (for PRACH)
	Section Type 4	7.4.6	NOT REQ
	Section Type 5	7.4.7	REQ (for DMRS-BF)
	Section Type 6	7.4.8	NOT REQ
	Section Type 7	7.4.9	N/A
	Section Type 8	7.4.10	NOT REQ
	Section Type 9	7.4.11	NOT REQ
	Section Type 10	7.4.12	NOT REQ
	Section Type 11	7.4.13	NOT REQ
	"symInc" flag	7.5.3.3	NOT REQ
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	REQ: Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	REQ
	Section extension 2 (beamforming attributes)	7.7.2	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	NOT REQ
	Section extension 4 (modulation compr. params)	7.7.4	O-RU REQ
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	PTC1, PTC2, PTC4, PTC5: NOT REQ PTC3, PTC6: O-RU REQ
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	PTC1, PTC2: NOT REQ PTC3, PTC4, PTC5, PTC6: O-RU REQ
	Section extension 7 (Multiple-eAxC designation)	7.7.7	NOT REQ
	Section extension 8 (regularization factor)	7.7.8	NOT REQ
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	NOT REQ
	Section extension 10 (Multiple ports grouping)	7.7.10	REQ (for DMRS-BF)
	Section extension 11 (Flexible BF weights)	7.7.11	PTC1, PTC2: NOT REQ PTC3, PTC4, PTC5, PTC6: O-RU REQ
	Section extension 12	7.7.12	PTC1, PTC2: NOT REQ PTC3, PTC4, PTC5, PTC6: O-RU REQ
	Section extension 13	7.7.13	NOT REQ
	Section extension 14	7.7.14	NOT REQ
	Section extension 15	7.7.15	NOT REQ
	Section extension 16	7.7.16	NOT REQ
	Section extension 17	7.7.17	NOT REQ
	Section extension 18	7.7.18	NOT REQ
	Section extension 19	7.7.19	NOT REQ
	Section extension 20	7.7.20	NOT REQ
	Section extension 21	7.7.21	NOT REQ
	Section extension 22	7.7.22	NOT REQ
	Section extension 23	7.7.23	PTC1, PTC2, PTC3, PTC4, PTC5: NOT REQ PTC6: O-RU REQ
	Section extension 24	7.7.24	REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 25	7.7.25	NOT REQ
	Section extension 26	7.7.26	O-RU REQ
	Section extension 27	7.7.27	PTC1, PTC2, PTC3, PTC4, PTC6: NOT REQ PTC5: REQ
DMRS-BF	Port reduced DMRS data sending	12.6.3.1	REQ
	User group self assembly	7.7.24	PTC1, PTC2, PTC3, PTC4, PTC5: NOT REQ (no self-assembly = noSA) PTC6: O-RU REQ: SAID (self assembly with id)
	Per-window symbol reordering with DMRS sending	7.7.25, 12.6.1.5	NOT REQ
	Per section tx-window reassignment	7.7.25, 12.6.1.5	NOT REQ
	symbolid meaning when reordering (up-symbolid-type-supporte)	7.7.25	N/A
	Transform precoding supported	7.7.24	O-RU REQ
	PUSCH and DMRS mux supported	7.7.24	O-RU REQ
	DMRS configuration Type (dType)	7.7.24, 7.7.24.14	REQ: Type 1 (dType=0)
	dmrsSymbolMask	7.7.24	REQ: 1+1 (symbol location: 2, 11)
	different-transform-precoding-in-user-group-supported	7.7.24	NOT REQ
	different-cdm-without-data-in-user-group-supported	7.7.24	NOT REQ
Measurements	SINR reporting	7.2.11, 7.4.11	N/A
	MEAS-UE-TAE (UE timing advance error)	9.2.1	NOT REQ. O-RU declares feature DMRS-BF-NEQ-UNALTERED-TAE.
	MEAS-UE-LAYER-POWER	9.2.1	NOT REQ
	MEAS-UE-FREQ-OFFSET	9.2.1	NOT REQ. O-RU declares feature DMRS-BF-NEQ-UNALTERED-FREQ-OFFSET.
	MEAS-IPN-ALLOC	9.2.1	NOT REQ
	MEAS-IPN-UNALLOC	9.2.1	NOT REQ
	MEAS-ANT-DMRS-SNR	9.2.1	NOT REQ
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	NOT REQ
	Local PRTC	5.2.3, 10.1, 11	N/A
	Topology configuration	11.2.2, 11.2.3 (for PLFS)	REQ: LLS-C1 or LLS-C3

Table A.3.1.11-2 lists “to be reported” and “to be validated” capabilities in radio layer 3 C-plane establishment and initial radio U-plane data transfer test. Table A.3.1.11-3 lists “to be reported” and “to be validated” capabilities in U-plane DL data transfer test. Table A.3.1.11-4 lists “to be reported” and “to be validated” capabilities in U-plane UL data transfer test.

Table A.3.1.11-2: Capability to be reported or validated for NR TDD IOT M-MIMO profile 8 in radio layer 3 C-Plane establishment and initial radio U-Plane data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW Fixed point 12 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC3	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, SE5, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC4	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer
PTC5	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, SE6, SE12, SE11, SE27, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer
PTC6	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, BFW BeamSpace II + BFP 8 bits, SE4, SE5, SE6, SE12, SE11, SE23, Selective RE sending, User group self assembly, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer

Table A.3.1.11-3: Capability to be reported or validated for NR TDD IOT M-MIMO profile 8 in U-Plane DL data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-NEQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW Fixed point 12 bits, SE4, eAxC-ID format (3238 or 2338)
PTC2	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-NEQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338)
PTC3	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-NEQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW BFP 8 bits, SE4, SE5, SE6, SE12, SE11, eAxC-ID format (3238 or 2338)
PTC4	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-NEQ for PUSCH, M-Plane static window control for SRS transfer, BFW BFP 8 bits, SE4, SE6, SE12, SE11, eAxC-ID format (3238 or 2338)
PTC5	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-NEQ for PUSCH, M-Plane static window control for SRS transfer, BFW BFP 8 bits, SE4, SE6, SE12, SE11, SE27, eAxC-ID format (3238 or 2338)
PTC6	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-NEQ for PUSCH, M-Plane static window control for SRS transfer, BFW BeamSpace II + BFP 8 bits, SE4, SE5, SE6, SE12, SE11, SE23, Selective RE sending, User group self assembly, eAxC-ID format (3238 or 2338)

Table A.3.1.11-4: Capability to be reported or validated for NR TDD IOT M-MIMO profile 8 in U-Plane data UL transfer test

PTCs	
PTC1	Capabilities to be validated: DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW Fixed point 12 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC3	Capabilities to be validated: DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, SE5, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC4	Capabilities to be validated: DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, SE6, SE12, SE11, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer
PTC5	Capabilities to be validated: DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, SE6, SE12, SE11, SE27, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer
PTC6	Capabilities to be validated: DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: WDBF or PDBF for DL, BFW Beamspace II + BFP 8 bits, SE4, SE5, SE6, SE12, SE11, SE23, Selective RE sending, User group self assembly, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer

A.3.1.12 NR TDD IOT M-MIMO Profile 9 - NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Reordering

Table A.3.1.12-1 shows the base profile. It is expected that re-ordering feature is used when using reordering PTCs for IOT testing. The following PTCs are defined.

- 1) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Reordering_PTC1_[8bitBFP.BFW-nondelay.SRS-PerWindows.reord-onair.upsymbolId-SE4.ModComp-noSE5-noJumbo]
- 2) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Reordering_PTC2_[8bitBFP.BFW-nondelay.SRS-PerWindow.reord-txwindow.upsymbolId-SE4.SE5.ModComp-noJumbo]
- 3) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Reordering_PTC3_[8bitBFP.BFW-MplaneTxWindow.SRS-PerWindows.reord-txwindow.upsymbolId-noModComp-noJumbo]

- 4) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Reordering_PTC4_[12bitFP.BFW-nondelay.SRS-PerSection.reord-onair.upsymbolId-SE4.ModComp-noSE5-Jumbo]
- 5) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Reordering_PTC5_[8bitBFP.BFW-MplaneTxWindow.SRS-PerSection.reord-onair.upsymbolId-SE4.ModComp-noSE5-Jumbo]

Table A.3.1.12-1: NR TDD IOT M-MIMO Profile 9: NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-EQ-Reordering

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	DDDSUUDDDD
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size		4096
	Total channel bandwidth	-	100MHz x 1 CC
	Number of user layers	-	DL: 2 – 16 UL: 2 – 8
	Fronthaul Ethernet link	-	25Gbps x 1 lane
	PRACH preamble format	-	O-RU REQ: F0
	O-RU category	4.2.1	Category B
	LAA	7.2.5	N/A
Delay management, general	Network delay determination	4.4.4	Defined Transport Method
	O-RU adaptation of delay profile information (based on IIS-CU delay profile and transport delay)	4.4.4.3	NOT REQ
	O-DU timing advance type	4.4.5-4.4.6, Annex B	REQ: Fixed timing Advance
	SRS transfer	4.4.7, 4.6.4	PTC3, PTC5: M-Plane static window control (STATIC-TRANSMISSION-WINDOW-CONTROL) REQ PTC1, PTC2, PTC4: Non-delay managed U-Plane REQ
	Delay Sets for timing parameters		REQ: Error! Reference source not found.

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Delay management, default	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
Delay management, non default	non-default beamforming methods #1	12.6.3	REQ: DMRS-BF-EQ
	Delay Sets for timing parameters		REQ: Error! Reference source not found.
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	PTC4, PTC5: O-RU REQ PTC1, PTC2, PTC3: NOT REQ
	Transport header	5.1.3	REQ: eCPRI
	eCPRI concatenation	5.1.3.2	NOT REQ
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 3 and 2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	O-RU REQ: 2 and 3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	O-RU REQ: 3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 8
	Fragmentation	5.5	Application layer fragmentation REQ
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority REQ
	Transport prioritization within U-Plane	5.3	NOT REQ
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID REQ
	Transport-based separation within C/U-Plane traffic	5.4	NOT REQ
Digital Power Scaling	UL gain_correction (not applicable to DMRS-BF-EQ)	8.1.3.2	0 dB
	equalization-data-scaling	12.6.3.5	REQ: Scaling-function1 or Scaling-function2
Beamforming	O-RU beamforming type	4.2.1	REQ: Digital beamforming
	Beamforming control method	7.2.1, 12.6, Annex J	DL: REQ: WDBF and PDBF UL: REQ: DMRS-BF for PUSCH, WDBF or PDBF for non-PUSCH

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	BFW IQ	7.7.1, Annex J	REQ: PTC4: Fixed point, 12 bits PTC1, PTC2, PTC3, PTC5: BFP, 8 bits
IQ compression	U-Plane data compression method	8, Annex A	DL: O-RU REQ PTC1, PTC2, PTC4, PTC5: Mod Comp, no sel RE PTC3: BFP UL: REQ: BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	DL: O-RU REQ PTC1, PTC2, PTC4, PTC5: 4 bits (Mod Comp) PTC3: 9 bits, BFP UL: REQ: 9 bits
	IQ data frame format not including udCompHdr field	8.3.3.13	REQ
C-Plane	Section Type 0	7.4.2	O-RU REQ
	Section Type 1	7.4.3	REQ
	Section Type 3	7.4.5	REQ (for PRACH)
	Section Type 4	7.4.6	NOT REQ
	Section Type 5	7.4.7	REQ (for DMRS-BF)
	Section Type 6	7.4.8	NOT REQ
	Section Type 7	7.4.9	N/A
	Section Type 8	7.4.10	NOT REQ
	Section Type 9	7.4.11	REQ
	Section Type 10	7.4.12	REQ
	Section Type 11	7.4.13	O-RU REQ
	"symInc" flag	7.5.3.3	NOT REQ
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	REQ: Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	REQ
	Section extension 2 (beamforming attributes)	7.7.2	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	NOT REQ
	Section extension 4 (modulation compr. params)	7.7.4	PTC3: NOT REQ PTC1, PTC2, PTC4, PTC5: O-RU REQ
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	PTC1, PTC3, PTC4, PTC5: NOT REQ PTC2: O-RU REQ
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	NOT REQ
	Section extension 7 (Multiple-eAxC designation)	7.7.7	NOT REQ
	Section extension 8 (regularization factor)	7.7.8	NOT REQ
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	NOT REQ
	Section extension 10 (Multiple ports grouping)	7.7.10	REQ (for DMRS-BF)
	Section extension 11 (Flexible BF weights)	7.7.11	O-RU REQ
	Section extension 12	7.7.12	NOT REQ
	Section extension 13	7.7.13	NOT REQ
	Section extension 14	7.7.14	NOT REQ
	Section extension 15	7.7.15	NOT REQ
	Section extension 16	7.7.16	NOT REQ
	Section extension 17	7.7.17	NOT REQ
	Section extension 18	7.7.18	NOT REQ
	Section extension 19	7.7.19	NOT REQ
	Section extension 20	7.7.20	NOT REQ
	Section extension 21	7.7.21	NOT REQ
	Section extension 22	7.7.22	NOT REQ
	Section extension 23	7.7.23	NOT REQ
	Section extension 24	7.7.24	REQ
	Section extension 25	7.7.25	REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 26	7.7.26	O-RU REQ
	Section extension 27	7.7.27	NOT REQ
DMRS-BF	Port reduced DMRS data sending	12.6.3.1	O-RU REQ
	User group self assembly	7.7.24	NOT REQ
	Per-window symbol reordering	7.7.25, 12.6.1.5	PTC4, PTC5: NOT REQ PTC1, PTC2, PTC3: REQ If "per-window symbol reordering" is used, then "per-section tx-window reassignment" is not used.
	Per section tx-window reassignment	7.7.25, 12.6.1.5	PTC1, PTC2, PTC3: NOT REQ PTC4, PTC5: REQ If "per-section tx-window reassignment" is used, then "per-window symbol reordering" is not used.
	symbolId meaning when reordering (up-symbolId-type-supported)	7.7.25	REQ PTC2, PTC3: Transmission window number PTC1, PTC4, PTC5: on-air symbol number
	Transform precoding supported	7.7.24	O-RU REQ
	PUSCH and DMRS mux supported	7.7.24	O-RU REQ
	DMRS configuration Type (dType)	7.7.24, 7.7.24.14	REQ: Type 1 (dType=0)
	dmrsSymbolMask	7.7.24	REQ: 1+1 (symbol location: 2, 11)
	different-transform-precoding-in-user-group-supported	7.7.24	NOT REQ
	different-cdm-without-data-in-user-group-supported	7.7.24	NOT REQ
Measurements	SINR reporting	7.2.11, 7.4.11	O-RU REQ
	MEAS-UE-TAE (UE timing advance error)	9.2.1	O-RU REQ
	MEAS-UE-LAYER-POWER	9.2.1	O-RU REQ
	MEAS-UE-FREQ-OFFSET	9.2.1	O-RU REQ
	MEAS-IPN-ALLOC	9.2.1	O-RU REQ
	MEAS-IPN-UNALLOC	9.2.1	O-RU REQ
	MEAS-ANT-DMRS-SNR	9.2.1	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	REQ
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	NOT REQ
	Local PRTC	5.2.3, 10.1, 11	N/A
	Topology configuration	11.2.2, 11.2.3 (for PLFS)	REQ: LLS-C1 or LLS-C3

Table A.3.1.12-2 lists “to be reported” and “to be validated” capabilities in radio layer 3 C-plane establishment and initial radio U-plane data transfer test. Table A.3.1.12-3 lists “to be reported” and “to be validated” capabilities in U-plane DL data transfer test. Table A.3.1.12-4 lists “to be reported” and “to be validated” capabilities in U-plane UL data transfer test.

Table A.3.1.12-2: Capability to be reported or validated for NR TDD IOT M-MIMO profile 9 in radio layer 3 C-Plane establishment and initial radio U-Plane data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression, Per-window symbol reordering using on-air symbol number
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, SE5, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC3	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer
PTC4	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression, Per-section tx-window reassignment using on-air symbol number
	Capabilities to be reported: PDBF for DL, BFW Fixed point 12 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC5	Capabilities to be validated: WDBF for DL, DMRS-BF-EQ for PUSCH, modulation compression, Per-section tx-window reassignment using on-air symbol number
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer

Table A.3.1.12-3: Capability to be reported or validated for NR TDD IOT M-MIMO profile 7 in U-Plane DL data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), Per-window symbol reordering using on-air symbol number
PTC2	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW BFP 8 bits, SE4, SE5, eAxC-ID format (3238 or 2338), Per-window symbol reordering using tx-window symbol number
PTC3	Capabilities to be validated: WDBF for DL
	Capabilities to be reported: PDBF for DL, DMRS-BF-EQ for PUSCH, M-Plane static window control for SRS transfer, BFW BFP 8 bits, eAxC-ID format (3238 or 2338), Per-window symbol reordering using tx-window symbol number
PTC4	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW Fixed point 12 bits, SE4, eAxC-ID format (3238 or 2338), Per-section tx-window reassignment using on-air symbol number
PTC5	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-EQ for PUSCH, M-Plane static window control for SRS transfer, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), Per-section tx-window reassignment using on-air symbol number

Table A.3.1.12-4: Capability to be reported or validated for NR TDD IOT M-MIMO profile 7 in U-Plane data UL transfer test

PTCs	
PTC1	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression, Per-window symbol reordering using on-air symbol number
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, SE5, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC3	Capabilities to be validated: DMRS-BF-EQ for PUSCH, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer
PTC4	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression, Per-section tx-window reassignment using on-air symbol number
	Capabilities to be reported: WDBF or PDBF for DL, BFW Fixed point 12 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC5	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression, Per-section tx-window reassignment using on-air symbol number
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer

A.3.1.13 NR TDD IOT M-MIMO Profile 10 - NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Reordering

Table A.3.1.13-1 describes the base profile. It is expected that reordering feature is used when using reordering PTCs for IOT testing. The following PTCs are defined.

- 1) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Reordering_PTC1_[8bitBFP.BFW-nondelay.SRS-PerWindows.reord-txwindow.upsymbolId-SE4.ModComp]
- 2) NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Reordering_PTC2_[12bitFP.BFW-MplaneTxWindow.SRS-PerWindows.reord-txwindow.upsymbolId-noModComp]

Table A.3.1.13-1: NR TDD IOT M-MIMO Profile 10: NR-TDD-FR1-CAT-B-mMIMO-WDBF-DMRS-BF-NEQ-Reordering

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	DDDSUUDDDD
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Nominal FFT size		4096
	Total channel bandwidth	-	100MHz x 1 CC
	Number of user layers	-	DL: 2 – 16 UL: 2 – 8
	Fronthaul Ethernet link	-	25Gbps x 1 lane
	PRACH preamble format	-	O-RU REQ: F0
	O-RU category	4.2.1	Category B
	LAA	7.2.5	N/A
Delay management, general	Network delay determination	4.4.4	Defined Transport Method
	O-RU adaptation of delay profile information (based on IIS-CU delay profile and transport delay)	4.4.4.3	NOT REQ
	O-DU timing advance type	4.4.5-4.4.6, Annex B	REQ: Fixed timing Advance
	SRS transfer	4.4.7, 4.6.4	PTC2: M-Plane static window control (STATIC-TRANSMISSION-WINDOW-CONTROL) REQ PTC1: Non-delay managed U-Plane REQ
Delay management, default	Delay Sets for timing parameters		REQ: Error! Reference source not found.
	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
Delay management, non default	non-default beamforming methods #1	12.6.3	REQ: DMRS-BF-NEQ
	Delay Sets for timing parameters		REQ: Error! Reference source not found.
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	NOT REQ
	Transport header	5.1.3	REQ: eCPRI
	eCPRI concatenation	5.1.3.2	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 3 and 2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	O-RU REQ: 2 and 3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	O-RU REQ: 3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 8
	Fragmentation	5.5	Application layer fragmentation REQ
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority REQ
	Transport prioritization within U-Plane	5.3	NOT REQ
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID REQ
	Transport-based separation within C/U-Plane traffic	5.4	NOT REQ
Digital Power Scaling	UL gain_correction (not applicable to DMRS-BF-EQ)	8.1.3.2	0 dB
	equalization-data-scaling	12.6.3.5	N/A
Beamforming	O-RU beamforming type	4.2.1	REQ: Digital beamforming
	Beamforming control method	7.2.1, 12.6, Annex J	DL: REQ: WDBF and PDBF UL: REQ: DMRS-BF for PUSCH, WDBF or PDBF for non-PUSCH
	BFW IQ	7.7.1, Annex J	REQ PTC2: Fixed point, 12 bits PTC1: BFP, 8 bits
IQ compression	U-Plane data compression method	8, Annex A	DL: O-RU REQ PTC1: Mod Comp, no sel RE PTC2: BFP UL: REQ BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	DL: O-RU REQ PTC1: 4 bits (Mod Comp) PTC2: 9 bits, BFP UL: REQ: 9 bits
	IQ data frame format not including udCompHdr field	8.3.3.13	REQ
C-Plane	Section Type 0	7.4.2	O-RU REQ
	Section Type 1	7.4.3	REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section Type 3	7.4.5	REQ (for PRACH)
	Section Type 4	7.4.6	NOT REQ
	Section Type 5	7.4.7	REQ (for DMRS-BF)
	Section Type 6	7.4.8	NOT REQ
	Section Type 7	7.4.9	N/A
	Section Type 8	7.4.10	NOT REQ
	Section Type 9	7.4.11	NOT REQ
	Section Type 10	7.4.12	NOT REQ
	Section Type 11	7.4.13	NOT REQ
	"symInc" flag	7.5.3.3	NOT REQ
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	REQ: Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	REQ
	Section extension 2 (beamforming attributes)	7.7.2	NOT REQ
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	NOT REQ
	Section extension 4 (modulation compr. params)	7.7.4	PTC2: NOT REQ PTC1: O-RU REQ
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	NOT REQ
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	NOT REQ
	Section extension 7 (Multiple-eAxC designation)	7.7.7	NOT REQ
	Section extension 8 (regularization factor)	7.7.8	NOT REQ
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	NOT REQ
	Section extension 10 (Multiple ports grouping)	7.7.10	REQ (for DMRS-BF)
	Section extension 11 (Flexible BF weights)	7.7.11	O-RU REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 12	7.7.12	NOT REQ
	Section extension 13	7.7.13	NOT REQ
	Section extension 14	7.7.14	NOT REQ
	Section extension 15	7.7.15	NOT REQ
	Section extension 16	7.7.16	NOT REQ
	Section extension 17	7.7.17	NOT REQ
	Section extension 18	7.7.18	NOT REQ
	Section extension 19	7.7.19	NOT REQ
	Section extension 20	7.7.20	NOT REQ
	Section extension 21	7.7.21	NOT REQ
	Section extension 22	7.7.22	NOT REQ
	Section extension 23	7.7.23	NOT REQ
	Section extension 24	7.7.24	REQ
	Section extension 25	7.7.25	REQ
	Section extension 26	7.7.26	O-RU REQ
	Section extension 27	7.7.27	NOT REQ
DMRS-BF	Port reduced DMRS data sending	12.6.3.1	REQ
	User group self assembly	7.7.24	NOT REQ
	Per-window symbol reordering	7.7.25, 12.6.1.5	REQ
	Per section tx-window reassignment	7.7.25, 12.6.1.5	NOT REQ
	symbolid meaning when reordering (up-symbolId-type-supported)	7.7.25	REQ: Transmission window number
	Transform precoding supported	7.7.24	O-RU REQ
	PUSCH and DMRS mux supported	7.7.24	O-RU REQ
	DMRS configuration Type (dType)	7.7.24, 7.7.24.14	REQ: Type 1 (dType=0)
	dmrsSymbolMask	7.7.24	REQ: 1+1 (symbol location: 2, 11)

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	different-transform-precoding-in-user-group-supported	7.7.24	NOT REQ
	different-cdm-without-data-in-user-group-supported	7.7.24	NOT REQ
Measurements	SINR reporting	7.2.11, 7.4.11	N/A
	MEAS-UE-TAE (UE timing advance error)	9.2.1	NOT REQ. O-RU declares feature DMRS-BF-NEQ-UNALTERED-TAE.
	MEAS-UE-LAYER-POWER	9.2.1	NOT REQ
	MEAS-UE-FREQ-OFFSET	9.2.1	NOT REQ. O-RU declares feature DMRS-BF-NEQ-UNALTERED-FREQ-OFFSET.
	MEAS-IPN-ALLOC	9.2.1	NOT REQ
	MEAS-IPN-UNALLOC	9.2.1	NOT REQ
	MEAS-ANT-DMRS-SNR	9.2.1	NOT REQ
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	REQ
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	NOT REQ
	Local PRTC	5.2.3, 10.1, 11	N/A
	Topology configuration	11.2.2, 11.2.3 (for PLFS)	REQ: LLS-C1 or LLS-C3

Table A.3.1.13-2 lists “to be reported” and “to be validated” capabilities in radio layer 3 C-plane establishment and initial radio U-plane data transfer test. Table A.3.1.13-3 lists “to be reported” and “to be validated” capabilities in U-plane DL data transfer test. Table A.3.1.13-4 lists “to be reported” and “to be validated” capabilities in U-plane UL data transfer test.

Table A.3.1.13-2: Capability to be reported or validated for NR TDD IOT M-MIMO profile 10 in radio layer 3 C-Plane establishment and initial radio U-Plane data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH, modulation compression, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: WDBF for DL, DMRS-BF-NEQ for PUSCH, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: PDBF for DL, BFW Fixed point 12 bits, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer

Table A.3.1.14-3: Capability to be reported or validated for NR TDD IOT M-MIMO profile 11 in U-Plane DL data transfer test

PTCs	
PTC1	Capabilities to be validated: WDBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, DMRS-BF-NEQ for PUSCH, Non-delay managed U-plane for SRS transfer, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), Per-window symbol reordering using tx-window symbol number
PTC2	Capabilities to be validated: WDBF for DL
	Capabilities to be reported: PDBF for DL, DMRS-BF-NEQ for PUSCH, M-Plane static window control for SRS transfer, BFW Fixed point 12 bits, eAxC-ID format (3238 or 2338), Per-window symbol reordering using tx-window symbol number

Table A.3.1.14-4: Capability to be reported or validated for NR TDD IOT M-MIMO profile 11 in U-Plane data UL transfer test

PTCs	
PTC1	Capabilities to be validated: DMRS-BF-NEQ for PUSCH, modulation compression, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: WDBF or PDBF for DL, BFW BFP 8 bits, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: DMRS-BF-NEQ for PUSCH, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: WDBF or PDBF for DL, BFW Fixed point 12 bits, eAxC-ID format (3238 or 2338), M-Plane static window control for SRS transfer

A.3.1.14 NR TDD IOT M-MIMO Profile 11 - NR-TDD-FR1-CAT-B-mMIMO-CIBF-DMRS-BF-EQ

Table A.3.1.14-1 shows the base profile. The following PTCs are defined.

- 1) NR-TDD-FR1-CAT-B-mMIMO-CIBF-DMRS-BF-EQ_PTC1_[SE4.ModComp-noSE25-noReorder]
- 2) NR-TDD-FR1-CAT-B-mMIMO-CIBF-DMRS-BF-EQ_PTC2_[SE4.ModComp-SE25-PerWindow.reord-txwindow.upsymbolId]
- 3) NR-TDD-FR1-CAT-B-mMIMO-CIBF-DMRS-BF-EQ_PTC3_[SE4.ModComp-SE25-PerSection.reord-onair.upsymbolId]

Table A.3.1.14-1: NR TDD IOT M-MIMO Profile 11: NR-TDD-FR1-CAT-B-mMIMO-CIBF-DMRS-BF-EQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	DDDSUUDDDD
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size		4096
	Total channel bandwidth	-	100MHz x 1 CC
	Number of user layers	-	DL: 2 – 16 UL: 2 – 8
	Fronthaul Ethernet link	-	25Gbps x 1 lane
	PRACH preamble format	-	O-RU REQ: F0
	O-RU category	4.2.1	Category B
	LAA	7.2.5	N/A
Delay management, general	Network delay determination	4.4.4	Defined Transport Method
	O-RU adaptation of delay profile information (based on IIs-CU delay profile and transport delay)	4.4.4.3	NOT REQ
	O-DU timing advance type	4.4.5-4.4.6, Annex B	REQ: Fixed timing Advance
	SRS transfer	4.4.7	REQ: Non-delay managed U-Plane
	Delay Sets for timing parameters		REQ: Error! Reference source not found.
	T12_max	4.4, Annex B	160us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Delay management, default	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
Delay management, non default	non-default beamforming methods #1	12.6.3	REQ: DMRS-BF-EQ
	Delay Sets for timing parameters		REQ: Error! Reference source not found.
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	O-RU REQ
	Transport header	5.1.3	REQ: eCPRI
	eCPRI concatenation	5.1.3.2	NOT REQ
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 3 and 2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	O-RU REQ: 2 and 3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	O-RU REQ: 3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 8
	Fragmentation	5.5	Application layer fragmentation REQ
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority REQ
	Transport prioritization within U-Plane	5.3	NOT REQ
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID REQ
	Transport-based separation within C/U-Plane traffic	5.4	NOT REQ
Digital Power Scaling	UL gain_correction (not applicable to DMRS-BF-EQ)	8.1.3.2	0 dB
	equalization-data-scaling	12.6.3.5	REQ: Scaling-function1 or Scaling-function2
Beamforming	O-RU beamforming type	4.2.1	REQ: Digital beamforming
	Beamforming control method	7.2.1, 12.6, Annex J	DL: REQ: CIBF and PDBF UL: REQ: DMRS-BF for PUSCH, PDBF for non-PUSCH
	BFW IQ	7.7.1, Annex J	N/A
	U-Plane data compression method	8, Annex A	DL: REQ: Modulation compression, no selective RE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
IQ compression			UL: REQ: BFP
	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	REQ for Mod Comp in DL: 4 bits for BFP in UL: 8 and 9 bits for DMRS-BF and 9 bits for other BF methods
	IQ data frame format not including udCompHdr field	8.3.3.13	REQ
C-Plane	Section Type 0	7.4.2	O-RU REQ
	Section Type 1	7.4.3	REQ
	Section Type 3	7.4.5	REQ (for PRACH)
	Section Type 4	7.4.6	NOT REQ
	Section Type 5	7.4.7	REQ
	Section Type 6	7.4.8	REQ
	Section Type 7	7.4.9	N/A
	Section Type 8	7.4.10	NOT REQ
	Section Type 9	7.4.11	REQ
	Section Type 10	7.4.12	REQ
	Section Type 11	7.4.13	O-RU REQ
	"symInc" flag	7.5.3.3	NOT REQ
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	REQ: Single C-Plane message
	Section extension 1 (beamforming weights)	7.7.1	NOT REQ
	Section extension 2 (beamforming attributes)	7.7.2	NOT REQ
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	NOT REQ
	Section extension 4 (modulation compr. params)	7.7.4	REQ
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	NOT REQ
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 7 (Multiple-eAxC designation)	7.7.7	NOT REQ
	Section extension 8 (regularization factor)	7.7.8	NOT REQ
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	NOT REQ
	Section extension 10 (Multiple ports grouping)	7.7.10	REQ
	Section extension 11 (Flexible BF weights)	7.7.11	NOT REQ
	Section extension 12	7.7.12	NOT REQ
	Section extension 13	7.7.13	NOT REQ
	Section extension 14	7.7.14	NOT REQ
	Section extension 15	7.7.15	NOT REQ
	Section extension 16	7.7.16	NOT REQ
	Section extension 17	7.7.17	NOT REQ
	Section extension 18	7.7.18	NOT REQ
	Section extension 19	7.7.19	NOT REQ
	Section extension 20	7.7.20	NOT REQ
	Section extension 21	7.7.21	NOT REQ
	Section extension 22	7.7.22	NOT REQ
	Section extension 23	7.7.23	NOT REQ
	Section extension 24	7.7.24	REQ
	Section extension 25	7.7.25	PTC1: NOT REQ PTC2, PTC3: REQ
	Section extension 26	7.7.26	O-RU REQ
	Section extension 27	7.7.27	NOT REQ
DMRS-BF	Port reduced DMRS data sending	12.6.3.1	O-RU REQ
	User group self assembly	7.7.24	NOT REQ
	Per-window symbol reordering	7.7.25, 12.6.1.5	PTC1, PTC3: NOT REQ PTC2: REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
			If "per-window symbol reordering" is used, then "per-section tx-window reassignment" is not used.
	Per section tx-window reassignment	7.7.25, 12.6.1.5	PTC1, PTC2: NOT REQ PTC3: REQ If "per-section tx-window reassignment" is used, then "per-window symbol reordering" is not used.
	symbolId meaning when reordering (up-symbolId-type-supported)	7.7.25	PTC1: N/A PTC2: Transmission window number REQ PTC3: on-air symbol number REQ
	Transform precoding supported	7.7.24	O-RU REQ
	PUSCH and DMRS mux supported	7.7.24	O-RU REQ
	DMRS configuration Type (dType)	7.7.24, 7.7.24.14	REQ: Type 1 (dType=0)
	dmrsSymbolMask	7.7.24	REQ: 1+1 (symbol location: 2, 11)
	different-transform-precoding-in-user-group-supported	7.7.24	NOT REQ
	different-cdm-without-data-in-user-group-supported	7.7.24	NOT REQ
Measurements	SINR reporting	7.2.11, 7.4.11	O-RU REQ
	MEAS-UE-TAE (UE timing advance error)	9.2.1	O-RU REQ
	MEAS-UE-LAYER-POWER	9.2.1	O-RU REQ
	MEAS-UE-FREQ-OFFSET	9.2.1	O-RU REQ
	MEAS-IPN-ALLOC	9.2.1	O-RU REQ
	MEAS-IPN-UNALLOC	9.2.1	O-RU REQ
	MEAS-ANT-DMRS-SNR	9.2.1	NOT REQ
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	REQ
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	NOT REQ
	Local PRTC	5.2.3, 10.1, 11	N/A
	Topology configuration	11.2.2, 11.2.3 (for PLFS)	REQ: LLS-C1 or LLS-C3

Table A.3.1.14-2 lists “to be reported” and “to be validated” capabilities in radio layer 3 C-plane establishment and initial radio U-plane data transfer test. Table A.3.1.14-3 lists “to be reported” and “to be validated” capabilities in U-plane DL data transfer test. Table A.3.1.14-4 lists “to be reported” and “to be validated” capabilities in U-plane UL data transfer test.

Table A.3.1.14-2: Capability to be reported or validated for NR TDD IOT M-MIMO profile 11 in radio layer 3 C-Plane establishment and initial radio U-Plane data transfer test

PTCs	
PTC1	Capabilities to be validated: CIBF for DL, DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: CIBF for DL, DMRS-BF-EQ for PUSCH, modulation compression, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: PDBF for DL, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC3	Capabilities to be validated: CIBF for DL, DMRS-BF-EQ for PUSCH, modulation compression, Per-section tx-window reassignment using on-air symbol number
	Capabilities to be reported: PDBF for DL, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer

Table A.3.1.14-3: Capability to be reported or validated for NR TDD IOT M-MIMO profile 11 in U-Plane DL data transfer test

PTCs	
PTC1	Capabilities to be validated: CIBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, SE4, DMRS-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits)
PTC2	Capabilities to be validated: CIBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, SE4, DMRS-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Per-window symbol reordering using tx-window symbol number
PTC3	Capabilities to be validated: CIBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, SE4, DMRS-BF-EQ for PUSCH, Non-delay managed U-plane for SRS transfer, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Per-section tx-window reassignment using on-air symbol number

Table A.3.1.14-4: Capability to be reported or validated for NR TDD IOT M-MIMO profile 11 in U-Plane data UL transfer test

PTCs	
PTC1	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression
	Capabilities to be reported: CIBF or PDBF for DL, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC2	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression, Per-window symbol reordering using tx-window symbol number
	Capabilities to be reported: CIBF or PDBF for DL, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer
PTC3	Capabilities to be validated: DMRS-BF-EQ for PUSCH, modulation compression, Per-section tx-window reassignment using on-air symbol number
	Capabilities to be reported: CIBF or PDBF for DL, SE4, eAxC-ID format (3238 or 2338), PUSCH data IQ bitwidth (8 bits or 9 bits), Non-delay managed U-plane for SRS transfer

A.3.1.15 NR TDD IOT M-MIMO Profile 12 - NR-TDD-FR1-CAT-B-mMIMO-CIBF-DMRS-BF-NEQ

Table A.3.1.15-1 describes the base profile. The following PTC is defined.

- 1) NR-TDD-FR1-CAT-B-mMIMO-CIBF-DMRS-BF-NEQ_PTC1_[SE4.ModComp-noReorder]

Table A.3.1.15-1: NR TDD IOT M-MIMO Profile 12: NR-TDD-FR1-CAT-B-mMIMO-CIBF-DMRS-BF-NEQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR TDD
	TDD configuration	-	DDDSUDDDDD
	Nominal sub-carrier spacing	-	30kHz
	SSB sub-carrier spacing	-	30kHz
	Nominal FFT size		4096
	Total channel bandwidth	-	100MHz x 1 CC
	Number of user layers	-	DL: 2 – 16 UL: 2 – 8
	Fronthaul Ethernet link	-	25Gbps x 1 lane
	PRACH preamble format	-	O-RU REQ: F0

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	O-RU category	4.2.1	Category B
	LAA	7.2.5	N/A
Delay management, general	Network delay determination	4.4.4	Defined Transport Method
	O-RU adaptation of delay profile information (based on IIs-CU delay profile and transport delay)	4.4.4.3	NOT REQ
	O-DU timing advance type	4.4.5-4.4.6, Annex B	REQ: Fixed timing Advance
	SRS transfer	4.4.7	REQ: Non-delay managed U-Plane
Delay management, default	Delay Sets for timing parameters		REQ: Error! Reference source not found.
	T12_max	4.4, Annex B	160us
	T12_min	4.4, Annex B	0us
	T34_max	4.4, Annex B	160us
	T34_min	4.4, Annex B	0us
Delay management, non default	non-default beamforming methods #1	12.6.3	REQ: DMRS-BF-NEQ
	Delay Sets for timing parameters		REQ: Error! Reference source not found.
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	O-RU REQ
	Transport header	5.1.3	REQ: eCPRI
	eCPRI concatenation	5.1.3.2	NOT REQ
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 3 and 2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	O-RU REQ: 2 and 3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	O-RU REQ: 3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	O-RU REQ: 8
	Fragmentation	5.5	Application layer fragmentation REQ
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS Priority REQ
	Transport prioritization within U-Plane	5.3	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID REQ
	Transport-based separation within C/U-Plane traffic	5.4	NOT REQ
Digital Power Scaling	UL gain_correction (not applicable to DMRS-BF-EQ)	8.1.3.2	0 dB
	equalization-data-scaling	12.6.3.5	N/A
Beamforming	O-RU beamforming type	4.2.1	REQ: Digital beamforming
	Beamforming control method	7.2.1, 12.6, Annex J	DL: REQ: CIBF and PDBF UL: REQ: DMRS-BF for PUSCH, PDBF for non-PUSCH
	BFW IQ	7.7.1, Annex J	N/A
	U-Plane data compression method	8, Annex A	DL: REQ: Modulation compression, no selective RE UL: REQ: BFP
IQ compression	U-Plane (DL/UL) data IQ bitwidth	8, Annex D	DL: REQ: 4 bits UL: REQ: 9 bits
	IQ data frame format not including udCompHdr field	8.3.3.13	REQ
C-Plane	Section Type 0	7.4.2	O-RU REQ
	Section Type 1	7.4.3	REQ
	Section Type 3	7.4.5	REQ (for PRACH)
	Section Type 4	7.4.6	NOT REQ
	Section Type 5	7.4.7	REQ
	Section Type 6	7.4.8	REQ
	Section Type 7	7.4.9	N/A
	Section Type 8	7.4.10	NOT REQ
	Section Type 9	7.4.11	NOT REQ
	Section Type 10	7.4.12	NOT REQ
	Section Type 11	7.4.13	NOT REQ
	"symInc" flag	7.5.3.3	NOT REQ
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	REQ: Single C-Plane message

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 1 (beamforming weights)	7.7.1	NOT REQ
	Section extension 2 (beamforming attributes)	7.7.2	NOT REQ
	Section extension 3 (DL Precoding configuration parameters and indications)	7.7.3	NOT REQ
	Section extension 4 (modulation compr. params)	7.7.4	REQ
	Section extension 5 (modulation compression additional scaling parameters)	7.7.5	NOT REQ
	Section extension 6 (Non-contiguous PRB allocation)	7.7.6	NOT REQ
	Section extension 7 (Multiple-eAxC designation)	7.7.7	NOT REQ
	Section extension 8 (regularization factor)	7.7.8	NOT REQ
	Section extension 9 (Dynamic Spectrum Sharing parameters)	7.7.9	NOT REQ
	Section extension 10 (Multiple ports grouping)	7.7.10	REQ
	Section extension 11 (Flexible BF weights)	7.7.11	NOT REQ
	Section extension 12	7.7.12	NOT REQ
	Section extension 13	7.7.13	NOT REQ
	Section extension 14	7.7.14	NOT REQ
	Section extension 15	7.7.15	NOT REQ
	Section extension 16	7.7.16	NOT REQ
	Section extension 17	7.7.17	NOT REQ
	Section extension 18	7.7.18	NOT REQ
	Section extension 19	7.7.19	NOT REQ
	Section extension 20	7.7.20	NOT REQ
	Section extension 21	7.7.21	NOT REQ
	Section extension 22	7.7.22	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section extension 23	7.7.23	NOT REQ
	Section extension 24	7.7.24	REQ
	Section extension 25	7.7.25	NOT REQ
	Section extension 26	7.7.26	O-RU REQ
	Section extension 27	7.7.27	NOT REQ
DMRS-BF	Port reduced DMRS data sending	12.6.3.1	REQ
	User group self assembly	7.7.24	NOT REQ
	Per-window symbol reordering with DMRS sending	7.7.25, 12.6.1.5	NOT REQ
	Per section tx-window reassignment	7.7.25, 12.6.1.5	NOT REQ
	symbolid meaning when reordering (up-symbolid-type-supported)	7.7.25	N/A
	Transform precoding supported	7.7.24	O-RU REQ
	PUSCH and DMRS mux supported	7.7.24	O-RU REQ
	DMRS configuration Type (dType)	7.7.24, 7.7.24.14	REQ: Type 1 (dType=0)
	dmrsSymbolMask	7.7.24	REQ: 1+1 (symbol location: 2, 11)
	different-transform-precoding-in-user-group-supported	7.7.24	NOT REQ
	different-cdm-without-data-in-user-group-supported	7.7.24	NOT REQ
Measurements	SINR reporting	7.2.11, 7.4.11	N/A
	MEAS-UE-TAE (UE timing advance error)	9.2.1	NOT REQ. O-RU declares feature DMRS-BF-NEQ-UNALTERED-TAE.
	MEAS-UE-LAYER-POWER	9.2.1	NOT REQ
	MEAS-UE-FREQ-OFFSET	9.2.1	NOT REQ. O-RU declares feature DMRS-BF-NEQ-UNALTERED-FREQ-OFFSET.
	MEAS-IPN-ALLOC	9.2.1	NOT REQ
	MEAS-IPN-UNALLOC	9.2.1	NOT REQ
	MEAS-ANT-DMRS-SNR	9.2.1	NOT REQ

Category	Item	Related O-RAN CUS-Plane specification section(s)	
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	REQ
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	NOT REQ
	Local PRTC	5.2.3, 10.1, 11	N/A
	Topology configuration	11.2.2, 11.2.3 (for PLFS)	REQ: LLS-C1 or LLS-C3

Table A.3.1.15-2 lists “to be reported” and “to be validated” capabilities in radio layer 3 C-plane establishment and initial radio U-plane data transfer test. Table A.3.1.15-3 lists “to be reported” and “to be validated” capabilities in U-plane DL data transfer test. Table A.3.1.15-4 lists “to be reported” and “to be validated” capabilities in U-plane UL data transfer test.

Table A.3.1.15-2: Capability to be reported or validated for NR TDD IOT M-MIMO profile 12 in radio layer 3 C-Plane establishment and initial radio U-Plane data transfer test

PTCs	
PTC1	Capabilities to be validated: CIBF for DL, DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: PDBF for DL, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer

Table A.3.1.15-3: Capability to be reported or validated for NR TDD IOT M-MIMO profile 12 in U-Plane DL data transfer test

PTCs	
PTC1	Capabilities to be validated: CIBF for DL, modulation compression
	Capabilities to be reported: PDBF for DL, SE4, DMRS-BF-NEQ for PUSCH, Non-delay managed U-plane for SRS transfer, eAxC-ID format (3238 or 2338)

Table A.3.1.15-4: Capability to be reported or validated for NR TDD IOT M-MIMO profile 12 in U-Plane data UL transfer test

PTCs	
PTC1	Capabilities to be validated: DMRS-BF-NEQ for PUSCH, modulation compression
	Capabilities to be reported: CIBF or PDBF for DL, SE4, eAxC-ID format (3238 or 2338), Non-delay managed U-plane for SRS transfer

A.3.2 NR FDD

A.3.2.1 NR FDD IOT Profile 1 - NR-FDD-FR1(15kHzSCS)-CAT-B-DBF

Profile Test Configurations:

- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[20MHz-8SS-PRACH0-9bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[30MHz-8SS-PRACH0-9bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[20MHz-16SS-PRACH0-9bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[30MHz-16SS-PRACH0-9bitIQ-llsC2]

- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[20MHz-8SS-PRACHC2-9bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[30MHz-8SS-PRACHC2-9bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[20MHz-16SS-PRACHC2-9bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[30MHz-16SS-PRACHC2-9bitIQ-llsC2]

- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[20MHz-8SS-PRACH0-12bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[30MHz-8SS-PRACH0-12bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[20MHz-16SS-PRACH0-12bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[30MHz-16SS-PRACH0-12bitIQ-llsC3]

- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[20MHz-8SS-PRACHC2-12bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[30MHz-8SS-PRACHC2-12bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[20MHz-16SS-PRACHC2-12bitIQ-llsC3]

- NR-FDD-FR1(15kHzSCS)-CAT-B-DBF_[30MHz-16SS-PRACHC2-12bitIQ-llsC3]

Table A.3.2.1-1: NR FDD IOT Profile 1 - NR-FDD-FR1(15kHzSCS)-CAT-B-DBF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR FDD
	TDD configuration	-	NA
	Nominal sub-carrier spacing	-	15 kHz
	SSB sub-carrier spacing	-	15 kHz
	Nominal FFT size	-	2048
	Total channel bandwidth	-	Entry1: 20MHz Entry2: 20MHz+10MHz
	Number of spatial/antenna streams	-	Entry1: 8 Entry2: 16
	Fronthaul Ethernet link	-	25Gbps x 1lane for 8 spatial streams, 25Gbps x 2lane for 16 spatial streams
	PRACH preamble format	-	Entry1: 0 Entry2: C2
	RU category	4.2.1	Category B
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 437us
	T1a_min_up	4.4, Annex B	More than or equal to 366us
	T2a_max_up	4.4, Annex B	More than or equal to 437us
	T2a_min_up	4.4, Annex B	Less than or equal to 206us
	Tcp_adv_dl	4.3.2, Annex B	125 us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Ta3_max_up	4.4, Annex B	Less than or equal to 232us
	Ta3_min_up	4.4, Annex B	More than or equal to 70us
	Ta4_max_up	4.4, Annex B	More than or equal to 392us
	Ta4_min_up	4.4, Annex B	Less than or equal to 70us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 356us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 356us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	8
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Digital Power Scaling	UL gain_correction	8.1.3.2	0dB
Beamforming	RU beamforming type	4.2.1 (10.1)	Digital beamforming
	Beamforming control method	7.3, Annex J	Beamforming ID based
IQ compression	U-Plane data compression method	8, Annex A	Block floating point
	U-Plane data IQ bitwidth	8, Annex D	Entry1: 9 Entry2: 12
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	FALSE
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C2 Entry2: lls-C3

A.3.2.2 NR FDD IOT Profile 2 - NR-FDD-FR1-CAT-B-DBF

Profile Test Configurations:

- NR-FDD-FR1-CAT-B-DBF_[20MHz-8SS-9bitIQ-llsC2]
- NR-FDD-FR1-CAT-B-DBF_[30MHz-8SS-9bitIQ-llsC2]
- NR-FDD-FR1-CAT-B-DBF_[20MHz-16SS-9bitIQ-llsC2]
- NR-FDD-FR1-CAT-B-DBF_[30MHz-16SS-9bitIQ-llsC2]
- NR-FDD-FR1-CAT-B-DBF_[20MHz-8SS-12bitIQ-llsC3]
- NR-FDD-FR1-CAT-B-DBF_[30MHz-8SS-12bitIQ-llsC3]
- NR-FDD-FR1-CAT-B-DBF_[20MHz-16SS-12bitIQ-llsC3]
- NR-FDD-FR1-CAT-B-DBF_[30MHz-16SS-12bitIQ-llsC3]

Table A.3.2.2-1: NR FDD IOT Profile 2 - NR-FDD-FR1-CAT-B-DBF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR FDD
	TDD configuration	-	NA
	Nominal sub-carrier spacing	-	30 kHz
	SSB sub-carrier spacing	-	30 kHz
	Nominal FFT size	-	1024
	Total channel bandwidth	-	Entry1: 20MHz Entry2: 20MHz+10MHz
	Number of spatial/antenna streams	-	Entry1: 8 Entry2: 16

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Fronthaul Ethernet link	-	25Gbps x 1lane for 8 spatial streams, 25Gbps x 2lane for 16 spatial streams
	PRACH preamble format	-	0
	RU category	4.2.1	Category B
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 345us
	T1a_min_up	4.4, Annex B	More than or equal to 294us
	T2a_max_up	4.4, Annex B	More than or equal to 345us
	T2a_min_up	4.4, Annex B	Less than or equal to 134us
	Tcp_adv_dl	4.3.2, Annex B	125 us
	Ta3_max_up	4.4, Annex B	Less than or equal to 171us
	Ta3_min_up	4.4, Annex B	More than or equal to 50us
	Ta4_max_up	4.4, Annex B	More than or equal to 331us
	Ta4_min_up	4.4, Annex B	Less than or equal to 50us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 336us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 336us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	8
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0dB
Beamforming	RU beamforming type	4.2.1..(10.1)	Digital beamforming
	Beamforming control method	7.3, Annex J	Beamforming ID based
IQ compression	U-Plane data compression method	8, Annex A	Block floating point
	U-Plane data IQ bitwidth	8, Annex D	Entry1: 9 Entry2: 12
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	FALSE
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C2 Entry2: lls-C3

A.3.2.3 NR FDD IOT Profile 3 - NR-FDD-FR1-CAT-A-NoBF

Profile Test Configurations:

- NR-FDD-FR1-CAT-A-NoBF_[20MHz-9bitIQ-llsC2]
- NR-FDD-FR1-CAT-A-NoBF_[30MHz-9bitIQ-llsC2]
- NR-FDD-FR1-CAT-A-NoBF_[20MHz-12bitIQ-llsC3]
- NR-FDD-FR1-CAT-A-NoBF_[30MHz-12bitIQ-llsC3]

Table A.3.2.3-1: NR FDD IOT Profile 3 - NR-FDD-FR1-CAT-A-NoBF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR FDD
	TDD configuration	-	NA
	Nominal sub-carrier spacing	-	30 kHz
	SSB sub-carrier spacing	-	30 kHz
	Nominal FFT size	-	1024
	Total channel bandwidth	-	Entry1: 20MHz Entry2: 20MHz+10MHz
	Number of spatial/antenna streams	-	4
	Fronthaul Ethernet link	-	10Gbps x 1 lane
	PRACH preamble format	-	0
	RU category	4.2.1	Category A
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 345us
	T1a_min_up	4.4, Annex B	More than or equal to 294us
	T2a_max_up	4.4, Annex B	More than or equal to 345us
	T2a_min_up	4.4, Annex B	Less than or equal to 134us
	Tcp_adv_dl	4.3.2, Annex B	125 us
	Ta3_max_up	4.4, Annex B	Less than or equal to 171us
	Ta3_min_up	4.4, Annex B	More than or equal to 50us
	Ta4_max_up	4.4, Annex B	More than or equal to 331us
	Ta4_min_up	4.4, Annex B	Less than or equal to 50us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 336us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 336us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	8
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0dB
Beamforming	RU beamforming type	4.2.1..(10.1)	No beamforming
	Beamforming control method	7.3, Annex J	Beamforming ID based (always "0")
IQ compression	U-Plane data compression method	8, Annex A	Block floating point

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	U-Plane data IQ bitwidth	8, Annex D	Entry1: 9 Entry2: 12
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	FALSE
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C2 Entry2: lls-C3

A.3.2.4 NR FDD IOT Profile 4 - NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF

Profile Test Configurations:

- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-4SS-PRACHF0-9bitIQ-llsC2]

- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[20MHz-4SS-PRACHF0-9bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-4SS-PRACHF0-12bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[20MHz-4SS-PRACHF0-12bitIQ-llsC2]

- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-4SS-PRACHF0-9bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[20MHz-4SS-PRACHF0-9bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-4SS-PRACHF0-12bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[20MHz-4SS-PRACHF0-12bitIQ-llsC3]

- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-2SS-PRACHF0-9bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[20MHz-2SS-PRACHF0-9bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-2SS-PRACHF0-12bitIQ-llsC2]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[20MHz-2SS-PRACHF0-12bitIQ-llsC2]

- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-2SS-PRACHF0-9bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[20MHz-2SS-PRACHF0-9bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-2SS-PRACHF0-12bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[20MHz-2SS-PRACHF0-12bitIQ-llsC3]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-2SS-PRACHF0-14bitIQ-llsC1C2]
- NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF_[10MHz-2SS-PRACHF1-14bitIQ-llsC1C2]

Table A.3.2.4-1: NR FDD IOT Profile 4 - NR-FDD-FR1(15kHzSCS)-CAT-A-NoBF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	NR FDD
	TDD configuration	-	NA
	Nominal sub-carrier spacing	-	15 kHz
	SSB sub-carrier spacing	-	15 kHz
	Nominal FFT size	-	1024 for 10MHz BW, 2048 for 20MHz BW
	Total channel bandwidth	-	Entry1: 10MHz Entry2: 20MHz
	Number of spatial/antenna streams	-	Entry1: 4 Entry2: 2
	Fronthaul Ethernet link	-	10Gbps x 1 lane
	PRACH preamble format	-	Entry1: Long preamble F0 Entry2: Long preamble F1
	RU category	4.2.1	Category A
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 437us
	T1a_min_up	4.4, Annex B	More than or equal to 366us
	T2a_max_up	4.4, Annex B	More than or equal to 437us
	T2a_min_up	4.4, Annex B	Less than or equal to 206us
	Tcp_adv_dl	4.3.2, Annex B	125 us
	Ta3_max_up	4.4, Annex B	Less than or equal to 232us
	Ta3_min_up	4.4, Annex B	More than or equal to 70us
	Ta4_max_up	4.4, Annex B	More than or equal to 392us
	Ta4_min_up	4.4, Annex B	Less than or equal to 70us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Ta3_max_up (scs=1.25kHz)	4.4, Annex B	Less than or equal to 1650us
	Ta3_min_up (scs=1.25kHz)	4.4, Annex B	More than or equal to 827us
	Ta4_max_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	More than or equal to 1810us
	Ta4_min_up (scs=1.25kHz)	4.4.3-4.4.4, Annex B	Less than or equal to 827us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 356us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 356us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	6
	eAxC ID CC_ID bitwidth	5.1.3.2.7	4
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	4
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Digital Power Scaling	UL gain_correction	8.1.3.2	0dB
Beamforming	RU beamforming type	4.2.1..(10.1)	No beamforming
	Beamforming control method	7.3, Annex J	Beamforming ID based (always "0")
IQ compression	U-Plane data compression method	8, Annex A	Block floating point
	U-Plane data IQ bitwidth	8, Annex D	Entry1: 9 Entry2: 12 Entry3: 14
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	FALSE
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C2 (can also apply lls-C1) Entry2: lls-C3

A.3.3 LTE FDD

A.3.3.1 LTE FDD IOT Profile 1 - LTE-FDD-FR1-CAT-B-DBF

Profile Test Configurations:

- LTE-FDD-FR1-CAT-B-DBF_[20MHz-8SS-9bitIQ-llsC2]
- LTE-FDD-FR1-CAT-B-DBF_[30MHz-8SS-9bitIQ-llsC2]
- LTE-FDD-FR1-CAT-B-DBF_[20MHz-16SS-9bitIQ-llsC2]
- LTE-FDD-FR1-CAT-B-DBF_[30MHz-16SS-9bitIQ-llsC2]
- LTE-FDD-FR1-CAT-B-DBF_[20MHz-8SS-12bitIQ-llsC3]
- LTE-FDD-FR1-CAT-B-DBF_[30MHz-8SS-12bitIQ-llsC3]
- LTE-FDD-FR1-CAT-B-DBF_[20MHz-16SS-12bitIQ-llsC3]
- LTE-FDD-FR1-CAT-B-DBF_[30MHz-16SS-12bitIQ-llsC3]

Table A.3.3.1-1: LTE FDD IOT Profile 1 - LTE-FDD-FR1-CAT-B-DBF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	LTE FDD
	TDD configuration	-	NA
	Nominal sub-carrier spacing	-	15 kHz
	SSB sub-carrier spacing	-	NA
	Nominal FFT size	-	2048
	Total channel bandwidth	-	Entry1: 20MHz Entry2: 20MHz+10MHz

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Number of spatial/antenna streams	-	Entry1: 8 Entry2: 16
	Fronthaul Ethernet link	-	25Gbps x 1lane for 8 spatial streams, 25Gbps x 2lane for 16 spatial streams
	PRACH preamble format	-	0
	RU category	4.2.1	Category B
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 437us
	T1a_min_up	4.4, Annex B	More than or equal to 366us
	T2a_max_up	4.4, Annex B	More than or equal to 437us
	T2a_min_up	4.4, Annex B	Less than or equal to 206us
	Tcp_adv_dl	4.3.2, Annex B	125 us
	Ta3_max_up	4.4, Annex B	Less than or equal to 232us
	Ta3_min_up	4.4, Annex B	More than or equal to 70us
	Ta4_max_up	4.4, Annex B	More than or equal to 392us
	Ta4_min_up	4.4, Annex B	Less than or equal to 70us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 356us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 356us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	8
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL_gain_correction	8.1.3.2	0dB
Beamforming	RU beamforming type	4.2.1..(10.1)	Digital beamforming
	Beamforming control method	7.3, Annex J	Beamforming ID based
IQ compression	U-Plane data compression method	8, Annex A	Block floating point
	U-Plane data IQ bitwidth	8, Annex D	Entry1: 9 Entry2: 12
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	TRUE
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C2 Entry2: lls-C3

A.3.3.2 LTE FDD IOT Profile 2 - LTE-FDD-FR1-CAT-A-NoBF

Profile Test Configurations:

- LTE-FDD-FR1-CAT-A-NoBF_[10MHz-2SS-9bitIQ-llsC1C2]
- LTE-FDD-FR1-CAT-A-NoBF_[20MHz-2SS-9bitIQ-llsC1C2]
- LTE-FDD-FR1-CAT-A-NoBF_[10MHz-2SS-12bitIQ-llsC1C2]
- LTE-FDD-FR1-CAT-A-NoBF_[20MHz-2SS-12bitIQ-llsC1C2]
- LTE-FDD-FR1-CAT-A-NoBF_[10MHz-4SS-9bitIQ-llsC3]

- LTE-FDD-FR1-CAT-A-NoBF_[20MHz-4SS-9bitIQ-llsC3]
- LTE-FDD-FR1-CAT-A-NoBF_[10MHz-4SS-12bitIQ-llsC3]
- LTE-FDD-FR1-CAT-A-NoBF_[20MHz-4SS-12bitIQ-llsC3]

- LTE-FDD-FR1-CAT-A-NoBF_[10MHz-2SS-9bitIQ-llsC3]
- LTE-FDD-FR1-CAT-A-NoBF_[15MHz-2SS-9bitIQ-llsC3]
- LTE-FDD-FR1-CAT-A-NoBF_[15MHz-4SS-9bitIQ-llsC3]
- LTE-FDD-FR1-CAT-A-NoBF_[20MHz-2SS-9bitIQ-llsC3]

- LTE-FDD-FR1-CAT-A-NoBF_[40MHz-2SS-12bitIQ-llsC1C2]
- LTE-FDD-FR1-CAT-A-NoBF_[20MHz-4SS-12bitIQ-llsC1C2]
- LTE-FDD-FR1-CAT-A-NoBF_[20MHz-4SS-12bitIQ-llsC1C2-Jumbo]
- LTE-FDD-FR1-CAT-A-NoBF_[20MHz-2SS-12bitIQ-llsC3-Jumbo]
- LTE-FDD-FR1-CAT-A-NoBF_[20MHz-4SS-12bitIQ-llsC3-Jumbo]
- LTE-FDD-FR1-CAT-A-NoBF_[40MHz-4SS-12bitIQ-llsC3-Jumbo]

- LTE-FDD-FR1-CAT-A-NoBF_[5MHz-2SS-16bitIQ-llsC1C2]
- LTE-FDD-FR1-CAT-A-NoBF_[60MHz-4SS-16bitIQ-llsC1C2-Jumbo]

Table A.3.3.2-1: LTE FDD IOT Profile 2 - LTE-FDD-FR1-CAT-A-NoBF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	LTE FDD

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	TDD configuration	-	NA
	Nominal sub-carrier spacing	-	15 kHz
	SSB sub-carrier spacing	-	NA
	Nominal FFT size	-	1024 for 10MHz BW, 1536 for 15MHz BW, 2048 for 20MHz BW
	Total channel bandwidth	-	Entry1: 5MHz Entry2: 10MHz Entry3: 15MHz Entry4: 20MHz Entry5: 20MHz+20MHz Entry6: 20MHz+20MHz+20MHz
	Number of spatial/antenna streams	-	Entry1: 2 Entry2: 4
	Fronthaul Ethernet link	-	10Gbps x 1lane
	PRACH preamble format	-	0
	RU category	4.2.1	Category A
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 437us
	T1a_min_up	4.4, Annex B	More than or equal to 366us
	T2a_max_up	4.4, Annex B	More than or equal to 437us
	T2a_min_up	4.4, Annex B	Less than or equal to 206us
	Tcp_adv_dl	4.3.2, Annex B	125 us
	Ta3_max_up	4.4, Annex B	Less than or equal to 232us
	Ta3_min_up	4.4, Annex B	More than or equal to 70us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Ta4_max_up	4.4, Annex B	More than or equal to 392us
	Ta4_min_up	4.4, Annex B	Less than or equal to 70us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 356us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 356us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	Entry1: FALSE Entry2: TRUE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	2
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	6
	eAxC ID CC_ID bitwidth	5.1.3.2.7	4
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	4
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0dB

Category	Item	Related O-RAN CUS-Plane specification section(s)	
Beamforming	RU beamforming type	4.2.1..(10.1)	No beamforming
	Beamforming control method	7.3, Annex J	Beamforming ID based (always "0")
IQ compression	U-Plane data compression method	8, Annex A	Entry1: Block floating point Entry2: No compression
	U-Plane data IQ bitwidth	8, Annex D	For Block floating point: Entry1: 9 Entry2: 12 For No compression: 16
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	FALSE
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C1 (can also apply lls-C2) Entry2: lls-C3

A.3.4 LTE TDD

A.3.4.1 LTE TDD IOT Profile 1 - LTE-TDD-FR1-CAT-A-DBF

Profile Test Configurations:

- LTE-TDD-FR1-CAT-A-DBF_[20MHz-9bitIQ-llsC2]
- LTE-TDD-FR1-CAT-A-DBF_[30MHz-9bitIQ-llsC2]
- LTE-TDD-FR1-CAT-A-DBF_[40MHz-9bitIQ-llsC2]

- LTE-TDD-FR1-CAT-A-DBF_[20MHz-16bitIQ-llsC2]
- LTE-TDD-FR1-CAT-A-DBF_[30MHz-16bitIQ-llsC2]
- LTE-TDD-FR1-CAT-A-DBF_[40MHz-16bitIQ-llsC2]

- LTE-TDD-FR1-CAT-A-DBF_[20MHz-9bitIQ-llsC3]
- LTE-TDD-FR1-CAT-A-DBF_[30MHz-9bitIQ-llsC3]
- LTE-TDD-FR1-CAT-A-DBF_[40MHz-9bitIQ-llsC3]

- LTE-TDD-FR1-CAT-A-DBF_[20MHz-16bitIQ-llsC3]
- LTE-TDD-FR1-CAT-A-DBF_[30MHz-16bitIQ-llsC3]
- LTE-TDD-FR1-CAT-A-DBF_[40MHz-16bitIQ-llsC3]

Table A.3.4.1-1: LTE TDD IOT Profile 1 - LTE-TDD-FR1-CAT-A-DBF

Category	Item	Related O-RAN CUS-Plane specification section(s)	
General	Radio access technology	-	LTE TDD
	TDD configuration	-	subframeAssignment sa2
	Nominal sub-carrier spacing	-	15 kHz
	SSB sub-carrier spacing	-	NA
	Nominal FFT size	-	1024 for 10MHz BW, 2048 for 20MHz BW
	Total channel bandwidth	-	Entry1: 20MHz Entry2: 20MHz+10MHz Entry3: 20MHz+20MHz
	Number of spatial/antenna streams	-	8
	Fronthaul Ethernet link	-	10Gbps x 1 lane for 20MHz 10Gbps x 2 lane for 20MHz+10MHz, 20MHz+20MHz
	PRACH preamble format	-	0
	RU category	4.2.1	Category A
	LAA	-	FALSE
Delay management	Network delay determination	4.4.4.2	Defined Transport Method
	RU adaptation of delay profile information (based on O-DU delay profile and transport delay)	4.4.4.3	FALSE
	O-DU timing advance type	4.4.5-4.4.6, Annex B	Fixed Timing Advance
	T1a_max_up	4.4, Annex B	Less than or equal to 437us
	T1a_min_up	4.4, Annex B	More than or equal to 366us
	T2a_max_up	4.4, Annex B	More than or equal to 437us
	T2a_min_up	4.4, Annex B	Less than or equal to 206us
	Tcp_adv_dl	4.3.2, Annex B	125 us
	Ta3_max_up	4.4, Annex B	Less than or equal to 232us
	Ta3_min_up	4.4, Annex B	More than or equal to 70us
	Ta4_max_up	4.4, Annex B	More than or equal to 392us

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Ta4_min_up	4.4, Annex B	Less than or equal to 70us
	T1a_max_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 356us
	T1a_min_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 285us
	T2a_max_cp_ul	4.4.3-4.4.4, Annex B	More than or equal to 356us
	T2a_min_cp_ul	4.4.3-4.4.4, Annex B	Less than or equal to 125us
	T12_max	4.4, Annex B	160 us
	T12_min	4.4, Annex B	0 us
	T34_max	4.4, Annex B	160 us
	T34_min	4.4, Annex B	0 us
	Non-delay managed U-Plane traffic	4.4.7	FALSE
C/U-Plane transport	Transport encapsulation	5.1.1-5.1.2	Ethernet
	Jumbo frames	5.1.2	FALSE
	Transport header	5.1.3	eCPRI
	eCPRI concatenation	5.1.3.1-5.1.3.2	FALSE
	eAxC ID DU_Port_ID bitwidth	5.1.3.2.7	4
	eAxC ID BandSector_ID bitwidth	5.1.3.2.7	3
	eAxC ID CC_ID bitwidth	5.1.3.2.7	3
	eAxC ID RU_Port_ID bitwidth	5.1.3.2.7	6
	Fragmentation	5.5	Application layer fragmentation
	Transport prioritization across C/U/S/M-Plane	5.3	Default L2 CoS priority
	Transport prioritization within U-Plane	5.3	False (Default U-Plane priority applies)-Plane
	Separation of C/U-Plane and M-Plane traffic	5.4	VLAN ID
	Transport-based separation within C/U-Plane traffic	5.4	FALSE
Digital Power Scaling	UL gain_correction	8.1.3.2	0dB
Beamforming	RU beamforming type	4.2.1..(10.1)	Digital beamforming

Category	Item	Related O-RAN CUS-Plane specification section(s)	
	Beamforming control method	7.3, Annex J	Beamforming weight based
IQ compression	U-Plane data compression method	8, Annex A	Entry1: Block floating point Entry2: No compression
	U-Plane data IQ bitwidth	8, Annex D	9 for Block floating point, 16 for No compression
	IQ data frame format not including udCompHdr field	8.3.3.13	TRUE
C-Plane	Section Type 0	7.4.2	TRUE
	Section Type 1	7.4.3	TRUE
	Section Type 3	7.4.5	TRUE
	Section Type 5	7.4.7	FALSE
	Section Type 6	7.4.8	FALSE
	Section Type 7	7.4.9	FALSE
	"symInc" flag	7.5.3.3	FALSE (always set to '0')
	C-Plane for PRACH formats with preamble repetition	7.2.3.4	Single C-Plane message
	Section extension 1	7.7.1	FALSE
	Section extension 2	7.7.2	FALSE
	Section extension 3	7.7.3	TRUE
	Section extension 4	7.7.4	FALSE
	Section extension 5	7.7.5	FALSE
S-Plane	PTP Full Timing Support (G.8275.1)	5.2.3, 10.1, 11	TRUE
	PTP Partial Timing Support (G.8275.2)	5.2.3, 10.1, 11	FALSE
	Local PRTC	5.2.3, 10.1, 11	FALSE
	Topology configuration	11.2.2 10, 11.2.3 (for PLFS)	Entry1: lls-C2 Entry2: lls-C3

A.4 Delay Sets

A.4.1 NR TDD FR1 Cat-B mMIMO WDBF Delay Sets for DMRS-BF profiles

Table A.4.1-1 defines the Delay Sets for NR TDD FR1 Cat-B mMIMO WDBF when used in DMRS-BF profiles. For IOT entry, both O-DU and O-RU shall share at least one common Delay Set.

Table A.4.1-1: NR TDD FR1 CAT-B mMIMO WDBF Delay Sets for DMRS-BF profile

C/U-Plane timing parameters	Range	DS-W1 (WDBF DS1)	DS-W2 (WDBF DS2)
T1a_max_up	Less than or equal to	345 us	475 us
T1a_min_up	More than or equal to	294 us	400 us
T1a_max_cp_dl	Less than or equal to	669 us	971 us
T1a_min_cp_dl	More than or equal to	419 us	550 us
T2a_max_up	More than or equal to	345 us	475 us
T2a_min_up	Less than or equal to	134 us	240 us
T2a_max_cp_dl	More than or equal to	669 us	971 us
T2a_min_cp_dl	Less than or equal to	259 us	390 us
Tcp_adv_dl		125 us	150 us
Ta3_max_up	Less than or equal to	171 us	280 us
Ta3_min_up	More than or equal to	50 us	50 us
Ta4_max_up	More than or equal to	331 us	440 us
Ta4_min_up	Less than or equal to	50 us	50 us
Ta3_max_up (scs=1.25kHz)	Less than or equal to	1650 us	1650 us
Ta3_min_up (scs=1.25kHz)	More than or equal to	827 us	827 us
Ta4_max_up (scs=1.25kHz)	More than or equal to	1810 us	1810 us
Ta4_min_up (scs=1.25kHz)	Less than or equal to	827 us	827 us
T1a_max_cp_ul	Less than or equal to	840 us	840 us
T1a_min_cp_ul	More than or equal to	660 us	660 us
T2a_max_cp_ul	More than or equal to	840 us	840 us
T2a_min_cp_ul	Less than or equal to	500 us	500 us

A.4.2 NR TDD FR1 Cat-B mMIMO CIBF Delay Sets

Delay Set for NR TDD FR1 Cat-B mMIMO CIBF is defined in Table A.4.2-1.

Table A.4.2-1: NR TDD FR1 CAT-B mMIMO CIBF Delay Sets

C/U-Plane timing parameters	Range	DS-C1 (CIBF DS1)
T1a_max_up	Less than or equal to	345 us
T1a_min_up	More than or equal to	294 us
T1a_max_cp_dl	Less than or equal to	820 us
T1a_min_cp_dl	More than or equal to	769 us
T2a_max_up	More than or equal to	345 us
T2a_min_up	Less than or equal to	134 us
T2a_max_cp_dl	More than or equal to	820 us
T2a_min_cp_dl	Less than or equal to	609 us
Tcp_adv_dl		475 us
Ta3_max_up	Less than or equal to	171 us
Ta3_min_up	More than or equal to	50 us
Ta4_max_up	More than or equal to	331 us
Ta4_min_up	Less than or equal to	50 us
Ta3_max_up (scs=1.25kHz)	Less than or equal to	1650 us
Ta3_min_up (scs=1.25kHz)	More than or equal to	827 us
Ta4_max_up (scs=1.25kHz)	More than or equal to	1810 us
Ta4_min_up (scs=1.25kHz)	Less than or equal to	827 us
T1a_max_cp_ul	Less than or equal to	336 us
T1a_min_cp_ul	More than or equal to	285 us
T2a_max_cp_ul	More than or equal to	336 us
T2a_min_cp_ul	Less than or equal to	125 us

A.4.3 NR TDD FR1 Cat-B mMIMO DMRS-BF Delay Sets

The Delay Sets specified in Table A.4.3-1 apply to both DMRS-BF-EQ and DMRS-BF-NEQ. For entry to IOT testing, both O-DU and O-RU shall share at least one common Delay Set. DS-D1 (DMRS-BF DS 1) is recommended for interoperability testing. DS-D2 (DMRS-BF DS 2) may alternatively be used for interoperability testing. For profiles in which measurements are not required, support for 2g timing parameters is optional.

Table A.4.3-1: NR TDD FR1 CAT-B mMIMO DMRS-BF Delay Sets

C/U-Plane timing parameters	Range	DS-D1 (DMRS-BF DS 1)	DS-D2 (DMRS-BF DS 2)
Ta3_max_2g	Less than or equal to	1280 us	1280 us
Ta3_min_2g	More than or equal to	571 us	571 us
Ta4_max_2g	More than or equal to	1440 us	1440 us
Ta4_min_2g	Less than or equal to	571 us	571 us
Ta3_max_up	Less than or equal to	1080 us	1180 us
Ta3_min_up	More than or equal to	823 us	886 us
Ta4_max_up	More than or equal to	1240 us	1340 us
Ta4_min_up	Less than or equal to	823 us	886 us
T1a_max_cp_ul (NOTE)	Less than or equal to	840 us	840 us
T1a_min_cp_ul (NOTE)	More than or equal to	660 us	660 us
T2a_max_cp_ul (NOTE)	More than or equal to	840 us	840 us
T2a_min_cp_ul (NOTE)	Less than or equal to	500 us	500 us
NOTE: This is applicable to all C-plane messages constrained by T2a_max_cp_ul and T2a_min_cp_ul sent to the DMRS-BF endpoint.			

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Annex (informative): Change history/Change request (history)

Date	Revision	Description
2019.09.05	01.00	First published version for Fronthaul Interoperability Test Specification (IOT)
2020.03.13	02.00	Second published version for Fronthaul Interoperability Test Specification (IOT)
2020.11.5	03.00	Implement change requests, minor editorials, slave replaced by subordinate CIS_2020 10 13-oRAN-WG4 IoT CR CIS-16 SSHv2 ciphers (3) O-RAN.WG4.IOT.v2.00_FM_RVG3 Includes text mandating both the IOT Profile and the Spec version are nominated to ensure the appropriate parameter values are referenced.
2021.03.18	04.00	Rename Profiles as Profile Test Configurations NOK-0052 M-Plane correction NEC-0009 rev1 M-Plane system recovery action for S-Plane degraded conditions NEC-0010 Update eexc_id to align with M-Plane and CUS-Plane description Update eexc_id to eexc-id to align with M-Plane Specification Include mMIMO IOT profile and associated test equipment update mMIMO Profile Test Configurations and FFS Includes values of Ta3_max-up, Ta3_min_up, Ta4_max_up, Ta4_min_up in the mMIMO profiles which were accidentally omitted.
2021.07.06	05.00	CRs and text proposals were implemented along with editorial updates Correct description of delay profile tests MAV-0017 Add mMIMO IOT profile test configurations VIA-CR0005 Update mMIMO IOT profile re section extension 1,11 and 4,5 MAV-0022 & 0026 Add mMIMO PRACH delay parameters VIA-CR0006 Updating IOT profile for S-plane and aligning them with CUS table 8-2 NOK-0075 Annex A text to clarify the intent and completeness of an IOT profile test
2021.11.14	06.00	CR and text proposals and editorial updates and updates to references IQ compression field correction for mMIMO IOR profiles MAV-0027 New mMIMO IOT profile timing parameters ERI-0030 Update of NR TDD IOT profile 1 CHM-0003 Update of NR TDD profile 2 DCM-0015 Update of NR FDD profile4 DCM-0016 M-Plane IOT profile for IPv6 and TLS RMI-0003 + IOT Profile Configurations
2022.03.08	07.00	CR and text proposals and editorial updates Correct M-Plane tests to use Ipv6/TLS VIA-0007 Add M-Plane hybrid mode profile Ipv6/TLS VIA-0008

2022.08.03	08.00	CR text proposals, update reference to CUS/M specs, editorial updates O-RAN FH IOT profile modifications for NR TDD DCM-0020 IOT Profile modifications for TDD configuration of UL heavy pattern DCM-0021 Removal of Annex ZZZ and related references
2022.11.08	09.00	CR text proposals, editorial updates Alignment to template VIA CR-0009 IOT-profile-terminology update QCM CR-0056
2023.03.17	10.00	CR text proposals, editorial updates Align to TSC drafting rules NEC CR-0020 Modification to delay management tests VIA CR-0009 Proposed set of parameters to assign as modifiable or forbidden for Customized IOT Profiles concept introduced into version 9.0 VIA CR-0008
2024.03.18	11.00	CR text proposals, editorial updates Clarification that it is mandatory to test at least 1 synchronization topology and provided clear test classifications for each test DTAG -0001
2024.07.03	12.00	CR text proposals, editorial updates Update that it is conditional mandatory to test M-Plane hybrid mode, conditional upon the O-DU supporting it. DCM-022
2024.12.03	12.00.01	CR text proposals, editorial updates Correction of references: ERI-0127 Update clause numbering ERI-0133
2025.03.04	12.00.02	Correction of clause numbering CR implementation after comments from Ericsson and Qualcomm
2025.03.11	12.00.03	Baseline version with changes from ERI-0127 and ERI-0133 accepted
2025.03.12	12.00.04	CR text proposals, editorial updates VIA-0010 DMRS-BF uplink throughput test v13 ERI-0134 IOT addition of ULPI profiles v10
2025.03.16	12.00.05	Correction of implementation after comments from NEC
2025.03.18	12.00.06	Correction of implementation after comments from ERI, QCM, NOK, and call to resolve issues
2025.03.19	12.00.07	Reinstate delay sets (accidentally removed)
2025.03.21	12.00.08	Minor updates from discussion in ad-hoc call, apply O-RAN formatting to tables
2025.03.23	13.00.00	Accept all changes and change title and file name to 13.00.00
2025.03.25	13.00	Update file name to 13.00 and minor editorial corrections